Are Frontier VLM Agents Ready to Be Robot Generalists?

An Empirical Study with the Embodied Agent Arena

Haojian Huang[1,3], **Pukun Zhao**[3], **Zexi Li**[2,3], **Yehang Zhang**[1,3], **Yangkai Wei**[3], **Wenqian Li**[3], **Han Yang**[3], **Kaiwen Zhou**[3], **Ying-Cong Chen**[1,3,†], **Yinchuan Li**[3,†]

[1] HKUST (GZ) [2] The Chinese University of Hong Kong [3] Knowin AI

Abstract. A robot generalist must turn perception and reasoning into actions that satisfy complete task goals. Frontier vision-language models (VLMs) perform many of these operations, but local competence does not establish complete-task capability. We introduce Embodied Agent Arena to evaluate ten VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation. Its 1,000 cases combine 32 established sources with GeoProbe, our 168-case benchmark for camera, motion, depth, and scale estimation. A minimal harness leaves task decisions to the model through source-native interfaces, while separate measures of precision, grounding, and goal satisfaction expose where progress stops. The comparison reveals complementary strengths: Astra leads rotation estimation and planning, Gemini 4 Argon leads spatial accuracy and region overlap, and Opus 5.5 leads manipulation success. Across TraceSpatial, HumanCLAW, CALVIN, and household manipulation, accurate paths, plausible contacts, and repeated actions can leave terminal conditions unsatisfied. The remaining challenge is sustained execution: reaching precise goal states, preserving achieved conditions, and recovering from failures within finite time and interaction budgets.

Project Page GitHub

1 Introduction

A robot generalist must turn scene understanding into actions that satisfy a complete goal. Locating an object, selecting a usable contact, and executing a grasp are intermediate achievements: the object must still reach its destination in the required state. Vision-language models (VLMs) increasingly address spatial relations (Ma et al., 2024, Yang et al., 2024) and action-relevant grounding (Wang et al., 2025a, Song et al., 2024). Learned visuomotor policies connect observations to robot actions (Kim et al., 2024, Black et al., 2024), while language-model controllers select skills or generate executable programs (Ahn et al., 2022, Liang et al., 2022). Perception support and visual action rehearsal further extend these systems (Fu et al., 2026, Guo et al., 2026, Li et al., 2026b, Zhang et al., 2026a). As execution competence expands, the central question becomes: *Are frontier VLM agents ready to be robot generalists?*

Answering this question requires distinguishing what an agent gets right locally from what it completes. A low trajectory error can coexist with a missed endpoint; a valid contact can coexist

[†]Corresponding authors.

Figure 1. Overview of Embodied Agent Arena. Each row illustrates six tasks in one capability domain, spanning camera and scene estimation, spatial relations, functional grounding, goal-directed planning, and robot manipulation. Fixed-input probes test estimates and judgments; interactive episodes test goal completion. Spatial-tracing overlays show reference paths and a 3D target region.

with an unsatisfied interaction goal. Conversely, a terminal failure can hide correct localization and substantial intermediate progress. Evaluating only isolated judgments or final success obscures this distinction. We therefore study the model's own perception, reasoning, and action selection across task requirements, connecting local measurements to the conditions that remain unsatisfied at termination.

We introduce Embodied Agent Arena, which evaluates ten VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation (Figure 1). The arena combines established benchmarks with GeoProbe, which varies camera motion, object motion, and scale references to diagnose geometric estimation. A minimal harness preserves source observations and operations while leaving task decisions to the model. Separate measures of metric precision, functional grounding, and native goal completion make it possible to examine both capability breadth and the failures hidden by aggregate scores.

The results show complementary strengths without consistent complete-task competence. Astra leads rotation estimation and planning, Gemini 4 Argon leads spatial accuracy and region overlap, and Opus 5.5 leads manipulation success. Within tasks, accurate paths miss required endpoints, agents reach seats without establishing a sitting state, and repeated actions leave household goals unfinished. Some Planning tasks defeat every model, while unsuccessful episodes consume a disproportionate share of tokens. Supplementary experiments on visual feedback and answer review examine whether additional information resolves these difficulties. The challenge is to turn useful local decisions into complete goals within the execution budget.

Our contributions are:

- **An evaluation centered on complete task requirements.** We connect geometric precision and functional grounding to terminal goal satisfaction across the five domains.
- **A unified arena with a new geometric benchmark, GeoProbe.** The 1,000-case arena combines 32 established sources with 168 GeoProbe cases, retaining native operations and success criteria in a shared model-directed evaluation framework.
- **A diagnosis of the gap between local competence and complete execution.** Ten-model comparisons and execution traces identify endpoint errors, unsatisfied interaction states, and repeated activity without task progress as concrete obstacles to embodied generalism.

2 Related Work

Embodied evaluation has developed along two complementary directions. Control suites measure execution and task generalization (James et al., 2019, Mees et al., 2021), while visual diagnostics isolate spatial inference (Ma et al., 2024, Yang et al., 2024) and interaction grounding (Song et al., 2024, Qian & Fouhey, 2023). Scene and task question answering (Majumdar et al., 2024, Sermanet et al., 2023) extends this diagnostic view to embodied decisions (Li et al., 2024b, Qi et al., 2025). These approaches test goal completion and the information needed to act. Evaluating both in the same agents connects local judgments to terminal outcomes.

The transition from judgment to action also depends on how control is represented. Learned policies map observations and instructions directly to actions (Brohan et al., 2023, Zitkovich et al., 2023), with broader task coverage motivating generalist policies (Ghosh et al., 2024, Kim et al., 2024, Black et al., 2024). Language-mediated control instead composes supplied skills (Ahn et al., 2022, Huang et al., 2023b) or executable programs (Liang et al., 2022, Singh et al., 2023). Programs expose control primitives, dependencies (Fu et al., 2026, Du et al., 2025), geometric feedback (Huang et al., 2023a), and reusable interfaces (Li et al., 2026a) to model-directed composition. Decisions still depend on visual evidence (Li et al., 2025, Wu et al., 2026) and its consistency over time (Huang et al., 2026b, Zhao et al., 2026b,a). Demonstrations (Jiang et al., 2023, Huang et al., 2026a), retained histories (Zhang et al., 2026b), and execution verification (Guo et al., 2024, Huang et al., 2026c) provide different forms of support. Task success thus reflects both model decisions and interface support.

Broader benchmarks bring perception and goal-directed interaction into one evaluation (Yang et al., 2025a, Cheng et al., 2025). OpenEQA (Majumdar et al., 2024), Embodied Arena, and Hard Vision, Easy Vision connect visual probes with navigation (Ni et al., 2025, Rasheed et al., 2026); VLABench also tests manipulation (Zhang et al., 2024). Table 1 compares their coverage, diagnostics, and scale. Embodied Agent Arena extends this evaluation perspective across metric geometry, spatial inference, functional grounding, planning, and manipulation, with GeoProbe isolating camera, motion, and scale requirements. It measures how far native VLM decisions support embodied generalism across diverse tasks.

3 The Embodied Agent Arena

Table 1. Evaluation coverage, diagnostic analysis, and sample scale. Fixed probes use supplied observations; region probes score image grounding; manipulation requires arm/gripper control; memory spans sessions. Calib./pose tests camera calibration or relative pose; Forms denotes task-type/condition breakdowns. Scale counts QA items or task episodes; NR: aggregate count not reported (Appendix G).

	Sample scale	Evaluation coverage					Diagnostic analysis	
Benchmark	Instances	Fixed	Region	Nav.	Manip.	Memory	Calib./pose	Forms
OpenEQA (2024)	1,636	✓	✗	✓	✗	✗	✗	✓
VLABench (2024)	NR	✓	✗	✗	✓	✗	✗	✓
BEAR (2025)	4,469	✓	✓	✗	✓*	✗	✗	✓
EmbodiedBench (2025a)	1,128	✗	✗	✓	✓	✗	✗	✓
EmbodiedEval (2025)	328	✗	✗	✓	✗	✗	✗	✓
Embodied Arena (2025)	NR	✓	✓	✓	✗	✗	✗	✓
CaP-X (2026)	700/tier	✗	✗	✓	✓	✗	✗	✓
WorldLines (2026b)	331	✓	✗	✗	✗	✓	✗	✓
Hard/Easy Vision (2026)	NR	✓	✓	✓	✗	✗	✗	✓
Embodied Agent Arena	**1,000**	✓	✓	✓	✓	✗	✓	✓

✓ included; ✗ not evaluated; *BEAR-Agent online validation. Scale details: Appendix G.

3.1 Benchmark Construction and Capability Domains

From local requirements to complete goals. The arena organizes 1,000 cases from 32 established sources and GeoProbe into five capability domains (Figures 1–2). Each case specifies observations, a query or goal, available operations, and scoring. Geometry, Spatial Reasoning, and Affordance isolate what the agent must estimate or ground; Task Planning and Manipulation test whether decisions produce the required environment state. GeoProbe adds 168 cases that vary camera motion, object motion, and scale references in controlled Blender scenes, complemented by real images. This construction makes metric ambiguities and incomplete execution separately observable. Appendix A details source allocation and adaptations.

Geometry: metric precision and feasibility. Agents estimate intrinsics, pose, depth, dimensions, or displacement from images. InFlux and Map-free test calibration and relative pose (Liang et al., 2025, Arnold et al., 2022); GeoProbe varies camera motion and scale references. TraceSpatial-Bench adds paths on 56 scenes (Zhou et al., 2026), pairing trajectory error with initiation, arrival, and collision constraints. These complementary measures distinguish a close geometric estimate from one that satisfies the task's spatial requirements.

Spatial Reasoning: relations and reference frames. Agents infer relations, viewpoints, configurations, and temporal order from images or video (Ma et al., 2024, Wang et al., 2025b, Yang et al., 2024). Comparing object distances, imagining another observer's view, and inferring camera motion require different reference-frame transformations. Questions specify the relevant frame and expect a category, quantity, or point set.

Affordance: contact and functional extent. Agents ground an intended action through a contact point or region. UMD tests tool functions such as cutting and scooping (Myers et al., 2015); instruction- and action-conditioned 3DOI queries test interaction contacts (Qian & Fouhey, 2023). Point validity measures whether a contact lies in the functional region. Region overlap measures how much of that region the model recovers, using native boxes and, in a separate experiment, direct polygons.

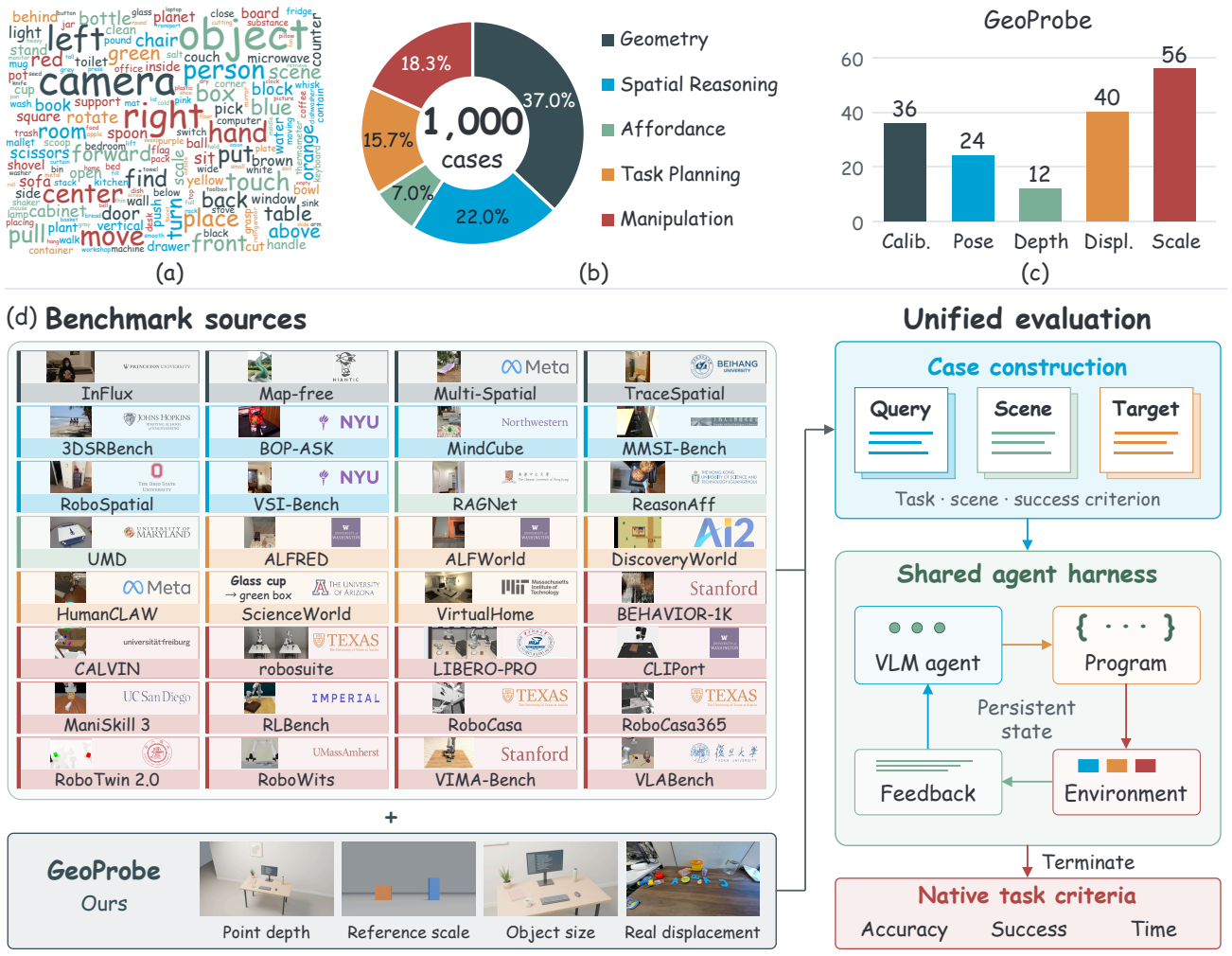


Figure 2. Construction of the 1,000-case arena. (a) Representative task vocabulary across sources. (b) Five-domain composition, including 112 spatial-tracing cases in Geometry. (c) GeoProbe estimation targets. (d) Source queries, scenes, and targets connect model interaction to termination and scoring. Colors identify domains; logos denote source institutions.

Task Planning: satisfying all goal conditions. Agents compose source-provided operations into household, navigation, or scientific procedures (Shridhar et al., 2019, Wang et al., 2022), using visual, textual, or symbolic feedback. Compound goals require several conditions to hold, such as heating an object and placing it at the destination. For terminal state s_T and required predicates $g_1, \ldots, g_K$, complete success is

$$\text{Success}(s_T, g) = \mathbf{1}\left[\bigwedge_{k=1}^{K} g_k(s_T)\right]. \tag{1}$$

Intermediate scores identify which predicates in Equation (1) remain unsatisfied.

Manipulation: executing and preserving progress. Agents execute robot motions or skills for placement, insertion, articulation, and coordinated handling (James et al., 2019, Chen et al., 2025a). Language-conditioned variants change the target relation or object (Zhang et al., 2024); sequential tasks require preserving progress across ordered goals. Native task success records completion, while condition-level scores expose achieved relations and control events.

3.2 Unified Agent Harness

Task decisions remain with the model. The harness provides persistent interaction through each source's observations and operations (Figure 2d). Models interpret observations, select operations, and construct action sequences from feedback. We use *native capability* to mean these model-directed decisions through the supplied source interfaces, including their helpers. The baseline adds no perception or control modules beyond those interfaces. Geometry, Spatial Reasoning, Affordance, and Planning use a Python runtime; Manipulation follows its native agent loop. Variables persist between Python rounds, and images produced during a round enter the next model request. Appendix B.1 specifies the interfaces; the supplementary review study adds specialist measurements (Section 4.4) (Zhang et al., 2025, Huang et al., 2026c).

Termination is distinct from success. Fixed-input submissions commit predictions. Interactive episodes end on agent termination, an environment terminal condition, or budget exhaustion. The evaluator then checks the submitted output or environment state independently of the agent's declaration. This separates an agent that stops from one that has completed the goal. Appendix B.2 specifies the round, token, action, and time limits.

3.3 Evaluation Protocol and Metrics

Completion and precision. Spatial Reasoning and Planning report Pass@1; categorical spatial accuracy is also reported separately. Manipulation pairs native success with consecutive goals and task-specific condition or control-evidence scores. Geometry and numerical spatial queries use MAE, RMSE, AbsRel/MAPE, displacement error, or pose error as appropriate to the target. Tracing pairs trajectory RMSE with offline 3D constraint satisfaction.

Functional grounding. The main table reports box–mask IoU for native rectangles and polygon–mask IoU for direct polygons on the same cases. Both compare rasterized predictions with the reference mask. Contact validity, box–box IoU, and strict box success provide complementary diagnostics in the appendix.

Variation and resource use. Repeated metrics report the equally weighted mean and sample standard deviation (SD) of run-level scores. Continuous scores average valid predictions within each run; binary success rates retain all assigned cases. Tokens, action calls, model rounds, and elapsed time describe different resources consumed before termination. Section 4.1 specifies repetitions, and Appendix B.3 defines aggregation and accounting.

4 Experiments

We first compare capability profiles, then examine where local successes stop short of complete tasks. We then test the effects of observation frequency and answer review.

4.1 Experimental Setup

Models. We evaluate Astra, Sol, Fable, Opus 5.5, Qwen-Max, Gemini 3.8 Flash, Gemini 4 Argon, Qwen-397B, Qwen-27B, and DeepSeek-V4.1-Flash-1017 (DeepSeek-1017). Appendix B.2 gives full identifiers and configurations. DeepSeek-1017 uses the `deepseek-flash` endpoint updated on 17 October 2026.

Repetitions and execution conditions. Non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance use five runs per case for nine models and three for Fable. We report

the mean and sample SD across run-level scores. Tracing, direct polygons, Planning, and Manipulation each use one episode per assignment. Polygon IoU covers the same 70 Affordance cases for all ten agents (Appendix C.5). Planning retains all 157 assignments, and Manipulation all 182 binary assignments; execution errors count as unsuccessful. All agents use native RGB observations and tools. The eight Baseline agents follow the source-native observation schedule; DeepSeek-1017 and Gemini 4 Argon receive current RGB at each Manipulation decision. Gemini 4 Argon uses the source-specific budgets detailed in Appendix J. Source-specific execution limits and repetition details appear in Appendices E.1, C.3, and C.4. Table 2 compares the resulting domain metrics.

4.2 Main Results

Capability leadership does not transfer uniformly across tasks. Different metrics favor different models (Table 2; Figure 3). Astra has the lowest rotation error and highest Planning success. Gemini 4 Argon leads Spatial Pass@1 and both region-overlap metrics, while Opus 5.5 leads Baseline Manipulation success. The contrast is especially clear for Qwen-Max: its Spatial Pass@1 is close to Astra's (68.9% versus 69.4%), yet Planning success is 35.7% versus 80.3%. Gemini 3.8 Flash has the lowest numerical spatial error, rather than the highest categorical or task success. These profiles distinguish the capabilities needed to reason about a scene from those needed to act through a complete task.

Strong local performance coexists with incomplete execution. Even Opus 5.5, the leading Baseline manipulation agent, completes only 47.8% of assignments. All ten models fail the five BEHAVIOR-1K tasks, leaving required final conditions unsatisfied within the execution limits. Domain-level rankings alone do not explain these failures. The task diagnostics below examine the gap directly, from accurate trajectories with invalid endpoints to successful target approach without the required interaction state.

Failures overlap across models and dominate token use. Within the nine-model cohort with execution-log analysis, at least one model completes 138 tasks, only 12 more than Astra alone; 19 of the 157 tasks defeat all nine. Unsuccessful episodes consume 55.5–86.9% of each model's recorded Planning tokens. For Astra, they account for 55.5% of tokens despite representing fewer than one fifth of its tasks. Thus, the remaining difficulty concerns both completing the goal and directing interaction toward useful progress. Appendix I relates these aggregate patterns to recorded actions and state changes.

4.3 From Local Competence to Complete Tasks

The five domains expose different ways in which an apparently useful prediction or action can leave the task unresolved. Figures 4–8 combine ten-model comparisons with focused task diagnostics; Appendices D, C.3, and C.4 provide the detailed results.

Geometry: low average error can conceal a failed endpoint. GeoProbe separates metric estimation from the ambiguities introduced by camera motion and scale. Among the original seven agents, Astra has the lowest errors on all five Blender estimation targets (Figure 4a). Its depth ordering reaches 96.3%, while oracle scale alignment reduces depth AbsRel from 0.104 to 0.038, locating much of the remaining error in metric scale. Viewpoint change introduces another difficulty: its object-displacement MAE rises from 0.24 to 1.13 cm between fixed and large-shift cameras (Figure 4b). These strengths depend on geometric cues: Sol has lower InFlux focal error, and mixed motion defeats the joint pose criterion across the original cohort.

Table 2. Performance across five domains. Box–mask and Polygon IoU use separate outputs on the same 70 cases. Planning covers 157 cases; Manipulation covers 182. Repeated metrics show mean ± run SD (five runs; Fable: three); the last three columns use one run. Best means are bold.

	Geometry		Spatial Reasoning		Affordance		Task Planning	Manipulation
Agent	Rot.↓ (°)	Trans.↓ (cm)	Pass@1↑ (%)	AbsRel↓	Box–mask↑ IoU	Polygon↑ IoU	Pass@1↑ (%)	Task SR↑ (%)
Astra	**17.2**±6.7	106.9±36.0	69.4±1.0	0.214±0.016	0.286±0.010	**0.379**	**80.3**	41.8
Sol	27.0±1.5	151.9±6.2	60.7±2.5	0.273±0.016	0.264±0.010	0.325	44.6	19.8
Fable	24.2±8.3	116.6±42.6	62.0±4.6	0.185±0.011	0.284±0.013	0.328	63.7	24.7
Qwen-Max	20.5±3.5	138.1±10.2	68.9±1.1	0.206±0.033	0.272±0.017	0.307	35.7	9.3
Gemini 3.8	23.7±5.6	138.0±41.7	65.4±1.3	**0.161**±0.028	0.251±0.081	0.291	48.4	18.7
Qwen-397B	38.4±1.1	249.8±10.2	50.6±1.8	0.251±0.027	0.116±0.014	0.092	17.8	9.9
Qwen-27B	38.4±3.6	213.7±12.6	48.0±0.9	0.288±0.023	0.013±0.006	0.051	19.1	11.0
Opus 5.5	23.4±3.7	135.8±12.7	67.5±5.1	0.184±0.040	0.311±0.020	0.397	48.4	**47.8**
DeepSeek-1017	30.2±3.9	182.0±29.0	51.6±1.5	0.309±0.035	0.217±0.011	0.249	31.2	20.9
Gemini 4 Argon	18.5±1.8	**105.6**±9.6	**71.6**±0.7	0.169±0.023	**0.356**±0.009	**0.441**	70.1	44.0

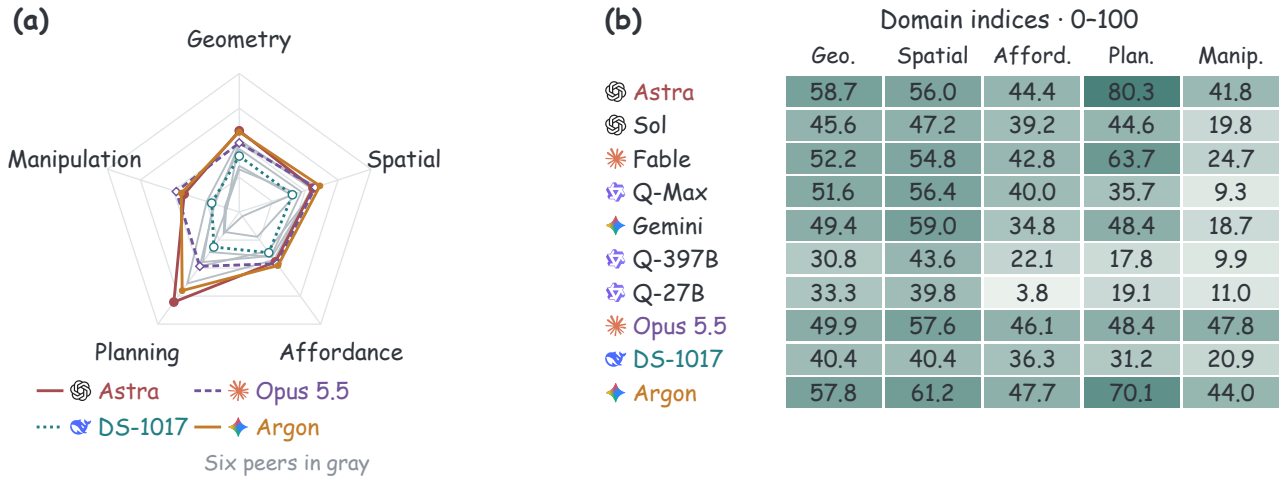


Figure 3. Capability breadth across ten agents. (a) Astra (red), Opus 5.5 (purple), DeepSeek-1017 (DS-1017, teal), Gemini 4 Argon (Argon, orange), and six peers (gray). (b) Domain indices on fixed 0–100 scales, defined in Appendix C.6; these summarize capability profiles rather than a common task success rate. DeepSeek-1017 and Argon receive per-round RGB in Manipulation; Appendix J gives Argon budgets. Planning includes all 157 cases. The Affordance index retains box–mask IoU and contact validity; Figure 6 separately examines direct polygons.

Tracing makes the distinction between precision and feasibility explicit. Gemini 4 Argon produces the lowest 3D trajectory RMSE, 0.310 m, yet satisfies joint constraints on only 42.9% of cases (Figure 4c). Astra likewise reaches 0.382 m RMSE but only 33.9% joint success; initiation or arrival fails in 36 of its 37 unsuccessful traces. Opus 5.5 and DeepSeek-1017 also have low joint success, at 8.9% and 14.3%. A path can closely follow the reference yet miss the initiation or arrival conditions required for success.

Spatial Reasoning: reference-frame demands change the ranking. Strong relational judgments do not imply equally strong viewpoint transformations. Astra reaches 77.0% on MMSI and ties Qwen-Max at 90.8% on MindCube, but trails Qwen-Max on MindCube camera shifts and Fable on 3DSRBench object-facing judgments (Figure 5a). These deficits persist in shared-run comparisons (Appendix D.2). Relating visible landmarks, inferring camera movement, and recovering an object's own axes place different demands on spatial reasoning.

The same distinction appears between depth ordering and object selection. Astra reaches 96–100% on BOP-ASK camera-depth questions but only 50% on VSI nearest-object selection, versus

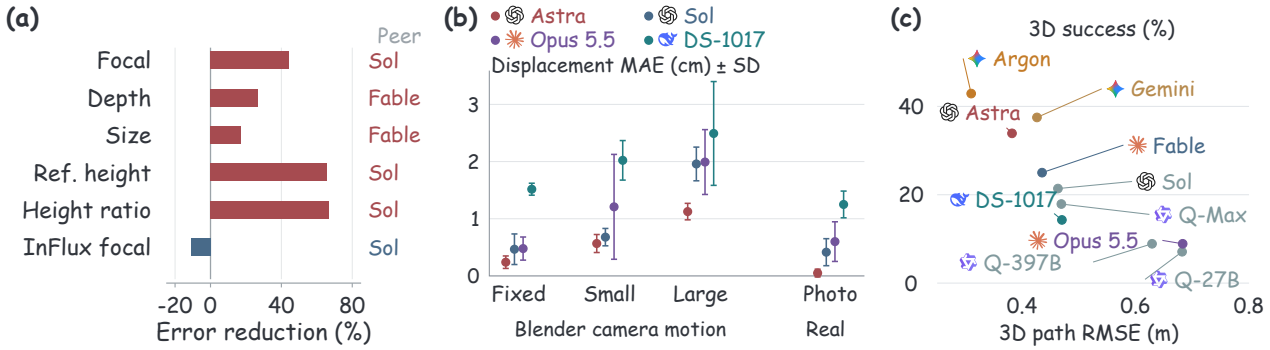


Figure 4. Geometric precision and task completion. (a) Astra's error reduction relative to the lowest-error original peer; the first five targets are GeoProbe. (b) Displacement under fixed, small-shift, and large-shift Blender cameras, plus separate real photos, for Astra, Sol, Opus 5.5, and DeepSeek-1017; mean ± sample SD of five run means. (c) Single-run 3D tracing error versus joint success for all ten agents; no repeated-run SD is available for this diagnostic. Appendix D.1 gives task-level comparisons; Appendices C.3 and C.4 detail the added models.

Qwen-Max's 76.7% (Figure 5b). Opus 5.5 and DeepSeek-1017 also score highly on BOP-ASK, at 95% and 88%, while their broader source profiles vary markedly (Figure 5c). Thus, success on a familiar spatial relation does not establish the ability to select or transform the reference frame required by another task.

Affordance: flexible boundaries cannot repair the wrong target. Direct polygons yield higher aggregate mask IoU than native boxes for nine of ten agents; Qwen-397B is the exception (Figure 6a). Gemini 4 Argon leads both representations, at 0.356 box–mask IoU and 0.441 polygon IoU. Polygons allow the model to express a more precise outline without an external segmenter, but performance remains strongly dependent on what must be grounded. Gemini 4 Argon reaches 73.1% polygon IoU on UMD tools, compared with 14.0% on RAGNet and 30.7% on ReasonAff-style scenes; Astra, Opus 5.5, and DeepSeek-1017 show the same disparity (Figure 6b).

The matched examples reveal two distinct errors: a model may select the correct part with the wrong extent, or accurately outline the wrong object (Figure A6). Representation flexibility addresses the first, while the second requires grounding the intended interaction. Contact validity adds a further distinction. Astra reaches 100% valid cutting contacts but only 0.361 box–mask IoU on the cutting region. Its pulling and two-handed contacts also differ sharply between ReasonAff and RAGNet (Figure 6c). Recognizing a usable point, recovering a functional region, and selecting the action-relevant object are separate achievements.

Task Planning: reaching a target is not the required state change. Astra's Planning advantage extends to goals that combine state changes and multiple objects. It completes 75% of ALFRED and 70% of ALFWorld tasks, compared with Fable's 35% and 30% (Figure 7a). Among the original agents, Astra alone solves ALFRED heating and dual placement (panel b). Its largest ScienceWorld gains over Fable concern linked scientific operations, including genetics and melting-point measurement; thermometer use reverses the comparison (panel c). Planning therefore requires composing a procedure and correctly executing its operations.

HumanCLAW exposes where that distinction becomes decisive. All ten agents fail all eight sitting tasks under ceilings of 101 code rounds, 2M tokens, and 180 minutes per episode (Table A6). Five Astra episodes reach the seat and issue a sitting action; Opus 5.5 finds and approaches within 20 cm in four, but neither establishes the required sitting state. These episodes separate successful approach from the requested change in interaction state.

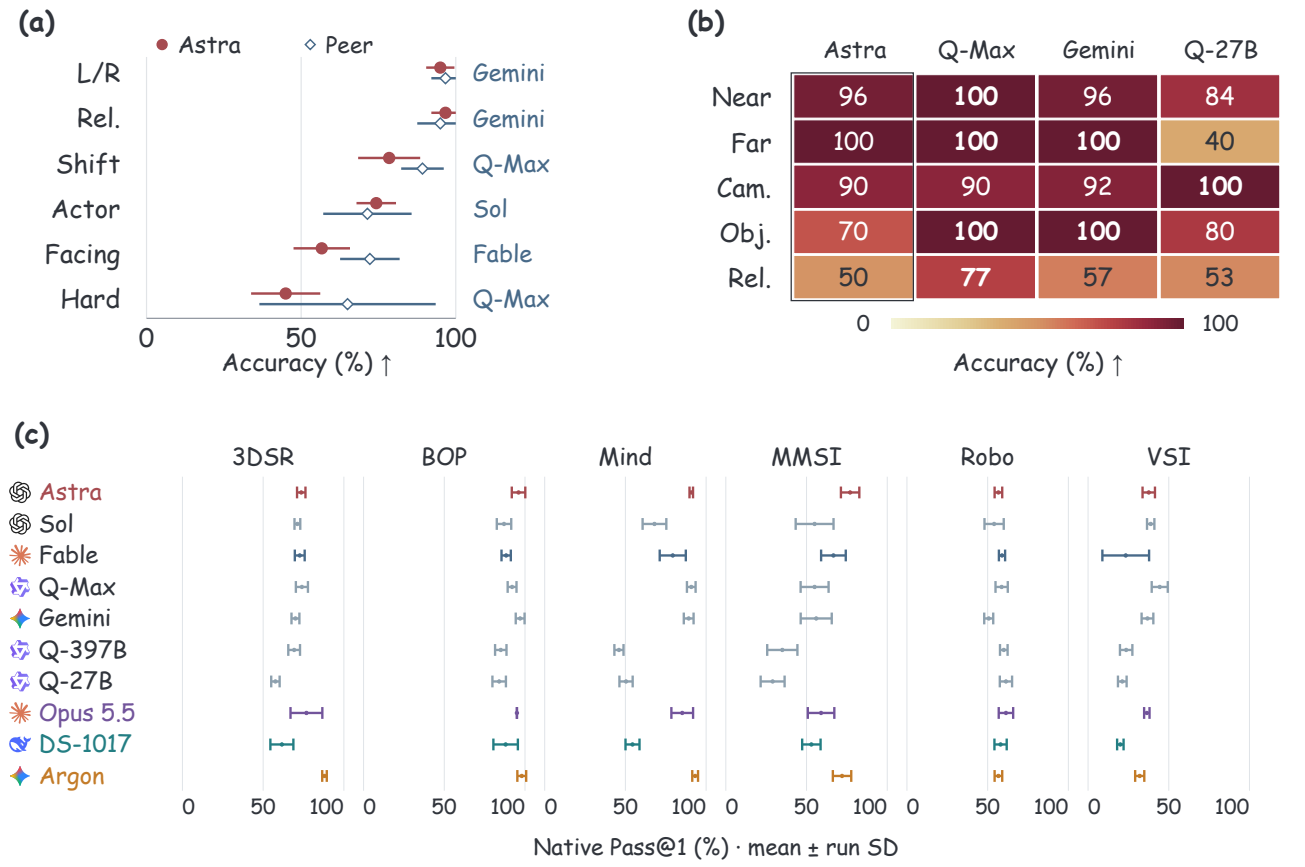


Figure 5. Spatial strengths depend on task and source. (a) Astra's relation and viewpoint accuracy versus the best original peer, with run SD. (b) Original-cohort mean accuracy on camera-relative depth (Near/Far/Cam.) and inter-object distance (Obj./Rel.). (c) Native source Pass@1 for all ten agents, including numerical questions; small dots mark means and capped whiskers span ± sample SD over five runs (three for Fable). Opus 5.5's BOP-ASK SD is zero. 3DSR, Mind, Robo, and VSI abbreviate 3DSRBench, MindCube, RoboSpatial-Home, and VSI-Bench.

Manipulation: local control must compose into sustained progress. The strongest Baseline agents have different operation profiles. Opus 5.5 ties Astra on CLIPort and ManiSkill, exceeds it on RoboTwin (46% versus 38%), and trails on RLBench (30% versus 60%; Figure 8a). Sequential control further separates partial from complete competence. Astra averages 3.10 consecutive CALVIN goals and Opus 5.5 2.40, but they complete only three and one of the ten full sequences, respectively (Appendices D.5 and C.3). Successful early actions therefore leave substantial room for failure before all five goals are achieved.

Household tasks require this progress to survive across navigation, articulated fixtures, and object handling (Nasiriany et al., 2024). Astra completes one of ten RoboCasa tasks and none in RoboCasa365; all ten agents fail the five BEHAVIOR-1K tasks within the source-specific budgets. These outcomes extend the endpoint and interaction-state failures above to multi-operation execution. A generalist must coordinate the whole sequence while preserving achieved conditions, rather than repeatedly demonstrating a subset of useful actions. Section 4.4 examines how feedback changes this process.

4.4 Feedback, Review, and Execution Progress

The main study evaluates native model decisions. We examine two ways of supplying more evidence. Execution traces then distinguish productive recovery from repetition.

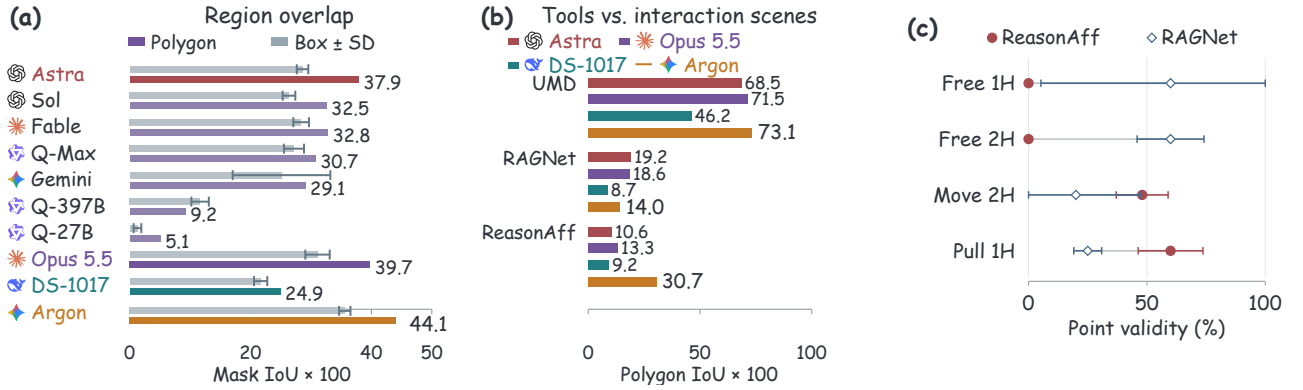


Figure 6. Flexible outlines help, but interaction scenes remain difficult. (a) Aggregate mask IoU for native boxes and direct polygons across all ten agents. Box whiskers show sample SD over five runs (three for Fable); polygon bars use one separate run on the same 70 cases. (b) Polygon IoU on tools (UMD) and interaction-conditioned scenes (RAGNet and ReasonAff) for Astra, Opus 5.5, DeepSeek-1017, and Gemini 4 Argon. (c) Astra's original contact validity by action, with run SD; 1H/2H denotes one/two hands. Invalid or absent polygon outputs score zero. Appendix C.5 defines the polygon protocol.

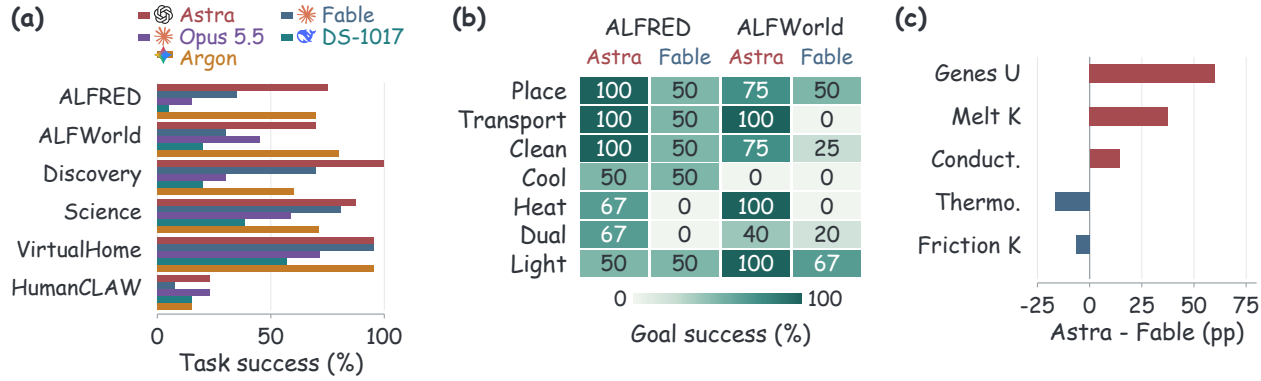


Figure 7. Planning success varies across sources and goals. (a) Source-level success for Astra, Fable, Opus 5.5, DeepSeek-1017, and Gemini 4 Argon. DeepSeek-1017 and Argon each complete 2/13 HumanCLAW tasks, both navigation. Opus 5.5 completes 3/13 HumanCLAW tasks, all navigation; none of its eight sitting tasks succeeds. (b) Astra/Fable success on seven household goal families. (c) Their five nonzero ScienceWorld gaps; thirteen families tie, and K/U denotes known/unknown conditions. Each assignment uses one run. Appendix D.4 gives the task-family breakdown; Table A11 gives Opus 5.5 coverage.

More visual feedback can change how control is organized. The paired observation experiment compares the original seven agents under their native observation schedule and with current RGB at every decision, keeping inference budgets and execution-step limits fixed. CALVIN success changes from 30% to zero for Astra and from zero to 40% for Fable (Table 3b). Astra's median input tokens rise from 0.32M to 1.21M while action calls fall from 1,795 to 481. On sequence 0557, its reusable feedback controller appears at round 12 rather than 5; expanded Fable constructs one at round 2 (Appendix E.1). In these trajectories, receiving more images does not ensure earlier establishment of effective control.

The effect also depends on the task. Across 153 common assignments, Astra remains at 63 successes while Fable rises from 34 to 65. Astra improves on RoboTwin, yet its ManiSkill condition score falls from 60% to zero (Figure 8c). Observation frequency affects how a model uses its execution budget, rather than providing a uniform increase in task competence.

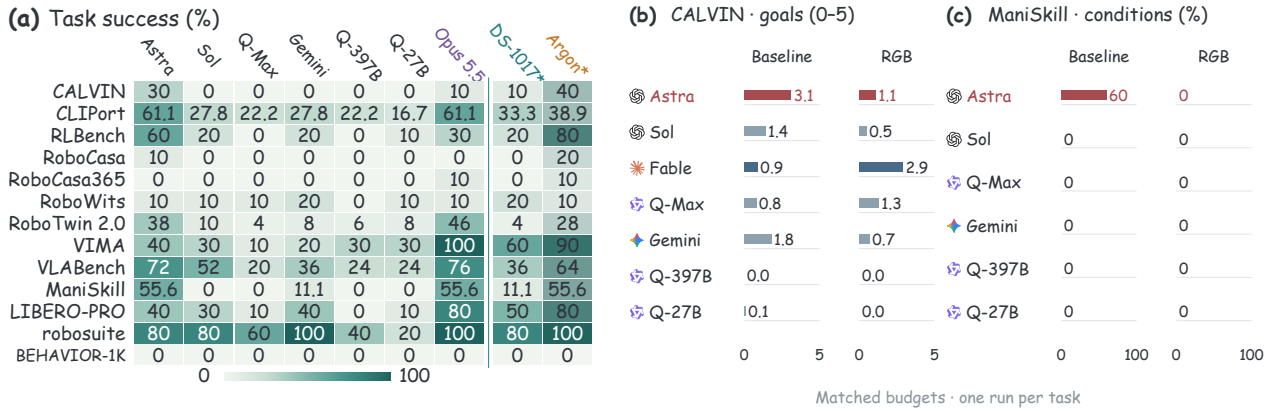


Figure 8. Manipulation across suites and execution settings. (a) Thirteen-suite success for nine agents with complete source records; *marks per-round RGB agents; Argon uses the source-specific budgets in Appendix J. (b) Mean consecutive CALVIN goals, out of five, for seven agents. (c) Native condition satisfaction on five ManiSkill tasks for six agents. Panels (b–c) include only agents with both observation settings, using matched budgets and one run per task. Fable's incomplete ManiSkill records exclude it from (a,c). Appendices D.5, C.3, and C.4 give detailed results.

Table 3. Multi-round review (a) and RGB-expanded Manipulation (b), using the original model cohort. Baseline and RGB-expanded share budgets and execution-step limits; RGB-expanded supplies per-round images. I/T/W counts improved/tied/worsened entries; SR_{50} requires box IoU≥ 0.5. Goals gives average consecutive completions (0–5); Common pools nine suites (153 tasks). Details: Appendix E.

(a)		Geometry	Spatial Reasoning	Affordance	
Model	Condition	I/T/W	Acc. (%)↑	Box IoU↑	SR_{50} (%)↑
Astra	Initial	–	**72.2**	0.2903	22.2
	Self-review	3/20/1	66.7	0.2914	22.2
	Crop review	4/19/1	66.7	0.2912	22.2
	Tool-assisted	5/15/4	**72.2**	**0.3174**	**33.3**
Qwen-27B	Initial	–	55.6	0.0336	0.0
	Self-review	6/15/3	50.0	**0.1524**	0.0
	Crop review	3/19/1	44.4	0.0937	0.0
	Tool-assisted	10/6/8	**61.1**	0.1452	**11.1**

(b)		CALVIN		ManiSkill	VIMA	Common
Model	Condition	SR (%)↑	Goals↑	SR (%)↑	SR (%)↑	SR (%)↑
Astra	Baseline	**30.0**	**3.10**	**55.6**	40.0	**41.2**
	RGB-expanded	0.0	1.10	11.1	**100.0**	**41.2**
Sol	Baseline	0.0	**1.40**	**0.0**	30.0	19.0
	RGB-expanded	**10.0**	0.50	**0.0**	**90.0**	**23.5**
Fable	Baseline	0.0	0.90	–	30.0	22.2
	RGB-expanded	**40.0**	**2.90**	55.6	**100.0**	**42.5**

Answer revision helps only when it repairs the relevant error. On 46 shared fixed-input cases, Self-review and Crop review allow two additional responses; Tool-assisted review combines six additional responses with specialist measurements (Appendix E.2). Tool assistance raises Astra's Affordance success from 22.2% to 33.3%, but improves five and worsens four Geometry entries (Table 3a). Its Spatial accuracy remains at 72.2%, while Qwen-27B rises from

55.6% to 61.1%. Qwen-27B's self-review produces higher mean box IoU than tool assistance, yet tool assistance alone yields a box above the success threshold. These results distinguish better average estimates from corrections that satisfy the task criterion; the tool condition combines additional measurements and inference.

Recovery is useful when it restores environment progress. Opus 5.5 completes 48 of the 105 Planning episodes containing an execution error, accounting for 48 of its 76 successes (Appendix I). On Archaeology Dating, it recovers from two initial code errors, executes the excavation and measurement procedure, and completes the goal. DeepSeek-1017 makes partial progress, then spends its last three rounds on code errors without advancing the environment.

Appendices F and H connect decisions to observations, actions, and outcomes. Table A38 compares completion, rounds, and elapsed time on shared tasks; Appendix E.3 discusses task composition and external execution support.

5 Conclusion

Embodied Agent Arena evaluates how far native VLM decisions support embodied generalism by connecting geometric precision, functional grounding, and complete task outcomes. The ten agents have complementary strengths, but none consistently carries those strengths through to terminal goal satisfaction. Accurate paths can miss required endpoints, valid contacts can leave interaction states unsatisfied, and repeated actions can consume the available budget without completing the task.

The resulting gap is concrete: current agents need to reach precise goal states, preserve achieved conditions, and use further interaction to resolve what remains. The best Baseline manipulation agent still leaves more than half of its assigned tasks incomplete; unsuccessful Planning episodes consume a disproportionate share of inference resources. Additional visual feedback can change which model succeeds, but does not uniformly resolve these failures. Progress toward embodied generalism therefore requires measuring complete goal satisfaction together with the time and interaction it costs, rather than treating local accuracy or continued activity as sufficient evidence of competence.

References

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, et al. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. *arXiv preprint arXiv:2204.01691*, 2022.

Eduardo Arnold, Jamie Wynn, Sara Vicente, Guillermo Garcia-Hernando, Áron Monszpart, Victor Adrian Prisacariu, Daniyar Turmukhambetov, and Eric Brachmann. Map-free Visual Relocalization: Metric Pose Relative to a Single Image. *arXiv preprint arXiv:2210.05494*, 2022.

Vineet Bhat, Sungsu Kim, Valts Blukis, Greg Heinrich, Prashanth Krishnamurthy, Ramesh Karri, Stan Birchfield, Farshad Khorrami, et al. BOP-ASK: Object-Interaction Reasoning for Vision-Language Models. *arXiv preprint arXiv:2511.16857*, 2025.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and

Ury Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint arXiv:2410.24164*, 2024.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S Ryoo, Grecia Salazar, Pannag R Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong Tran, Vincent Vanhoucke, Steve Vega, Quan H Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: Robotics Transformer for Real-World Control at Scale. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.025.

Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu, Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollár, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. SAM 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.

Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, et al. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. *arXiv preprint arXiv:2506.18088*, 2025a.

Zeren Chen, Xiaoya Lu, Zhijie Zheng, Pengrui Li, Lehan He, Yijin Zhou, Jing Shao, Bohan Zhuang, and Lu Sheng. Geometrically-constrained agent for spatial reasoning. *arXiv preprint arXiv:2511.22659*, 2025b. doi: 10.48550/arXiv.2511.22659.

Dongzhou Cheng, Taoran Yi, Ye Fang, Xingwu Zhang, Fan Feng, Yixuan Li, Gengxiong Zhuang, Rongze Wang, Shuai Yang, Wei Song, Weizhi Xue, Minyan Wu, Jie Gui, Jiaqi Wang, and Tong Wu. In-Context Robot Learning with VLM Agents. *arXiv preprint arXiv:2609.19138*, 2026.

Zhili Cheng, Yuge Tu, Ran Li, Shiqi Dai, Jinyi Hu, Shengding Hu, Jiahao Li, Yang Shi, et al. EmbodiedEval: Evaluate Multimodal LLMs as Embodied Agents. *arXiv preprint arXiv:2501.11858*, 2025.

Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, and Min-Hung Chen. SpatialClaw: Rethinking action interface for agentic spatial reasoning. Preprint, 2026.

Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2018.

Junjia Du, Yadi Liu, Hongcheng Guo, Jiawei Wang, Haojian Huang, Yunyi Ni, and Zhoujun Li. DependEval: Benchmarking LLMs for Repository Dependency Understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 7150–7179. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.373.

Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, Huang Huang, Wenli Xiao, Guanzhi Wang, et al. CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation. *arXiv preprint arXiv:2603.22435*, 2026.

Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An Open-Source Generalist Robot Policy. In *Proceedings of Robotics: Science and Systems*, 2024. doi: 10.15607/RSS.2024.XX.090.

Pengyuan Guo, Zhonghao Mai, Zhengtong Xu, Kaidi Zhang, Quan Khanh Luu, Heng Zhang, Zichen Miao, Arash Ajoudani, Zachary Kingston, Qiang Qiu, and Yu She. PLanAR: Planning-Language-Grounded Agentic Reasoning for Robot Manipulation. *arXiv preprint arXiv:2602.01662*, 2026.

Yanjiang Guo, Yen-Jen Wang, Lihan Zha, and Jianyu Chen. DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

Haojian Huang, Zexi Li, Junhao Guo, Yehang Zhang, Wenxuan Peng, Bohan Zhou, Weilin Ruan, Leyi Wu, Chenxu Wang, Jianchong Su, Binghui Xie, Wosong Chen, Yingjie Xu, Tianhao Zhou, Suzeyu Chen, Pukun Zhao, Jiaqi He, Xinyi Li, Runze Li, Peiran Dong, Shaoxiang Dang, Jing Huang, Yingbing Chen, Yifan Chang, Tianyi Zhang, Shiyuan Deng, Haozhi Wang, Yangkai Wei, Wenqian Li, Han Yang, Kaiwen Zhou, Huaping Liu, James Cheng, Rui Shao, Donglin Wang, Yaochu Jin, Jianye Hao, Ying-Cong Chen, and Yinchuan Li. In-context learning for robots: Methods and applications. *arXiv preprint arXiv:2609.36012*, 2026a.

Haojian Huang, Chuanyu Qin, Yinchuan Li, and Yingcong Chen. Find, Fix, Reason: Context Repair for Video Reasoning. In *Proceedings of the International Conference on Machine Learning*, 2026b.

Haojian Huang, Jiahao Shi, Yinchuan Li, and Yingcong Chen. Affordance agent harness: Verification-gated skill orchestration. *arXiv preprint arXiv:2605.00663*, 2026c.

Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 540–562. PMLR, 2023a.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models. In *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pp. 1769–1782. PMLR, 2023b.

Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. RLBench: The Robot Learning Benchmark & Learning Environment. *arXiv preprint arXiv:1909.12271*, 2019.

Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DISCOVERYWORLD: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents. *arXiv preprint arXiv:2406.06769*, 2024.

Yunfan Jiang, Agrim Gupta, Zichen Zhang, Guanzhi Wang, Yongqiang Dou, Yanjun Chen, Li Fei-Fei, Anima Anandkumar, Yuke Zhu, and Linxi Fan. VIMA: Robot Manipulation with Multimodal Prompts. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 14975–15022. PMLR, 2023.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, et al. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv preprint arXiv:2406.09246*, 2024.

Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, et al. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. *arXiv preprint arXiv:2403.09227*, 2024a.

Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, et al. Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making. *arXiv preprint arXiv:2410.07166*, 2024b.

Qiang Li, Qingsen Yan, Haojian Huang, Peng Wu, Haokui Zhang, and Yanning Zhang. Text-Visual Semantic Constrained AI-Generated Image Quality Assessment. In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025. doi: 10.1145/3746027.3755471.

Zexi Li, Yehang Zhang, Haojian Huang, Bohan Zhou, Wenqian Li, Chenxu Wang, Yifan Chang, Yangkai Wei, Tianyi Zhang, Ying-Cong Chen, Kaiwen Zhou, Yinchuan Li, and James Cheng. RACaP: Agentic Reasoning, Acting, and Coding as Policies for Evolvable Robot Learning. *arXiv preprint arXiv:2609.29394*, 2026a.

Zexi Li, Yehang Zhang, Wenqian Li, Haojian Huang, Chenxu Wang, Shiyuan Deng, Yangkai Wei, Tianyi Zhang, Binghui Xie, Bohan Zhou, Yifan Chang, Kaiwen Zhou, Ying-Cong Chen, James Cheng, and Yinchuan Li. Robo-Harness K1: Harnessing Robot-Use Agents via Perception Augmentation. *arXiv preprint arXiv:2609.29389*, 2026b.

Erich Liang, Roma Bhattacharjee, Sreemanti Dey, Rafael Moschopoulos, Caitlin Wang, Michel Liao, Grace Tan, Andrew Wang, et al. InFlux: A Benchmark for Self-Calibration of Dynamic Intrinsics of Video Cameras. *arXiv preprint arXiv:2510.23589*, 2025.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as Policies: Language Model Programs for Embodied Control. *arXiv preprint arXiv:2209.07753*, 2022.

Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. LightGlue: Local feature matching at light speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

Wufei Ma, Haoyu Chen, Guofeng Zhang, Yu-Cheng Chou, Jieneng Chen, Celso M de Melo, and Alan Yuille. 3DSRBench: A Comprehensive 3D Spatial Reasoning Benchmark. *arXiv preprint arXiv:2412.07825*, 2024.

Arjun Majumdar, Anurag Ajay, Xiaohan Zhang, Pranav Putta, Sriram Yenamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Alexander Sax, and Aravind Rajeswaran. OpenEQA: Embodied Question Answering in the Era of

Foundation Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16488–16498, 2024.

Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *arXiv preprint arXiv:2112.03227*, 2021.

Austin Myers, Ching L. Teo, Cornelia Fermüller, and Yiannis Aloimonos. Affordance Detection of Tool Parts from Geometric Features. In *IEEE International Conference on Robotics and Automation*, 2015.

Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-Scale Simulation of Everyday Tasks for Generalist Robots. *arXiv preprint arXiv:2406.02523*, 2024.

Fei Ni, Min Zhang, Pengyi Li, Yifu Yuan, Lingfeng Zhang, Yuecheng Liu, Peilong Han, Longxin Kou, et al. Embodied Arena: A Comprehensive, Unified, and Evolving Evaluation Platform for Embodied AI. *arXiv preprint arXiv:2509.15273*, 2025.

Xavier Puig, Kevin Ra, Marko Boben, Jiaman Li, Tingwu Wang, Sanja Fidler, and Antonio Torralba. VirtualHome: Simulating Household Activities via Programs. *arXiv preprint arXiv:1806.07011*, 2018.

Yu Qi, Haibo Zhao, Ziyu Guo, Siyuan Ma, Ziyan Chen, Yaokun Han, Renrui Zhang, Zitiantao Lin, et al. Dissecting Embodied Abilities in Multimodal Language Models through Skill-level Evaluation and Diagnosis. *arXiv preprint arXiv:2510.08759*, 2025.

Shengyi Qian and David F. Fouhey. Understanding 3D Object Interaction from a Single Image. *arXiv preprint arXiv:2305.09664*, 2023.

Hanoona Rasheed, Mohammed Irfan Kurpath, Bin Ren, Hisham Cholakkal, Fahad Shahbaz Khan, and Salman Khan. Hard vision, easy vision: What GPT-6 Astra reveals across computer vision. *arXiv preprint arXiv:2609.35718*, 2026.

Pierre Sermanet, Tianli Ding, Jeffrey Zhao, Fei Xia, Debidatta Dwibedi, Keerthana Gopalakrishnan, Christine Chan, Gabriel Dulac-Arnold, Sharath Maddineni, Nikhil J Joshi, Pete Florence, Wei Han, Robert Baruch, Yao Lu, Suvir Mirchandani, Peng Xu, Pannag Sanketi, Karol Hausman, Izhak Shafran, Brian Ichter, and Yuan Cao. RoboVQA: Multimodal Long-Horizon Reasoning for Robotics. *arXiv preprint arXiv:2311.00899*, 2023.

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks. *arXiv preprint arXiv:1912.01734*, 2019.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. *arXiv preprint arXiv:2010.03768*, 2020.

Mohit Shridhar, Lucas Manuelli, and Dieter Fox. CLIPort: What and Where Pathways for Robotic Manipulation. *arXiv preprint arXiv:2109.12098*, 2021.

Ruiyang Si, Jianxin Bi, Shunyu Yang, Rui Ni, Wenbo Huang, Qiang Wang, Shulong Jiang, Duomin Wang, Xiuyu Li, Haiwen Feng, Zhen Dong, and Daquan Zhou. Fewer Tokens, Better Action: GPT-6 Astra Robot Agents with 14% Higher Success Rate but 65% Fewer Tokens. *arXiv preprint arXiv:2610.01939*, 2026.

Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11523–11530, 2023. doi: 10.1109/ICRA48891.2023.10161317.

Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield. RoboSpatial: Teaching Spatial Understanding to 2D and 3D Vision-Language Models for Robotics. *arXiv preprint arXiv:2411.16537*, 2024.

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, et al. ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI. *arXiv preprint arXiv:2410.00425*, 2024.

Alexander Veicht, Paul-Edouard Sarlin, Philipp Lindenberger, and Marc Pollefeys. GeoCalib: Single-image calibration with geometric optimization. In *European Conference on Computer Vision*, 2024.

Hanqing Wang, Shaoyang Wang, Yiming Zhong, Zemin Yang, Jiamin Wang, Zhiqing Cui, Jiahao Yuan, Yifan Han, Mingyu Liu, and Yuexin Ma. Affordance-R1: Reinforcement Learning for Generalizable Affordance Reasoning in Multimodal Large Language Model. *arXiv preprint arXiv:2508.06206*, 2025a.

Qineng Wang, Baiqiao Yin, Pingyue Zhang, Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, et al. MindCube: Spatial Mental Modeling from Limited Views. *arXiv preprint arXiv:2506.21458*, 2025b.

Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. ScienceWorld: Is your Agent Smarter than a 5th Grader? *arXiv preprint arXiv:2203.07540*, 2022.

Dongming Wu, Yanping Fu, Saike Huang, Yingfei Liu, Fan Jia, Nian Liu, Feng Dai, Tiancai Wang, Rao Muhammad Anwer, Fahad Shahbaz Khan, and Jianbing Shen. Ragnet: Large-scale reasoning-based affordance segmentation benchmark towards general grasping. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11980–11990, October 2025.

Leyi Wu, Yifan Zhao, Jinjie Zhang, Suzeyu Chen, Wosong Chen, Zhifei Chen, Tianshuo Xu, Qingchun He, Hongxin Hu, Haojian Huang, Yangkai Wei, Wenqian Li, Yinchuan Li, and Ying-Cong Chen. RoboStressBench: Benchmarking VLM Robustness to Physical Visual Stress in Embodied Scenes. *arXiv preprint arXiv:2606.00828*, 2026.

Runsen Xu, Weiyao Wang, Hao Tang, Xingyu Chen, Xiaodong Wang, Fu-Jen Chu, Matt Feiszli, and Kevin J. Liang. Multi-SpatialMLLM: Multi-Frame Spatial Understanding with Multi-Modal Large Language Models. *arXiv preprint arXiv:2505.17015*, 2025.

Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024.

Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, et al. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents. *arXiv preprint arXiv:2502.09560*, 2025a.

Sihan Yang, Runsen Xu, Yiman Xie, Sizhe Yang, Mo Li, Jingli Lin, Chenming Zhu, Xiaochen Chen, et al. MMSI-Bench: A Benchmark for Multi-Image Spatial Intelligence. *arXiv preprint arXiv:2505.23764*, 2025b.

Shiduo Zhang, Zhe Xu, Peiju Liu, Xiaopeng Yu, Yuan Li, Qinghui Gao, Zhaoye Fei, Zhangyue Yin, et al. VLABench: A Large-Scale Benchmark for Language-Conditioned Robotics Manipulation with Long-Horizon Reasoning Tasks. *arXiv preprint arXiv:2412.18194*, 2024.

Yehang Zhang, Haojian Huang, Yifan Chang, Jianchong Su, Bohan Zhou, Yingjie Xu, Wosong Chen, Tianhao Zhou, Chenxu Wang, Tianyi Zhang, Yangkai Wei, Wenqian Li, Shiyuan Deng, Yinchuan Li, Ying-Cong Chen, and Zexi Li. World Action Agent: Harnessing VLMs for Robot Manipulation via World Action Rehearsal. *arXiv preprint arXiv:2609.29964*, 2026a.

Yehang Zhang, Jianchong Su, Haojian Huang, Yifan Chang, Tianhao Zhou, Xinli Xu, Yingjie Xu, Yinchuan Li, Zexi Li, and Ying-Cong Chen. WorldLines: Benchmarking and Modeling Long-Horizon Stateful Embodied Agents. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026b.

Zixin Zhang, Kanghao Chen, Hanqing Wang, Hongfei Zhang, Harold Haodong Chen, Chenfei Liao, Litao Guo, and Ying-Cong Chen. A4-Agent: An agentic framework for zero-shot affordance reasoning. *arXiv preprint arXiv:2512.14442*, 2025.

Pukun Zhao, Longxiang Wang, Chen Chen, Peicheng Wang, Fanqing Zhou, Runze Li, and Haojian Huang. STAR: Mitigating Cascading Errors in Spatial Reasoning via Turn-point Alignment and Segment-level DPO. In *IEEE International Conference on Multimedia and Expo (ICME)*, 2026a.

Pukun Zhao, Longxiang Wang, Miaowei Wang, Chen Chen, Fanqing Zhou, and Haojian Huang. EvoEmpirBench: Dynamic Spatial Reasoning with Agent-ExpVer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 36564–36572, 2026b. doi: 10.1609/aaai.v40i43.40979.

Enshen Zhou, Yibo Li, Jingkun An, Jiayuan Zhang, Shanyu Rong, Mengzhen Liu, Yi Han, Yuheng Ji, Huajie Tan, Jiawei He, Pengwei Wang, Zhongyuan Wang, Cheng Chi, Lu Sheng, and Shanghang Zhang. Towards spatial trace with reasoning in vision-language models for robotics. *arXiv preprint arXiv:2512.13660*, 2026.

Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

Appendix: Supporting Evidence and Protocols

Reading guide. Appendices A–B define task construction, interfaces, and scoring. Appendix C expands Table 2, including source-composition sensitivity and dispersion for Opus 5.5 and DeepSeek-1017. Appendix D follows the five domains of Section 4.3, from full task comparisons to Manipulation's auxiliary metric definitions. Appendix E specifies the execution-protocol and multi-round review comparisons, and Appendix F illustrates recorded actions and terminal outcomes, with additional examples across all five capabilities in Appendix H. Appendix I analyzes shared failures, resources, and recovery using the complete Planning records and linked execution traces. Appendix G gives the evaluation scales and public resources behind Table 1.

A Benchmark Construction and Source Inventory

Embodied Agent Arena contains 1,000 cases from 32 established sources and our GeoProbe benchmark (Table A1). A case is one query or episode; multiple queries can share a scene. Benchmark-specific allocation uses seed 42 and targets diagnostic coverage of the five capabilities in Section 3.1.

Table A1. The unified 1,000-case arena, drawn from 32 established sources and our GeoProbe benchmark. Detailed source allocations follow below.

Domain	Cases	Share (%)	Task coverage
Geometry	370	37.0	Calibration, pose, scale, tracing
Spatial Reasoning	220	22.0	Relations, views, video
Affordance	70	7.0	Functional parts, contact
Task Planning	157	15.7	Household, navigation, science
Manipulation	183	18.3	Robot operations, sequences
Total	1,000	100.0	660 fixed-input; 340 interactive

Query vocabulary. Figure 2a counts each term once per query, balances weights across source groups and domains, and includes English translations of GeoProbe queries. GeoProbe's ten protocols form one source group.

Geometry (370 cases). InFlux-Real (Liang et al., 2025) contributes 30 single-frame intrinsic-estimation cases, of which four have invalid reference annotations and are excluded from continuous-error calculations. Map-free (Arnold et al., 2022) contributes 30 relative-pose image pairs with known camera intrinsics. MultiSPA-derived uses Map-free images and poses with the camera-motion question formulation associated with Multi-SpatialMLLM (Xu et al., 2025): 12 angle, ten distance, and eight displacement-vector questions. This local adaptation retains the Map-free image and pose references. The pose transformation and camera-center displacement use distinct coordinate conventions and reference targets.

GeoProbe comprises ten diagnostic protocols: 168 queries over 130 unique RGB images, predominantly rendered in Blender, with four real-photo displacement cases. Shared scenes separate calibration, pose, depth, displacement, and scale requirements (Table A2).

TraceSpatial-Bench (Zhou et al., 2026), introduced with RoboTracer, adds paired 2D and 3D queries on 56 scenes. These 112 Geometry cases predict offline object trajectories, scored for reference agreement and geometric constraints.

Table A2. GeoProbe's ten diagnostic protocols, with shared images and scenes.

Diagnostic	Cases	Diagnostic	Cases
Object displacement, fixed camera	12	Full camera intrinsics	24
Object displacement, small view change	12	Focal length, centered principal point	12
Object displacement, large view change	12	Relative camera pose	24
Real tabletop displacement	4	Sparse depth	12
Object dimensions	12	Reference-based scale	44

Spatial Reasoning (220 cases). 3DSRBench (Ma et al., 2024) contributes 50 relation questions; 28 flip-tagged entries were displayed unflipped. Scores use corrected references for five affected answers under the unflipped presentation. MindCube (Wang et al., 2025b) contributes 50 multi-view cases: 39 among-view, six around-view, and five rotation. VSI-Bench (Yang et al., 2024) contributes 50 video questions: nine absolute-distance, nine object-size, fourteen relative-direction, six relative-distance, five counting, four room-size, two appearance-order, and one route-planning. Eight frames form its initial observation. RoboSpatial-Home (Song et al., 2024) contributes ten cases each for compatibility, configuration, and context. BOP-ASK (Bhat et al., 2025) contributes twelve relative-position and eight relative-depth questions. MMSI-Bench (Yang et al., 2025b) contributes 20 cross-image questions. Together these comprise 183 categorical, 27 numerical, and ten point-set questions, scored as defined in Appendix B.3.

Affordance (70 cases). UMD (Myers et al., 2015) contributes 30 views spanning five tool classes and six functions: grasp, cut, scoop, contain, pound, and support. Multiple views can depict the same object. The 20 ReasonAff-style cases adapt instruction-conditioned queries motivated by ReasonAff (Wang et al., 2025a) to 3DOI (Qian & Fouhey, 2023) interaction annotations. Another 20 RAGNet-3DOI cases use explicit action-conditioned queries (Wu et al., 2025). The two subsets use disjoint images. These images and annotations support native point and region evaluation.

Task Planning (157 cases). ALFRED (Shridhar et al., 2019) and ALFWorld (Shridhar et al., 2020) contribute 20 visual cases each, covering placement, heating, cooling, cleaning, containers, multiple objects, and illumination. ScienceWorld (Wang et al., 2022) contributes 73 text-based cases across eighteen topics. VirtualHome (Puig et al., 2018) contributes 21 symbolic programs in seven scenes. The composition analysis removes these 94 cases together. DiscoveryWorld (Jansen et al., 2024) contributes ten tasks with structured observations. HumanCLAW contributes thirteen Habitat navigation and sitting episodes: five sofa, three toilet, two chair, two plant, and one television target. Observations are visual, textual, or symbolic.

Table A3. Manipulation tasks across 13 sources, with identical assignments for all agents.

Benchmark	Cases	Task coverage
CALVIN (Mees et al., 2021)	10	Ten sequences, each with five ordered subgoals
CLIPort (Shridhar et al., 2021)	18	Eighteen task and color-generalization configurations
RLBench (James et al., 2019)	10	Ten task configurations
RoboCasa (Nasiriany et al., 2024)	10	Seven atomic and three composite tasks
RoboCasa365	10	Three atomic and seven composite tasks
RoboWits	10	Ten official task configurations
RoboTwin 2.0 (Chen et al., 2025a)	50	Fifty distinct manipulation tasks
VIMA-Bench (Jiang et al., 2023)	10	Six placement-generalization and four new-task cases
VLABench (Zhang et al., 2024)	25	8 ID, 3 category, 4 commonsense, 7 semantic, 3 texture
ManiSkill (Tao et al., 2024)	10	Nine binary tabletop tasks; one quadruped reward task
LIBERO-PRO (Fu et al., 2026)	10	Ten object, spatial, goal, and multi-object configurations
robosuite	5	Two lift seeds; stacking, restacking, and wiping
BEHAVIOR-1K (Li et al., 2024a)	5	Five multi-object household activities

Manipulation (183 cases). Table A3 lists 182 binary tasks and one continuous-reward task. LIBERO-PRO, robosuite, and BEHAVIOR-1K use benchmark-specific code interfaces (Fu et al., 2026). RoboCasa365 uses variations designated for pretraining in the source inventory. Appendix B.1 specifies the native assistance available to agents.

B Evaluation Protocol and Reproducibility

B.1 Observations and Model-Directed Interaction

Geometry, Spatial Reasoning, and Affordance inspect fixed observations; Planning and Manipulation interact with an environment. All domains except Manipulation use a persistent Python runtime; Manipulation uses its native agent loop. Both enforce episode-level budgets and termination. Variables and environment state persist between code rounds, and images produced by a cell reach the next model request after that cell ends.

Table A4. Model-visible observations, actions, and feedback. Private scoring annotations are unavailable to the agent.

Domain	Observation support	Actions and feedback
Geometry	Image or image pair; intrinsics for Map-free	One final numerical submission
Spatial Reasoning	Images, crops, and video-frame queries	Evidence inspection and answer submission
Affordance	Image and interaction instruction	Point and box submission; polygons in the added experiment
Task Planning	Images, text, or symbolic state	Navigation, interaction, programs, public progress
Manipulation	Protocol-specific scene/state observations and native grounding helpers	Motion or skill calls; execution and task checks

ALFRED and visual ALFWorld expose observation, search, navigation, interaction, and memory primitives. CLIPort and VIMA expose instance inspection and pick-and-place actions.

RoboTwin provides actor localization, pose inspection, planning probes, and execution. LIBERO-PRO, robosuite, and BEHAVIOR-1K expose benchmark-specific observation and action interfaces; RoboCasa includes fixture and contact localization. The model makes decisions using these native helpers; the baseline adds no specialist perception or verification pipeline. Geometry requests direct visual estimates and excludes pose-solver implementation.

A runtime exception can be followed by another code round if budget remains; prior state changes are not automatically undone. Model requests, code rounds, primitive calls, and environment steps count different units. Stored visualization frames are also distinct from images actually sent to the model. Qualitative figures sample the recorded camera streams.

B.2 Models and Execution Budgets

Table A5. Models and configured identifiers for all ten agents in the five domains.

Model label	Identifier
GPT-6 Astra	`gpt-6-astra`
GPT-6 Sol	`gpt-6-sol`
Claude Fable 5.1	`claude-fable-5-1`
Claude Opus 5.5	`claude-opus-5-5`
DeepSeek-V4.1-Flash-1017	`deepseek-flash`
Gemini 3.8 Flash	`gemini-3.8-flash`
Gemini 4 Argon	`gemini-4-argon`
Qwen 3.8 Max	`qwen3.8-max`
Qwen 3.5 397B-A17B	`qwen3.5-397b-a17b`
Qwen 3.5 27B	`qwen3.5-27b`

Gemini 4 Argon (Argon in compact figures and tables) uses the recorded API identifier `gemini-4-argon`. Gemini elsewhere in the original-cohort diagnostics denotes Gemini 3.8 Flash. Appendix J details Argon's coverage and execution settings.

DeepSeek-1017 (DS-1017 in compact figures and tables) denotes the 17 October 2026 release DeepSeek-V4.1-Flash-1017, served by `deepseek-flash`; this alias is not an immutable model snapshot. Its evaluation was recorded on 18–19 October 2026. DeepSeek-1017 Spatial runs use no YOLOE, and 3DSRBench inputs use the corrected image flips and matching native labels. The original fixed-input and Planning configurations use temperature zero. Opus 5.5 uses the recorded identifier above and the same Baseline Manipulation protocol as the original seven agents. Baseline request limits are 1,800 output tokens and 360 seconds; fixed-input profiles also allow up to 16,384 tokens and 900 seconds. On the configured CLI interface, reasoning effort is maximum for Sol and high for Astra and Fable; API requests omit this parameter. Astra's first repetition on the original 90-case Geometry core uses the high-effort CLI configuration. Manipulation's request settings and shared execution limits are specified in Appendix E.1.

Table A6. Baseline episode budgets. Tokens accumulate within each case; rounds are code interactions, not repetitions. Dashes denote unspecified fields. Appendix B.4 lists fixed-input settings.

Entry	Code rounds	Tokens	Env. steps	Primitives	Episode seconds	Task seconds
Geometry	12	120k	–	–	–	960
Spatial: non-VSI	6	96k	–	–	900	960
Spatial: VSI	8	240k	–	–	1,200	1,500
ALFRED, ALFWorld	8	320k	120	540	1,200	1,500
DiscoveryWorld	8	320k	100	400	1,200	1,500
HumanCLAW	101	2,000k	100	400	10,800	11,000
ScienceWorld	8	320k	100	500	1,200	1,500
VirtualHome	8	320k	16	120	1,200	1,500
Affordance	12	120k	–	–	–	960

Execution and native-call time limits are clipped by remaining budgets. Appendix E.1 gives Manipulation's shared execution limits, camera presets, and native step-limit semantics.

B.3 Scoring and Repeated Evaluation

Submission and termination. A final submission commits a fixed-input prediction. Interactive runs end when the agent stops, the environment reaches a terminal condition, or a source-specific round, token, action, or time limit is reached. The evaluator scores the submitted output or recorded state independently of the agent's declaration. A complete episode includes all its internal reasoning, actions, and recovery; these steps are not separate attempts.

Task scores and aggregation. Categorical questions use answer accuracy and interactive tasks use native goal success. Invalid or absent predictions remain in success denominators. Invalid reference annotations are excluded before scoring. Continuous errors average valid predictions with compatible metrics and units within each run. Repeated metrics then weight the run-level scores equally, with valid coverage reported separately. Numerical Spatial Reasoning uses exact-match success, $|\hat{y} - y| < 10^{-6}$, alongside continuous errors; this differs from VSI-Bench's official mean relative accuracy. Point-set questions retain their local set criterion.

Pass@1 and repeated scores. For N assigned cases and R repetitions, let $s_{ir} \in \{0, 1\}$ denote success on case i in run r. The average single-episode success rate is

$$\text{Pass@1} = \frac{1}{R} \sum_{r=1}^{R} \left(\frac{1}{N} \sum_{i=1}^{N} s_{ir} \right). \tag{2}$$

This estimates success in one complete episode under the specified condition. Fresh agent state is used for formal repetitions. For any per-run score M_r, we compute

$$\bar{M} = \frac{1}{R} \sum_{r=1}^{R} M_r, \qquad v_M = \frac{1}{R-1} \sum_{r=1}^{R} (M_r - \bar{M})^2, \qquad \text{SD}_M = \sqrt{v_M}. \tag{3}$$

Tables using $\bar{M} \pm \text{SD}_M$ report variation between run-level scores. Variance is supplied separately in squared metric units. Figure 4 averages valid-case errors within each run and then weights the recorded runs equally. Its coverage sensitivity instead fixes the cases valid in every shared repetition for each model pair; those common-cohort estimates are reported

separately. Figure 5a–b uses per-run categorical accuracies; panel c uses full-source native Pass@1. The original pairwise gaps average accuracy differences over shared run indices, with the comparator fixed per subtask. Run indices align reporting repetitions; seeds and tool configurations can differ. Run-to-run SD requires $R > 1$.

Scalar estimates and depth. For J valid targets, $e_j = \hat{y}_j - y_j$ gives $\mathrm{MAE} = J^{-1}\sum_j |e_j|$ and $\mathrm{RMSE} = \sqrt{J^{-1}\sum_j e_j^2}$. Absolute relative error is $\mathrm{AbsRel} = J^{-1}\sum_j |e_j|/|y_j|$ on nonzero targets, and $\mathrm{MAPE} = 100\,\mathrm{AbsRel}$ is expressed in percent. Size and focal-length estimates use absolute and relative errors. Depth additionally reports RMSE in meters. Depth errors are computed over positive finite predictions within each image, then averaged across images, with valid-point coverage reported. The fraction within 10% relative error and $\delta_1 = J_{\mathrm{all}}^{-1}\sum_j \mathbf{1}[\max(\hat{y}_j/y_j, y_j/\hat{y}_j) < 1.25]$ use all queried points as their denominator; invalid predictions contribute zero. Ordinal accuracy is the fraction of eligible point pairs with the correct depth ordering, with invalid predictions counted as incorrect. Scale alignment removes the global depth factor.

Displacement and camera pose. For a nonzero scalar or vector target y, motion relative error is $\|\hat{y} - y\|_2/\|y\|_2$. The MultiSPA-derived tolerance is 20%; a zero target requires zero absolute error. GeoProbe object-displacement diagnostics separately test movement status, direction, absolute distance error, and direction agreement with distance error at most 1 cm. Stationary cases are evaluated without division by zero. Camera translation error is $e_t = \|\hat{\mathbf{t}} - \mathbf{t}\|_2$, reported in centimeters. Rotation error is $e_R = \arccos[(\mathrm{tr}(R^{\mathsf{T}}\hat{R}) - 1)/2]$, in degrees, clipping the argument to $[-1, 1]$. GeoProbe reports component errors and joint success within 2° and 2 cm.

Camera calibration. Focal MAPE averages the absolute relative errors of f_x and f_y per case. Principal-point error is $e_c = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$ in pixels, also normalized by the image diagonal D. Joint calibration requires both focal components to satisfy 10% relative error and $e_c \le 0.02D$. Main-table pose errors, complementary continuous metrics, and Figure 4 use equal-run means and run SD, as specified above.

Spatial tracing. For each case, predicted and reference paths are independently resampled to $L = 100$ equally spaced arc-length positions. Trajectory error is $\mathrm{RMSE}_{\mathrm{trace}} = \sqrt{L^{-1}\sum_{\ell=1}^{L}\|\hat{\mathbf{p}}_\ell - \mathbf{p}_\ell\|_2^2}$. We average this per-case value over valid scenes. The 2D metric uses normalized image coordinates; the 3D metric uses meters. The two metrics retain their respective units. The supplied variance is the population variance of the 56 per-scene RMSE values for a single run.

Offline 3D success requires valid initiation, target arrival, and collision checks to pass together. The predicted start must lie within 20 cm of the object point cloud or the reference start. At least one of the final three predicted points must lie within 20 cm of the target 3D box; for two-point paths, only the endpoint is checked. Translating the object point cloud to each checked predicted position must yield at most 20% overlap with the environment. Collision is checked at the sampled positions. Missing or invalid trajectories count as failures. 2D tracing reports continuous error. Qwen-27B trajectory outputs undergo format normalization that preserves all predicted coordinate values.

Native affordance grounding. For a directly predicted bounding box B and ground-truth mask G, the box interior is rasterized and $\mathrm{IoU}_{B,G} = |B \cap G|/|B \cup G|$. Contact validity is $\mathbf{1}[p \in G]$ for the submitted point p, with a reference-box fallback only where a mask is unavailable. A separate diagnostic uses $\mathrm{IoU}_{\mathrm{box}} = |B \cap B_G|/|B \cup B_G|$ against the reference box B_G. Strict box success requires $\mathrm{IoU}_{\mathrm{box}} \ge 0.5$; contact validity is scored independently. The source field named `giou` contains ordinary box–mask IoU, not generalized IoU. The separate polygon protocol rasterizes the directly submitted polygons into a region P and scores $|P \cap G|/|P \cup G|$. It adds no

external segmenter or iterative segmentation refinement; Appendix C.5 gives output validation and coverage.

Resource accounting. Inference tokens, action calls, model rounds, and elapsed time measure different aspects of execution. The Manipulation comparison reports token usage and action calls from the supplied execution analysis; protocol-matched per-case durations are unavailable. Request, trial, and native-call time limits are specified separately in Appendix E.1.

B.4 Repetition Protocol and Result Accounting

Repetitions. In the original seven-agent cohort, non-tracing Geometry, Spatial Reasoning, and Affordance use five repetitions per case for six models and three for Fable. Opus 5.5 uses five runs for non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance. DeepSeek-1017 uses five runs for non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance. Tracing, Planning, Manipulation, and direct-polygon prediction use one run per assignment for all models. Run-to-run SD is reported only for repeated measurements. Manipulation's observation schedules appear in Appendices E.1 and C.4.

Additional execution configurations. Fixed-input evaluation also uses per-case limits of up to 48 code rounds and 480k tokens for Geometry and Affordance, 24 rounds and 384k tokens for non-VSI Spatial Reasoning, and 32 rounds and 960k tokens for VSI. Task time limits reach 7,200 seconds, or 10,800 seconds for VSI. Budgets vary by source and execution configuration.

Result selection. In the original cohort, fixed-input and Planning trial slots use the first valid outcome, independent of correctness. Manipulation uses the fixed adopted baseline and RGB-expanded manifests. The original Baseline CSV covers 182 binary assignments. Table 2 compares eight Baseline agents and DeepSeek-1017, whose per-round RGB Manipulation setting uses the limits in Table A15. The Opus 5.5 Manipulation score uses the same Baseline source interfaces and is included in the main-table comparison. Appendix E.1 defines missing-record accounting, shared-current sources, and the common subset used in Table 3.

Coverage and scoring denominators. Table A7 reports the original cohort and DeepSeek-1017 coverage. Opus 5.5 metric-specific valid counts are in Table A12; no whole-domain valid count is inferred from overlapping summary metrics. Opus 5.5 HumanCLAW uses the final completed record for each of the 13 assignments: navigation succeeds at 20 cm proximity, while sitting requires the native interaction condition. Two RLBench exceptions and one RoboTwin exception count as unsuccessful assignments; Opus 5.5 Manipulation remains 87/182 (47.8%). InFlux excludes four invalid references. Binary criteria retain all assigned tasks. Reported success requires the native evaluator, rather than an agent declaration. Superseded HumanCLAW attempts are not counted as additional assignments. Manipulation covers 182 binary tasks, including all five BEHAVIOR-1K assignments per model; AnymalC-Spin is excluded from this binary comparison. Continuous metrics retain finite valid predictions.

Table A7. Original seven agents and DeepSeek-1017 (DS-1017). Valid submission coverage for fixed-input domains and recorded terminal outcomes for Planning. Assigned denominators include repetitions and invalid-reference slots; invalid references are excluded from metric scoring. Manipulation uses the fixed allocation in Table A3.

Domain	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Geometry	1382/1402	1382/1402	819/886	1382/1402	1314/1402	1382/1402	1376/1402	1356/1402
Spatial Reasoning	1060/1100	1100/1100	586/660	1097/1100	1077/1100	1033/1100	982/1100	954/1100
Affordance	350/350	350/350	208/210	350/350	335/350	349/350	349/350	348/350
Task Planning	157/157	157/157	157/157	157/157	157/157	157/157	157/157	157/157

Manipulation diagnostics. CALVIN reports average consecutive goals completed. Other auxiliary metrics average task-normalized native criteria or control evidence over fixed applicable subsets. They differ in their scoring items and evidence time windows; Appendix D.5 defines the aggregation, coverage, and missing-record treatment.

Main-table metric scope. Geometry rotation and translation average valid predictions from the 30 Map-free and 24 GeoProbe pose cases within each run, in degrees and centimeters; the main table reports the mean and sample SD across runs. Spatial Pass@1 covers all 220 cases; numerical AbsRel covers 27, and categorical accuracy separately covers 183. Original Affordance box–mask IoU and contact validity cover all three sources, with 70 cases. Polygon IoU evaluates the same allocation under the direct-polygon protocol. Planning uses terminal success across 157 cases for every model. Manipulation uses task-weighted native success across 182 binary assignments; Tables A9 and A11 report each source separately.

Dispersion and valid subsets. Continuous metrics first average valid predictions within each run, then report the equal-weight mean and sample SD across runs (Equation 3). This SD measures variation between repeated evaluations, not variation among individual cases. Common-valid-cohort checks appear in Appendices D.1 and D.2.

C Complete Model Comparisons

This section supports Section 4.2 at three resolutions: complementary native metrics, differences among source benchmarks, and the metric profiles shown in Figure 3. Task-level explanations follow in Appendix D.

C.1 Metrics and Source-Level Results

Table A8 gives complementary native metrics. Tables A9 and A11 resolve the results by source, using the denominators and repetitions in Appendix B.4. These breakdowns distinguish broad domain performance from strengths concentrated in particular task collections. **Reading the metric families.** GeoProbe focal MAPE combines full-intrinsics and focal-only queries; displacement MAE combines its four camera/environment conditions. Scale MAPE combines object dimensions and numerical reference-scale queries, while qualitative scale judgments retain Pass@1. TraceSpatial reports 2D and 3D errors separately because their coordinates and units differ. Each metric retains the target's physical interpretation.

Table A8. Complementary native metrics across four domains; DS-1017 abbreviates DeepSeek-V4.1-Flash-1017. Repeated metrics show the mean above ± run SD (five runs; three for Fable). Each run averages valid continuous predictions or scores all assigned binary cases; runs receive equal weight. Polygon IoU uses a separate single run for all ten models. Planning uses all 157 cases for every model. N counts eligible cases. Argon denotes Gemini 4 Argon and G3.8 denotes Gemini 3.8 Flash. GP denotes GeoProbe; Manipulation source results appear in Tables A9 and A11.

Metric	N	Astra	Sol	Fable	Q-Max	G3.8	Q-397B	Q-27B	Opus 5.5	DS-1017	Argon
Geometry (370 cases)											
InFlux focal MAPE (%)↓	26	53.02 ±2.15	47.79 ±2.00	52.65 ±2.22	55.50 ±1.04	52.40 ±3.40	67.82 ±1.82	65.86 ±2.08	60.27 ±1.49	62.92 ±1.57	34.84 ±1.63
InFlux principal error (px)↓	26	3.82 ±0.00	3.82 ±0.00	3.82 ±0.00	3.82 ±0.00	3.88 ±0.09	3.82 ±0.00	3.82 ±0.00	3.82 ±0.00	3.55 ±0.46	3.82 ±0.00
Angle relative L2↓	12	0.760 ±0.130	0.915 ±0.442	1.318 ±0.046	1.410 ±0.281	0.961 ±0.083	1.298 ±0.172	2.037 ±0.331	1.166 ±0.113	1.497 ±0.352	0.735 ±0.046
Distance relative L2↓	10	0.294 ±0.024	0.418 ±0.130	0.785 ±0.115	0.560 ±0.090	0.419 ±0.077	0.411 ±0.059	0.331 ±0.095	0.455 ±0.069	0.461 ±0.196	0.313 ±0.017
Vector relative L2↓	8	0.425 ±0.092	0.591 ±0.155	0.625 ±0.094	0.454 ±0.122	0.643 ±0.071	1.018 ±0.080	1.031 ±0.060	0.516 ±0.042	0.743 ±0.254	0.529 ±0.120
Map-free rotation (°)↓	30	30.08 ±12.15	46.60 ±2.25	50.81 ±8.99	35.11 ±6.25	45.19 ±7.64	63.03 ±1.61	64.94 ±6.89	40.95 ±8.10	52.68 ±6.56	31.75 ±3.52
Map-free translation (cm)↓	30	188.98 ±65.21	265.85 ±10.41	248.74 ±50.12	243.06 ±18.41	272.95 ±59.81	430.82 ±18.13	374.96 ±22.44	243.13 ±35.82	324.19 ±46.48	185.06 ±14.78
Map-free direction (°)↓	30	26.43 ±12.73	48.91 ±2.14	54.20 ±7.11	38.92 ±5.54	50.72 ±14.17	95.15 ±6.59	78.78 ±9.15	44.29 ±9.06	64.63 ±6.45	27.47 ±2.95
GP focal MAPE (%)↓	36	6.92 ±4.90	9.57 ±1.25	31.77 ±1.05	14.67 ±2.72	26.59 ±4.73	33.39 ±2.99	38.56 ±2.95	16.48 ±6.13	33.99 ±5.33	28.39 ±0.91
GP principal error (px)↓	24	51.95 ±8.15	110.62 ±18.22	62.81 ±8.54	78.41 ±13.77	66.83 ±9.70	58.23 ±1.02	59.05 ±1.01	60.46 ±4.13	75.40 ±9.96	56.05 ±1.89
Depth MAE (m)↓	12	0.307 ±0.089	0.633 ±0.068	0.416 ±0.088	0.490 ±0.042	0.587 ±0.067	0.934 ±0.072	0.945 ±0.043	0.510 ±0.295	0.688 ±0.089	0.927 ±0.024
Depth RMSE (m)↓	12	0.338 ±0.087	0.753 ±0.086	0.522 ±0.093	0.598 ±0.041	0.697 ±0.064	1.053 ±0.080	1.059 ±0.044	0.599 ±0.302	0.814 ±0.122	0.973 ±0.024
Depth AbsRel↓	12	0.104 ±0.029	0.229 ±0.027	0.142 ±0.027	0.163 ±0.013	0.192 ±0.021	0.312 ±0.028	0.292 ±0.013	0.169 ±0.108	0.229 ±0.025	0.299 ±0.006
Displacement MAE (cm)↓	40	0.585 ±0.092	0.973 ±0.056	1.488 ±0.163	1.515 ±0.160	2.278 ±0.170	2.480 ±0.071	2.935 ±0.082	1.164 ±0.475	1.934 ±0.318	1.309 ±0.050
GP rotation (°)↓	24	1.17 ±0.38	2.54 ±1.02	2.80 ±0.94	2.21 ±0.18	3.10 ±0.48	7.55 ±0.77	5.26 ±0.97	3.05 ±0.96	3.75 ±0.57	2.71 ±0.36
GP translation (cm)↓	24	5.36 ±2.11	9.50 ±2.71	9.24 ±2.05	6.99 ±0.96	7.85 ±0.90	23.59 ±1.80	12.05 ±0.52	11.13 ±1.48	14.52 ±1.80	10.14 ±0.98
Scale MAPE (%)↓	44	1.90 ±0.28	2.84 ±0.21	3.76 ±0.05	3.77 ±0.16	3.93 ±0.46	16.07 ±1.73	16.45 ±0.48	3.91 ±2.74	7.62 ±1.13	2.21 ±0.09
Scale judgment Pass@1 (%)↑	12	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	93.3 ±7.0	100.0 ±0.0	98.3 ±3.7	100.0 ±0.0
2D trace RMSE↓	56	0.120	0.132	0.118	0.118	0.127	0.155	0.284	0.180	0.141	0.107
3D trace RMSE (m)↓	56	0.382	0.463	0.435	0.469	0.426	0.629	0.682	0.683	0.470	0.310
3D constraint success (%)↑	56	33.9	21.4	25.0	17.9	37.5	8.9	7.1	8.9	14.3	42.9
Spatial Reasoning (220 cases)											
Native Pass@1 (%)↑	220	69.4 ±1.0	60.7 ±2.5	62.0 ±4.6	68.9 ±1.1	65.4 ±1.3	50.6 ±1.8	48.0 ±0.9	67.5 ±5.1	51.6 ±1.5	71.6 ±0.7
Categorical Pass@1 (%)↑	183	82.2 ±0.6	71.3 ±3.0	72.7 ±5.2	81.2 ±1.3	76.7 ±1.9	60.0 ±2.2	56.8 ±0.7	79.2 ±5.6	61.3 ±1.7	84.2 ±1.3
Numerical AbsRel↓	27	0.214 ±0.016	0.273 ±0.016	0.185 ±0.011	0.206 ±0.033	0.161 ±0.028	0.251 ±0.027	0.288 ±0.023	0.184 ±0.040	0.309 ±0.035	0.169 ±0.023
Affordance (70 cases)											
BBox–GT box IoU↑	70	0.390 ±0.005	0.366 ±0.012	0.386 ±0.014	0.382 ±0.022	0.324 ±0.098	0.146 ±0.015	0.021 ±0.008	0.432 ±0.021	0.278 ±0.015	0.484 ±0.012
BBox–GT mask IoU↑	70	0.286 ±0.010	0.264 ±0.010	0.284 ±0.013	0.272 ±0.017	0.251 ±0.081	0.116 ±0.014	0.013 ±0.006	0.311 ±0.020	0.217 ±0.011	0.356 ±0.009
Polygon–GT mask IoU↑	70	0.379	0.325	0.328	0.307	0.291	0.092	0.051	0.397	0.249	0.441
BBox success@0.5 (%)↑	70	46.6 ±1.3	43.7 ±1.6	46.2 ±0.8	45.7 ±1.7	33.1 ±11.8	5.4 ±3.7	0.3 ±0.6	51.4 ±3.0	28.9 ±3.7	56.3 ±2.2
Contact-point validity (%)↑	70	60.3 ±2.6	52.0 ±3.6	57.1 ±2.5	52.9 ±4.2	44.6 ±13.0	32.6 ±4.7	6.3 ±1.6	61.1 ±1.9	50.9 ±4.2	59.7 ±1.2
Task Planning (157 cases)											
Native Pass@1 (%)↑	157	80.3	44.6	63.7	35.7	48.4	17.8	19.1	48.4	31.2	70.1

Table A9. Source results for the original seven agents and DeepSeek-1017 (DS-1017); Opus 5.5 appears in Table A11. Spatial/Affordance: mean above ± run SD (%); Planning: solved/assigned cases; Manipulation: success (%). N counts eligible cases.

Source	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Spatial Reasoning: categorical accuracy (%)									
3DSRBench	50	**73.6** ±2.6	**71.2** ±1.8	**72.7** ±3.1	**74.0** ±3.7	**70.0** ±2.4	**69.2** ±3.6	**57.6** ±2.6	**61.6** ±7.1
BOP-ASK	20	**96.0** ±4.2	**87.0** ±4.5	**88.3** ±2.9	**92.0** ±2.7	**97.0** ±2.7	**85.0** ±3.5	**84.0** ±4.2	**88.0** ±7.6
MindCube	50	**90.8** ±1.1	**68.0** ±7.3	**79.3** ±8.1	**90.8** ±2.7	**89.2** ±3.0	**46.0** ±2.8	**50.4** ±4.1	**54.4** ±4.3
MMSI-Bench	20	**77.0** ±5.7	**55.0** ±11.7	**66.7** ±7.6	**55.0** ±8.7	**56.0** ±9.6	**35.0** ±9.4	**29.0** ±7.4	**53.0** ±5.7
RoboSpatial	20	**85.0** ±3.5	**81.0** ±8.9	**88.3** ±2.9	**88.0** ±5.7	**76.0** ±4.2	**90.0** ±3.5	**92.0** ±5.7	**87.0** ±5.7
VSI-Bench	23	**72.2** ±7.3	**70.4** ±5.7	**36.2** ±28.9	**83.5** ±8.4	**65.2** ±5.3	**44.3** ±6.4	**39.1** ±3.1	**37.4** ±5.0
Affordance: strict box success at IoU ≥ 0.5 (%)									
ReasonAff-style	20	**19.0** ±5.5	**6.0** ±4.2	**16.7** ±7.6	**13.0** ±5.7	**5.0** ±0.0	**0.0** ±0.0	**0.0** ±0.0	**3.0** ±4.5
RAGNet-3DOI	20	**14.0** ±4.2	**15.0** ±5.0	**13.3** ±2.9	**7.0** ±2.7	**6.0** ±2.2	**4.0** ±2.2	**1.0** ±2.2	**5.0** ±6.1
UMD	30	**86.7** ±3.3	**88.0** ±4.5	**87.8** ±5.1	**93.3** ±2.4	**70.0** ±27.0	**10.0** ±7.8	**0.0** ±0.0	**62.0** ±6.9
Task Planning: terminal success (single run)									
ALFRED	20	15/20	3/20	7/20	3/20	2/20	1/20	0/20	1/20
ALFWorld	20	14/20	4/20	6/20	2/20	4/20	3/20	4/20	4/20
DiscoveryWorld	10	10/10	4/10	7/10	3/10	5/10	1/10	1/10	2/10
HumanCLAW	13	3/13	2/13	1/13	0/13	3/13	0/13	0/13	2/13
ScienceWorld	73	64/73	38/73	59/73	29/73	45/73	11/73	14/73	28/73
VirtualHome	21	20/21	19/21	20/21	19/21	17/21	12/21	11/21	12/21
Total	157	126/157	70/157	100/157	56/157	76/157	28/157	30/157	49/157
Manipulation: task success (%)									
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7	33.3
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0	20.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0	4.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0	20.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0	60.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0	36.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0	11.1
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0	50.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0	80.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	182	41.8	19.8	24.7†	9.3	18.7	9.9	11.0	20.9

†See Appendix E.1 for coverage and aggregation.

C.2 Sensitivity to Source Composition

Table A10. Task-composition analysis (success %). Spatial excludes VSI-Bench from the categorical set (160 remaining cases); Planning removes ScienceWorld and VirtualHome; Affordance removes UMD and uses strict box success. All columns retain the assigned-case denominator of the indicated subset. Spatial and Affordance show mean ± run SD; Planning uses one run.

Agent	Spatial without VSI	Planning without text/symbolic	Affordance without UMD
Astra	83.6±1.2	66.7	16.5±1.4
Sol	71.4±3.3	20.6	10.5±2.7
Fable	77.9±3.1	33.3	15.0±5.0
Qwen-Max	80.9±1.4	12.7	10.0±3.5
Gemini	78.4±2.2	22.2	5.5±1.1
Qwen-397B	62.2±1.6	7.9	2.0±1.1
Qwen-27B	59.4±0.9	7.9	0.5±1.1
Opus 5.5	81.4±6.9	28.6	19.0±4.9
DeepSeek-1017	64.8±1.6	14.3	4.0±3.8

Removing ScienceWorld and VirtualHome changes Astra's Planning success from 80.3% to 66.7%, and Qwen-Max's from 35.7% to 12.7%. The same table isolates categorical Spatial Reasoning without VSI-Bench and strict affordance-box success without UMD. These subset comparisons measure sensitivity to the task mixture.

Equal source weighting. To measure sensitivity to source allocation in the original seven-agent cohort, we average complete source scores with equal weight, $M_{\text{source}} = B^{-1} \sum_{b=1}^{B} M_b$, retaining every assigned case within each source. Across all six Spatial sources, Astra scores 71.9% versus Qwen-Max's 69.1%; across all three Affordance sources, contact validity is 54.2% versus Fable's 50.9%; across all six Planning sources, success is 75.2% versus Fable's 53.1%. These are the highest non-Astra macro scores within the original seven-agent cohort. The ranking therefore persists under both case weighting and equal source weighting. These analyses use the complete domain allocations, separate from the 46-case multi-round review diagnostic in Section 4.4.

C.3 Opus 5.5 Repetitions, Dispersion, and Coverage

Opus 5.5 has five runs on 258 non-tracing Geometry cases, 220 Spatial Reasoning cases, and 70 point-and-box Affordance cases (2,740 assigned records). Four invalid InFlux references are excluded from eligible scoring. Repeated results use the equal-weight mean of per-run metrics and sample SD (divisor four). Tracing, direct polygons, Planning, and Manipulation each retain one run. Table A12 gives metric-specific coverage.

Planning and Manipulation. Planning includes all 157 assignments. The five non-HumanCLAW sources contribute 73 successes, and HumanCLAW adds 3/13, for 76/157 (48.4%). On HumanCLAW, Opus 5.5 succeeds on 3/5 navigation tasks and 0/8 sitting tasks. It finds the target and reaches within 20 cm on 7/13 cases, including 4/8 sitting tasks; proximity does not establish interaction success. Opus 5.5 Manipulation uses the same Baseline protocol as the original main table; its 87/182 score counts two RLBench exceptions and one RoboTwin exception as unsuccessful assignments. The original paired-protocol analysis is retained

separately. The supplemental recordings in Appendix H illustrate individual episodes; they do not establish the per-round image-input condition used in the paired protocol comparison.

Sequential progress and final conditions. Opus 5.5 completes at least one through five consecutive CALVIN goals on 80%, 70%, 40%, 40%, and 10% of the ten assignments. Summing these survival rates gives 2.40 consecutive goals on average. On the five ManiSkill diagnostic tasks, it satisfies all final goal conditions on PickCube, PlaceSphere, and StackCube, and none on PegInsertionSide or PlugCharger; their mean condition score is 60%. These Baseline diagnostics supplement the main evaluation; Opus 5.5 is outside the paired observation experiment. Opus 5.5 3D tracing has 8.9% joint success across 56 assignments, alongside the 0.683 m RMSE in Table A8.

Table A11. Opus 5.5 source results. Planning includes all six sources. Manipulation retains all binary assignments, including execution errors. Spatial values cover all questions in each source and show mean ± run SD across five repetitions.

Domain	Source	Assigned	Success (%)
Spatial	3DSRBench	50	76.8 ± 9.9
Spatial	BOP-ASK	20	95.0 ± 0.0
Spatial	MindCube	50	85.2 ± 6.7
Spatial	MMSI-Bench	20	59.0 ± 8.2
Spatial	RoboSpatial-Home	30	61.3 ± 4.5
Spatial	VSI-Bench	50	36.4 ± 1.7
Planning	ALFRED	20	15.0
Planning	ALFWorld	20	45.0
Planning	DiscoveryWorld	10	30.0
Planning	ScienceWorld	73	58.9
Planning	VirtualHome	21	71.4
Planning	HumanCLAW	13	23.1
Planning	All assigned	157	48.4
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	61.1
Manipulation	RLBench	10	30.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	10.0
Manipulation	RoboWits	10	10.0
Manipulation	RoboTwin 2.0	50	46.0
Manipulation	VIMA-Bench	10	100.0
Manipulation	VLABench	25	76.0
Manipulation	ManiSkill	9	55.6
Manipulation	LIBERO-PRO	10	80.0
Manipulation	Robosuite	5	100.0
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	All assigned	182	47.8

Table A12. Opus 5.5 repeated metrics as mean ± run SD. Coverage counts valid continuous predictions or all assigned binary predictions across five runs.

Metric	Mean ± SD	Scored/eligible
InFlux focal MAPE	60.27±1.49	130/130
InFlux principal error	3.82±0.00	130/130
Angle relative L2	1.166±0.113	60/60
Distance relative L2	0.455±0.069	50/50
Vector relative L2	0.516±0.042	40/40
Map-free rotation	40.95±8.10	142/150
Map-free translation	243.13±35.82	142/150
Map-free direction	44.29±9.06	142/150
GP focal MAPE	16.48±6.13	180/180
GP principal error	60.46±4.13	120/120
Depth MAE	0.510±0.295	60/60
Depth RMSE	0.599±0.302	60/60
Depth AbsRel	0.169±0.108	60/60
Displacement MAE	1.164±0.475	200/200
GP rotation	3.05±0.96	120/120
GP translation	11.13±1.48	120/120
Scale MAPE	3.91±2.74	220/220
Scale judgment Pass@1	100.0±0.0	60/60
Numerical AbsRel	0.184±0.040	130/135
BBox–GT box IoU	0.432±0.021	350/350
BBox–GT mask IoU	0.311±0.020	350/350
BBox success@0.5	51.4±3.0	350/350
Contact-point validity	61.1±1.9	350/350
Spatial Native Pass@1	67.5±5.1	1100/1100
Categorical Pass@1	79.2±5.6	915/915

C.4 DeepSeek-1017: Protocol, Dispersion, and Task Outcomes

DeepSeek-1017 denotes DeepSeek-V4.1-Flash-1017, the 17 October 2026 release served by `deepseek-flash`. Its main evaluation contains 3,192 assigned case–run slots covering all 1,000 cases; the added polygon experiment contributes 70 separate slots. The supplied archive fixes one adopted record per task and run; superseded transport and adapter attempts remain provenance records rather than additional trials. Non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance have five runs; tracing, Planning, and Manipulation have one. Four invalid InFlux references are excluded from eligible scoring. Failed or absent binary predictions retain their assigned denominator; continuous errors use finite valid predictions. Five-run SDs measure variation between run means.

Observations and execution budgets. Spatial runs use no YOLOE; 3DSRBench's corrected flipped inputs use matching native labels. In Manipulation, both Baseline and RGB-expanded use native RGB observations and tools. RGB-expanded supplies current views at every decision, whereas Baseline follows each source's native observation schedule. DeepSeek-1017 uses the former and completes 38/182 tasks. Its source-specific round, token, and in-episode time ceilings (Table A15) match those in Table A33. The paired observation experiment covers the original seven agents. Original Affordance uses native points and rectangles. A separate

direct-polygon run scores all 70 assigned cases, yielding 0.249 IoU; its 67 valid outputs and three zero-scored failures are detailed in Appendix C.5.

Local evidence and complete task outcomes. The source breakdown in Table A14 connects aggregate performance to task demands. DeepSeek-1017 succeeds on 2/5 HumanCLAW navigation tasks and 0/8 sitting tasks. It solves 49/157 Planning assignments, falling to 9/63 when text/symbolic ScienceWorld and VirtualHome are removed. CALVIN averages 1.40 consecutive goals, while only 1/10 full five-goal sequences succeeds. In the five-task ManiSkill diagnostic, none of the final condition sets is fully satisfied. BEHAVIOR-1K has no complete success in five assignments: pre-termination logs confirm localization in four and a grasp candidate in three. Only stages confirmed before termination enter the diagnostic.

Table A13. DeepSeek-1017 mean ± run SD and metric-specific coverage. Continuous metrics average valid predictions within each run; binary rates retain all assigned cases. Polygon, Planning, and Manipulation use one run.

Metric	Mean	Run SD	Scored/eligible
Pose rotation (°)	30.2	3.9	259/270
Pose translation (cm)	182.0	29.0	259/270
Spatial Pass@1 (%)	51.6	1.5	1100/1100
Numerical AbsRel	0.309	0.035	120/135
Box–mask IoU	0.217	0.011	348/350
Contact validity (%)	50.9	4.2	350/350
Polygon–mask IoU	0.249	—	70/70
Planning success (%)	31.210	—	157/157
Manipulation success (%)	20.879	—	182/182

Table A14. DeepSeek-1017 native source outcomes. Spatial includes all question types; Affordance uses the original point-or-box criterion. Manipulation measures complete-task success.

Domain	Source	Cases	Success (%)
Spatial	3DSRBench	50	61.6 ± 7.1
Spatial	BOP-ASK	20	88.0 ± 7.6
Spatial	MMSI-Bench	20	53.0 ± 5.7
Spatial	MindCube	50	54.4 ± 4.3
Spatial	RoboSpatial-Home	30	58.0 ± 3.8
Spatial	VSI-Bench	50	20.0 ± 2.0
Affordance	RAGNet-3DOI	20	39.0 ± 7.4
Affordance	ReasonAff-style	20	15.0 ± 3.5
Affordance	UMD	30	86.0 ± 6.4
Planning	ALFRED	20	5.0
Planning	ALFWorld	20	20.0
Planning	DiscoveryWorld	10	20.0
Planning	HumanCLAW	13	15.4
Planning	ScienceWorld	73	38.4
Planning	VirtualHome	21	57.1
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	33.3
Manipulation	LIBERO-PRO	10	50.0
Manipulation	ManiSkill	9	11.1

Domain	Source	Cases	Success (%)
Manipulation	RLBench	10	20.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	0.0
Manipulation	RoboTwin2	50	4.0
Manipulation	RoboWits	10	20.0
Manipulation	Robosuite	5	80.0
Manipulation	VIMA-Bench	10	60.0
Manipulation	VLABench	25	36.0

Table A15. DeepSeek-1017 Manipulation ceilings. Tokens are cumulative; minutes are in-episode wall-clock limits. Process wrappers allow additional startup and teardown time. Each assigned task is one episode. ManiSkill success counts nine binary tasks; its additional locomotion case has a continuous reward.

Source	Model rounds	Tokens	Minutes
CALVIN	40	1280k	120
BEHAVIOR-1K	24	640k	180
LIBERO-PRO	24	640k	180
Robosuite	24	640k	180
CLIPort	24	960k	60
ManiSkill	32	640k	60
RLBench	16	640k	60
RoboCasa	24	1120k	60
RoboCasa365	24	1120k	60
RoboTwin2	16	640k	60
RoboWits	24	640k	60
VIMA-Bench	16	640k	60
VLABench	16	640k	60

Table A16. DeepSeek-1017 execution stages. Fractions count confirmed events per assignment, with zero credit for unverified events. BEHAVIOR-1K uses pre-termination evidence.

Source	Diagnostic	Confirmed / assigned
CALVIN	All five ordered goals completed	1/10
ManiSkill	All final conditions on the five-task subset	0/5
RoboCasa	Approach; grasp and lift; stable release	3/4; 3/4; 0/4
RoboCasa365	Approach; grasp and lift; stable release	3/7; 3/7; 0/7
BEHAVIOR-1K	Localization; grasp candidate	4/5; 3/5
	Arrival; holding; held motion; release	0/5 each

C.5 Direct Polygon Grounding

The ten-model Affordance experiment retains the same 70 images and interaction instructions, with 20 RAGNet, 20 ReasonAff-style/3DOI, and 30 UMD cases. Each model directly predicts polygons in image coordinates alongside its point and bounding box. Rasterizing the polygons yields a predicted mask for IoU and Dice; no SAM, detector, or iterative segmentation refinement is added. The separate polygon run tests a more flexible output representation on the same tasks. Table 2 compares box and polygon overlap, while Table A8 reports point validity.

Coverage and aggregation. Table A17 reports single-run polygon scores by source. Invalid polygons and absent final submissions score zero. The original point/box pass flag is not treated as polygon success. No run-to-run SD is inferred from one run. DeepSeek-1017 yields 67 valid polygons: two outputs have vertices outside the image, and one case reaches the 12-round limit without a submission. All three contribute zero to the 70-case mean (24.92%). Its existing thinking-enabled, high-effort setting uses a 1,800-token visible-response ceiling; the 120k-token case budget also counts reasoning. The other limits remain 12 rounds and 900 seconds per case. Each of the 70 cases contributes one adopted attempt.

Table A17. Polygon–mask IoU (%), with one run on 20 RAGNet, 20 ReasonAff-style, and 30 UMD cases. Invalid or absent submissions score zero; all 70 cases enter the overall mean.

Model	RAGNet	ReasonAff	UMD	Overall
Astra	19.19	10.60	68.52	37.88
Sol	13.22	7.40	62.17	32.54
Fable	12.95	17.00	56.51	32.77
Qwen-Max	5.64	7.01	63.28	30.74
Gemini	10.92	4.21	57.71	29.06
Qwen-397B	5.86	2.45	16.02	9.24
Qwen-27B	3.98	0.30	9.05	5.10
Opus 5.5	18.55	13.29	71.49	39.74
DeepSeek-1017	8.67	9.22	46.21	24.92
Gemini 4 Argon	14.03	30.71	73.08	44.10

Gemini 4 Argon has the highest aggregate at 44.10%, followed by Opus 5.5 at 39.74% and Astra at 37.88%; all three score far lower on the interaction-conditioned scenes than on UMD tools (Figure 6b).

C.6 Interpreting the Capability Profiles

Figure 3 places complementary metrics on fixed visualization scales. Geometry is $50[\exp(-r/45) + \exp(-t/150)]$, where r and t are the mean per-run rotation error in degrees and translation error in centimeters. Spatial is $[p + 100\exp(-a/0.25)]/2$, where p is native Pass@1 (%) and a is numerical AbsRel. Affordance is $[100u + c]/2$, where u is original box–mask IoU and c is contact validity (%). Planning and Manipulation use their task success rates (%). G, S, A, P, and M denote these five domains. These monotone scales summarize complementary metrics; they are not native task tolerances or success probabilities.

Opus 5.5 includes all 157 Planning assignments, scoring 76/157 (48.4%), and all 182 Baseline Manipulation assignments. The Affordance index uses box–mask IoU and contact validity from Table A8; polygon IoU is reported as an additional metric.

DeepSeek-1017 uses the same transformations, giving G/S/A/P indices of 40.4/40.4/36.3/31.2 and a Manipulation score of 20.9. Its Manipulation axis marks the per-round RGB schedule. Gemini 4 Argon gives 57.8/61.2/47.7/70.1/44.0 and also uses per-round RGB in Manipulation. Panel a plots all ten profiles, and panel b reports their domain indices separately.

D Task Requirements and Model Comparisons

These diagnostics explain the task requirements behind Section 4.3 and Figures 4–8. Focused comparisons examine Astra against the original seven-agent cohort, while task tables include DeepSeek-1017 where matching records are available. Appendix C.3 provides Opus 5.5 results. Each domain connects the aggregate score to specific estimation, grounding, or execution conditions; Appendix B.4 defines repetitions and valid coverage.

D.1 Geometry: Task Requirements and Source Conditions

Task-level comparisons. Table A18a supplies the absolute errors behind Figure 4a. Its relative reduction is $100(1 - e_{\text{Astra}}/e_{\text{peer}})$, using the lowest-error non-Astra peer in the original cohort for each target. Equal-run means retain task-specific units: MAPE for focal length, size, and height, and $100 \times \text{AbsRel}$ for depth. Astra leads the five GeoProbe targets; Sol leads on InFlux focal estimation.

Coverage sensitivity. Table A18b restricts each Astra–peer comparison to cases valid in every shared run. All six comparison directions remain unchanged. The ranking is thus stable on the common valid cohort.

Table A18. Geometric errors (%, lower is better), mean ± run SD. (a) Original seven agents plus DeepSeek-1017 (DS-1017); GP denotes GeoProbe. (b) Common-valid cohorts with matched cases and runs.

(a) All valid cases

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
GP focal-only	6.2±2.1	11.2±5.5	29.5±1.8	13.8±3.1	23.7±12.9	38.6±3.2	38.4±5.0	38.1±10.5
InFlux focal	53.0±2.1	47.8±2.0	52.6±2.2	55.5±1.0	52.4±3.4	67.8±1.8	65.9±2.1	62.9±1.6
GP depth	10.4±2.9	22.9±2.7	14.2±2.7	16.3±1.3	19.2±2.1	31.2±2.8	29.2±1.3	22.9±2.5
GP object size	6.3±1.0	8.5±0.7	7.6±0.2	9.6±0.8	8.3±0.3	11.4±0.6	9.4±0.3	8.9±0.7
GP ref. height	0.31±0.11	0.90±0.36	2.31±0.08	1.69±0.72	2.73±0.75	17.55±1.22	13.69±0.86	5.71±1.49
GP height ratio	0.32±0.15	0.95±0.41	2.23±0.17	2.17±0.22	2.67±0.28	20.35±0.87	17.08±2.70	8.35±5.02

(b) Matched common-valid cohorts

Task	Cases	Rounds	Peer	Astra	Peer
GP focal-only	12	5	Sol	6.21±2.10	11.16±5.47
InFlux focal	26	5	Sol	53.02±2.15	47.79±2.00
GP depth	9	3	Fable	10.16±3.49	14.22±2.88
GP object size	12	3	Fable	6.44±1.36	7.63±0.18
GP ref. height	12	5	Sol	0.31±0.11	0.90±0.36
GP height ratio	12	5	Sol	0.32±0.15	0.95±0.41

Camera motion and real photographs. The rendered displacement subsets share 12 object interventions while changing the camera condition. Table A19 compares Astra with Sol, the lowest-MAE original non-Astra peer under each displacement condition. The fixed-camera advantage persists when the two stationary Blender cases are excluded. Under the native direction-and-1 cm criterion, both agents achieve 96.7% fixed-camera success: the continuous error distinguishes precision that this tolerance does not resolve.

The four real-photo pairs form a separate tabletop subset. Their displacement labels are supplied annotations, without independent camera or metric calibration. The Blender comparison uses different scenes and object motions.

Table A19. Displacement MAE (cm), mean ± run SD. Sol is the strongest original non-Astra comparator; DeepSeek-1017 is added under the same conditions. Moving-only excludes stationary cases.

Condition	Cases	Astra	Sol	DeepSeek-1017
Blender: fixed camera	12	0.240±0.109	0.467±0.267	1.517±0.105
Blender: fixed, moving-only	10	0.288±0.131	0.560±0.321	1.780±0.076
Blender: small view change	12	0.567±0.158	0.678±0.152	2.022±0.346
Blender: large view change	12	1.125±0.144	1.958±0.295	2.492±0.908
Real photographs	4	0.050±0.068	0.415±0.236	1.250±0.234

Coupled motion and metric scale reveal distinct estimation errors. Table A20 separates the GeoProbe pose modes: mixed rotation and translation produce 14.16 cm translation error and zero joint success, despite much lower errors on simpler motions. The best non-Astra translation estimate on the mixed subset is Qwen-Max at 16.77 cm; none of the original seven agents achieves joint success on this subset. Translation-only cases have zero target rotation. Depth also separates ordering from metric scale: Astra's ordinal accuracy is 96.3%, whereas raw AbsRel is 0.104; oracle scale alignment reduces it to 0.038. Alignment uses reference depth.

Explicit scale references and regular rendered geometry support these estimates.

Table A20. GeoProbe camera-pose modes. Errors and joint success show mean ± run SD. Joint success uses 2°/2 cm tolerances; failed predictions remain in its denominator.

Motion	Model	Cases	Rotation (°)	Translation (cm)	Joint success (%)
Stationary	Astra	4	0.00±0.00	0.00±0.00	100.0±0.0
	DeepSeek-1017	4	0.00±0.00	0.00±0.00	100.0±0.0
Rotation only	Astra	6	0.35±0.78	1.00±2.24	96.7±7.5
	DeepSeek-1017	6	5.43±1.57	13.82±5.69	13.3±13.9
Translation only	Astra	6	0.00±0.00	1.55±1.59	76.7±25.3
	DeepSeek-1017	6	1.18±0.91	11.03±3.22	20.0±13.9
Rotation + translation	Astra	8	3.26±0.75	14.16±3.94	0.0±0.0
	DeepSeek-1017	8	6.36±0.54	25.16±2.73	0.0±0.0

TraceSpatial-Bench: path matching and boundary constraints. Astra has lower 3D path RMSE than Gemini (0.382 versus 0.426 m), but fewer jointly valid paths (19/56 versus 21/56; Table A8). Among Astra's trajectories, 27 fail initiation, 24 fail arrival, and ten fail collision checks; these sets overlap. Of its 37 joint failures, 36 violate initiation or arrival, while only one satisfies both boundaries and fails collision avoidance alone. This supports the boundary-anchoring diagnosis in Figure 4c. The offline tolerances and sampled-position collision rule are defined in Appendix B.3.

D.2 Spatial Reasoning: Cross-Source Task Contrasts

Complete task coverage. Table A22 covers all 43 groups across six sources. Source categories are retained; MindCube uses task axes and question templates, prioritizing composed rotation and turn–translation before individual motion and relation queries.

Figure labels and comparisons. Figure 5a shows BOP-ASK left/right (L/R), MindCube relations (Rel.) and camera translation (Shift), 3DSRBench actor-relative side (Actor) and object orientation (Facing), and VSI hard directions (Hard). Panel b compares BOP camera depth (Near/Far), 3DSR camera/object distance (Cam./Obj.), and VSI relative distance (Rel.); each displayed peer leads or ties on a row. Source scenes are retained; the Obj. row has two cases.

Shared-round gaps. The original diagnostic below selects one task per source and its highest-mean original non-Astra peer; Figure 5c shows native source scores for all nine agents. Gaps and SD then use shared run indices (Table A21; protocol in Appendix B.3). On object-facing queries, Fable's three-run mean is 72.2% versus Astra's 50.0%; Astra averages 56.7% across all five runs.

Table A21. Original source-specific Astra–peer comparisons. Accuracy and gaps use shared available runs and are in percentages and percentage points. Run-index alignment follows Appendix B.3.

Task	Peer	Rounds	Astra	Peer	Gap ± SD
MMSI camera–region	Fable	3	100.0	77.8	+22.2±19.2
VSI medium direction	Qwen-Max	5	100.0	83.3	+16.7±11.8
BOP-ASK left/right	Gemini	5	95.0	96.7	−1.7±7.0
MindCube camera shift	Qwen-Max	5	78.5	89.2	−10.8±14.0
3DSR object facing	Fable	3	50.0	72.2	−22.2±9.6
RoboSpatial configuration	Q-397B	5	78.0	100.0	−22.0±4.5

Numerical error and acceptance are complementary. The numerical rows in Table A22 use the exact-match criterion from Appendix B.3. Table A23a instead measures continuous error on valid predictions. Fixing Astra's cohort across runs preserves low object-size AbsRel (0.091), while absolute-distance coverage is only 25/45 submissions (Table A23b). The table reports error together with valid-prediction coverage.

Table A22. All Spatial task groups, with DeepSeek-1017 (DS-1017) added to the original cohort. Entries are mean ± run SD (%); bold means mark the highest score in each row, including ties. Astra's column is shaded. N counts distinct cases; each assigned case remains in the success denominator.

Task	N	**Astra**	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
3DSRBench									50 cases
World height	4	60.0±13.7	50.0±0.0	**66.7**±14.4	55.0±11.2	25.0±0.0	50.0±0.0	50.0±0.0	25.0±0.0
Above / below	7	**77.1**±7.8	60.0±12.0	71.4±0.0	**77.1**±12.8	74.3±12.0	74.3±6.4	68.6±12.0	60.0±21.2
Camera depth	10	90.0±7.1	98.0±4.5	93.3±5.8	90.0±0.0	92.0±4.5	96.0±5.5	**100.0**±0.0	92.0±8.4
Next to	1	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	**60.0**±54.8	0.0±0.0	0.0±0.0
Object distance	2	70.0±27.4	80.0±27.4	83.3±28.9	**100.0**±0.0	**100.0**±0.0	50.0±0.0	80.0±27.4	50.0±0.0
Parallel / perpendicular	1	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	20.0±44.7	**100.0**±0.0	80.0±44.7	40.0±54.8
Same/different facing	4	55.0±11.2	50.0±0.0	50.0±0.0	50.0±0.0	50.0±0.0	**75.0**±17.7	35.0±13.7	40.0±13.7
Object-facing side	6	56.7±9.1	53.3±13.9	**72.2**±9.6	66.7±0.0	66.7±0.0	36.7±18.3	23.3±9.1	56.7±19.0
In front of actor	4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	95.0±11.2	90.0±13.7
Actor left/right	7	**74.3**±6.4	71.4±14.3	61.9±8.2	71.4±10.1	65.7±7.8	45.7±12.0	5.7±7.8	**74.3**±6.4
Object viewpoint	4	**70.0**±11.2	**70.0**±11.2	50.0±0.0	60.0±13.7	**70.0**±11.2	**70.0**±11.2	65.0±28.5	30.0±11.2
BOP-ASK									20 cases
Closer to camera	5	96.0±8.9	96.0±8.9	86.7±11.5	**100.0**±0.0	96.0±8.9	80.0±14.1	84.0±8.9	84.0±16.7
Farther from camera	3	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	66.7±0.0	40.0±14.9	86.7±18.3
Left / right	12	95.0±4.6	80.0±4.6	86.1±4.8	86.7±4.6	**96.7**±4.6	91.7±0.0	95.0±4.6	90.0±7.0
MindCube									50 cases
Behind observer	7	97.1±6.4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	88.6±12.0	80.0±16.3	65.7±16.3	**100.0**±0.0
Camera translation	13	78.5±10.0	64.6±8.8	56.4±16.0	**89.2**±6.9	84.6±5.4	20.0±6.9	35.4±12.9	27.7±14.0
Imagined observer	2	**90.0**±22.4	70.0±27.4	50.0±0.0	80.0±27.4	70.0±27.4	30.0±27.4	30.0±27.4	60.0±22.4
Object relations	12	**96.7**±4.6	63.3±12.6	91.7±8.3	85.0±3.7	95.0±7.5	31.7±7.0	35.0±3.7	45.0±9.5
Composed rotation	5	**100.0**±0.0	96.0±8.9	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	96.0±8.9	96.0±8.9
Turn then translate	11	**90.9**±6.4	43.6±7.6	75.8±10.5	**90.9**±6.4	87.3±10.4	49.1±13.8	58.2±10.4	47.3±4.1
MMSI									20 cases
Attribute (Appr.)	2	**90.0**±22.4	70.0±27.4	50.0±0.0	60.0±22.4	50.0±0.0	70.0±27.4	40.0±22.4	60.0±41.8

Table A22 continued. Values are mean ± run SD (%).

Task	N	**Astra**	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Attribute (Meas.)	2	**100.0**±0.0	50.0±0.0	66.7±57.7	40.0±22.4	50.0±0.0	0.0±0.0	40.0±22.4	80.0±27.4
MSR	2	40.0±41.8	30.0±27.4	**50.0**±0.0	40.0±22.4	30.0±27.4	30.0±44.7	40.0±22.4	**50.0**±0.0
Motion (Cam.)	1	80.0±44.7	80.0±44.7	**100.0**±0.0	**100.0**±0.0	60.0±54.8	0.0±0.0	0.0±0.0	60.0±54.8
Motion (Obj.)	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	80.0±44.7	80.0±44.7	80.0±44.7	20.0±44.7	80.0±44.7
Camera–camera relation	3	**100.0**±0.0	73.3±27.9	**100.0**±0.0	66.7±0.0	73.3±14.9	26.7±14.9	33.3±0.0	80.0±18.3
Camera–object relation	1	0.0±0.0	40.0±54.8	**66.7**±57.7	0.0±0.0	60.0±54.8	40.0±54.8	0.0±0.0	20.0±44.7
Camera–region relation	3	**100.0**±0.0	60.0±14.9	77.8±19.2	66.7±0.0	73.3±14.9	66.7±23.6	53.3±18.3	46.7±18.3
Object–object relation	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	60.0±54.8	20.0±44.7	80.0±44.7	60.0±54.8	80.0±44.7
Object–region relation	4	**50.0**±0.0	30.0±11.2	25.0±25.0	45.0±20.9	**50.0**±30.6	5.0±11.2	0.0±0.0	15.0±22.4
RoboSpatial									30 cases
Placement compatibility	10	92.0±4.5	92.0±8.4	**96.7**±5.8	90.0±0.0	64.0±8.9	80.0±7.1	90.0±7.1	92.0±13.0
Configuration	10	78.0±4.5	70.0±10.0	80.0±0.0	86.0±11.4	88.0±4.5	**100.0**±0.0	94.0±8.9	82.0±8.4
Free-space point set	10	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
VSI-Bench									50 cases
Appearance order	2	40.0±22.4	80.0±27.4	50.0±50.0	**100.0**±0.0	40.0±41.8	40.0±22.4	80.0±27.4	10.0±22.4
Absolute distance	9	2.2±5.0	2.2±5.0	3.7±6.4	0.0±0.0	**6.7**±6.1	0.0±0.0	2.2±5.0	0.0±0.0
Object count	5	32.0±11.0	20.0±14.1	**40.0**±20.0	36.0±16.7	36.0±16.7	28.0±11.0	12.0±11.0	16.0±8.9
Direction: easy	4	**100.0**±0.0	85.0±22.4	33.3±28.9	**100.0**±0.0	85.0±13.7	30.0±20.9	45.0±11.2	50.0±17.7
Direction: hard	4	45.0±11.2	30.0±27.4	25.0±43.3	**65.0**±28.5	60.0±22.4	30.0±32.6	5.0±11.2	0.0±0.0
Direction: medium	6	**100.0**±0.0	73.3±9.1	38.9±38.5	83.3±11.8	66.7±11.8	40.0±14.9	20.0±13.9	43.3±14.9
Relative distance	6	50.0±20.4	**76.7**±22.4	38.9±9.6	**76.7**±9.1	56.7±14.9	60.0±19.0	53.3±13.9	46.7±21.7
Object size	9	4.4±6.1	**22.2**±7.9	11.1±0.0	13.3±5.0	11.1±7.9	2.2±5.0	8.9±5.0	6.7±6.1
Room area	4	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
Route planning	1	**100.0**±0.0	**100.0**±0.0	33.3±57.7	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0

Table A23. VSI-Bench numerical estimation. (a) AbsRel, equal-run mean ± SD on each model's valid predictions; lower is better. (b) Astra's coverage and fixed-cohort sensitivity: common cases have valid predictions in every recorded round.

(a) Numerical error across agents

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Object count	0.35±0.05	0.40±0.11	0.20±0.11	0.23±0.11	0.31±0.08	0.41±0.06	0.42±0.04	0.34±0.09
Absolute distance	0.28±0.11	0.42±0.05	0.28±0.07	0.26±0.02	0.12±0.03	0.29±0.09	0.31±0.08	0.48±0.09
Object size	0.09±0.01	0.09±0.02	0.10±0.01	0.12±0.02	0.05±0.01	0.13±0.02	0.19±0.02	0.14±0.07
Room area	0.24±0.03	0.19±0.03	0.21±0.04	0.26±0.07	0.27±0.04	0.22±0.05	0.26±0.10	0.35±0.12

(b) Astra coverage and common-cohort error

Task	Cases	Valid / assigned	Common cases	Common AbsRel
Object count	5	25/25	5	0.350±0.053
Absolute distance	9	25/45	2	0.279±0.083
Object size	9	42/45	8	0.091±0.017
Room area	4	20/20	4	0.240±0.029

Other task requirements. RoboSpatial separates placement compatibility (92.0%), configuration (78.0%), and free-space point sets (0.0%) for Astra. All agents in this table fail the point-set criterion. Compatibility and configuration judgments succeed more often than point-set predictions.

D.3 Affordance: Contact and Region Extent

From valid contact to functional-region extent. Table A24 compares Astra and DeepSeek-1017 across all fifteen action/hand groups, including zero-scoring conditions. UMD contact validity averages 96.7% for Astra and 94.4% for Fable. Cutting combines 100% valid contacts and 96% strict box success with box–mask IoU of 0.361; the box IoU is 0.694. This separates successful localization from the limitations of rectangular mask coverage. Containment adds a same-representation comparison: Astra's strict box success is 44%, versus Qwen-Max's 96% (Table A25).

Table A24. Astra and DeepSeek-1017 Affordance metrics, mean ± run SD. IoU compares the native box to the GT mask; strict box success compares it to the GT box at IoU ≥ 0.5. Point and box success are percentages.

Action / hands	Model	Cases	Box–mask IoU	Point (%)	Box (%)
RAGNet-3DOI					
free movement / two hands	Astra	2	0.046±0.063	20.0±27.4	20.0±27.4
	DeepSeek-1017	2	0.179±0.070	50.0±0.0	30.0±27.4
free / one hand	Astra	1	0.384±0.351	60.0±54.8	60.0±54.8
	DeepSeek-1017	1	0.241±0.291	40.0±54.8	20.0±44.7
free / two hands	Astra	5	0.249±0.070	60.0±14.1	36.0±16.7
	DeepSeek-1017	5	0.083±0.030	28.0±11.0	4.0±8.9
pull / one hand	Astra	12	0.008±0.002	25.0±5.9	0.0±0.0
	DeepSeek-1017	12	0.053±0.019	41.7±8.3	0.0±0.0
ReasonAff-style					
free movement / two hands	Astra	5	0.270±0.055	48.0±11.0	48.0±11.0
	DeepSeek-1017	5	0.026±0.028	0.0±0.0	0.0±0.0
free / one hand	Astra	4	0.000±0.000	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.035±0.040	0.0±0.0	0.0±0.0
free / two hands	Astra	4	0.001±0.002	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.004±0.008	0.0±0.0	0.0±0.0
pull / one hand	Astra	4	0.098±0.033	60.0±13.7	0.0±0.0
	DeepSeek-1017	4	0.086±0.047	55.0±11.2	0.0±0.0
push / one hand	Astra	3	0.318±0.221	46.7±29.8	46.7±29.8
	DeepSeek-1017	3	0.146±0.201	26.7±27.9	20.0±29.8
UMD					
contain	Astra	5	0.472±0.067	100.0±0.0	44.0±16.7
	DeepSeek-1017	5	0.593±0.045	96.0±8.9	84.0±16.7
cut	Astra	5	0.361±0.009	100.0±0.0	96.0±8.9
	DeepSeek-1017	5	0.223±0.019	48.0±30.3	52.0±17.9
grasp	Astra	5	0.491±0.007	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.377±0.063	92.0±11.0	64.0±29.7
pound	Astra	5	0.637±0.009	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.490±0.034	96.0±8.9	44.0±16.7
scoop	Astra	5	0.504±0.054	80.0±0.0	80.0±0.0
	DeepSeek-1017	5	0.337±0.059	76.0±16.7	52.0±17.9
support	Astra	5	0.638±0.017	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.466±0.056	88.0±17.9	76.0±16.7

Contact, extent, and shape give different comparisons. Astra leads box–mask IoU on UMD cutting, grasping, scooping, and support, while Sol leads containment and pounding. In the interaction-conditioned sources, one-handed pulling has zero strict box success for every original model despite nonzero contact validity. Table A25 identifies the best peer separately for each reported criterion. ReasonAff and RAGNet retain their original scenes and interaction annotations.

Table A25. Metric-specific Affordance peers. Values are mean ± run SD, using the repetition counts in Appendix B.4. Peers are selected from the original cohort; DS-1017 adds DeepSeek-1017 under the same criterion.

Task / criterion	Astra	Original peer	Model	DS-1017
UMD cutting: box–mask IoU	0.361±0.009	0.341±0.022	Fable	0.223±0.019
UMD cutting: strict box (%)	96.0±8.9	100.0±0.0	Fable	52.0±17.9
UMD containment: box–mask IoU	0.472±0.067	0.673±0.069	Sol	0.593±0.045
UMD containment: strict box (%)	44.0±16.7	96.0±8.9	Qwen-Max	84.0±16.7
ReasonAff pulling: point (%)	60.0±13.7	50.0±0.0	Fable	55.0±11.2
RAGNet pulling: point (%)	25.0±5.9	31.7±7.0	Qwen-Max	41.7±8.3
ReasonAff two-hand movement: point (%)	48.0±11.0	26.7±11.5	Fable	0.0±0.0
RAGNet two-hand movement: point (%)	20.0±27.4	50.0±0.0	Sol / Fable	50.0±0.0

D.4 Task Planning: Goal Families and Terminal States

Compound goals and source-specific requirements. Table A26 covers all 56 native families across the six sources. Each row compares Astra with the highest-success original peer for that family. Figure 7a compares four models by source; panels b–c hold Fable fixed for household and scientific goal comparisons. Heating and dual placement test state change and multiple-object goal retention; ScienceWorld distinguishes experimental topics and instrument use. This preserves the task-specific advantages and reversals behind the aggregate Planning lead.

Table A26. Task Planning by native goal family, with one episode per assigned case. Entries are solved/assigned cases. One highest-success peer from the original cohort is shown per row, with DeepSeek-1017 (DS-1017) added separately; * denotes a tie with another peer. Source-level results appear in Table A9.

Goal family	Astra	Original peer	Model	DS-1017
ALFRED				
Illumination	1/2	1/2	Fable*	0/2
Placement	4/4	2/4	Sol*	0/4
Container transport	2/2	1/2	Fable*	1/2
Cleaning	2/2	1/2	Sol*	0/2
Cooling	2/4	2/4	Fable	0/4
Heating	2/3	0/3	Sol*	0/3
Dual placement	2/3	0/3	Sol*	0/3
ALFWorld				
Illumination	3/3	2/3	Fable*	2/3
Placement	3/4	2/4	Sol*	1/4
Container transport	2/2	1/2	Gemini*	1/2
Cleaning	3/4	1/4	Sol*	0/4
Cooling	0/1	0/1	Sol*	0/1
Heating	1/1	0/1	Sol*	0/1
Dual placement	2/5	1/5	Sol*	0/5
DiscoveryWorld				
Archaeology dating	1/1	1/1	Sol*	0/1
Combinatorial chemistry	1/1	1/1	Sol*	0/1
It's not rocket science	1/1	1/1	Fable	0/1
Lost in translation	1/1	1/1	Fable	0/1
Plant nutrients	1/1	0/1	Sol*	0/1
Discovery feed	1/1	1/1	Sol*	1/1
Doors	1/1	1/1	Qwen-Max	0/1
Navigation in a house	1/1	1/1	Fable*	0/1
Pick and give	1/1	1/1	Sol*	1/1
Space sick	1/1	1/1	Fable	0/1

Continued from previous page.

Goal family	Astra	Original peer	Model	DS-1017
HumanCLAW				
Navigate to contact	3/5	3/5	Gemini	2/5
Sit on target	0/8	0/8	Sol*	0/8
ScienceWorld				
Find animal	3/3	3/3	Fable	0/3
Find living thing	2/2	2/2	Sol*	2/2
Find non living thing	3/3	3/3	Sol*	3/3
Find plant	3/3	3/3	Sol*	2/3
Grow fruit	2/3	2/3	Fable	0/3
Inclined plane determine angle	2/2	2/2	Fable	2/2
Inclined plane friction named surfaces	15/16	16/16	Fable	8/16
Inclined plane friction unnamed surfaces	1/1	1/1	Fable*	0/1
Lifespan longest lived	2/2	2/2	Sol*	2/2
Lifespan longest lived then shortest lived	1/1	1/1	Sol*	1/1
Lifespan shortest lived	1/1	1/1	Sol*	1/1
Measure melting point known substance	8/8	6/8	Gemini	5/8
Measure melting point unknown substance	1/1	1/1	Sol*	0/1
Mendelian genetics known plant	1/1	1/1	Sol*	1/1
Mendelian genetics unknown plant	3/5	1/5	Qwen-Max*	0/5
Test conductivity	6/7	6/7	Gemini	0/7
Test conductivity of unknown substances	6/8	6/8	Fable	1/8
Use thermometer	4/6	5/6	Fable*	0/6
VirtualHome				
Grab	1/1	1/1	Sol*	0/1
Greet	1/1	1/1	Sol*	1/1
Lie	1/1	1/1	Sol*	1/1
Navigation or observation	1/1	1/1	Sol*	1/1
Read	2/2	2/2	Fable*	0/2
Rinse	1/1	1/1	Sol*	0/1
Sit	3/3	3/3	Sol*	1/3
Switchoff	1/1	1/1	Sol*	1/1
Switchon	7/7	7/7	Sol*	5/7
Touch	1/1	1/1	Sol*	1/1
Type	1/1	0/1	Sol*	0/1
Wash	0/1	1/1	Sol*	1/1

Attempted actions and terminal goals. All eleven Astra household failures exhaust their assigned budget. ScienceWorld's nine failures have scored task-failure outcomes with no recorded parser-no-match, invalid-action, or code-exception events; negative score sentinels are not physical-progress measures. HumanCLAW identifies a different limit (Table A27): six Astra sitting episodes find the target, five approach within 20 cm and issue a sit action, yet none succeeds. Fable records no target-finding success on these episodes. The shared zero terminal rate conceals different progress before the required body–object condition.

D.5 Manipulation: Control and Protocol Dependence

Baseline performance and coverage. Table A28 supports Table 2 and Figure 8a with all thirteen sources and the original seven agents. DeepSeek-1017 source results and execution limits are in Appendix C.4; its setting has no matched Baseline run. The baseline aggregate combines ten-source baseline results with the three sources shared across conditions. Astra records 76 successes out of 182 assignments; Fable records 45. Appendix E.1 defines denominator and coverage.

Table A27. HumanCLAW sitting diagnostics for eight assigned cases per model. Counts pool independently recorded conditions. † denotes lower bounds because process fields are missing for two Gemini cases; dashes denote unavailable process metrics. Terminal success retains all eight assignments. Opus 5.5 uses its final single-run records; near-plus-sit action counts are not supplied. DeepSeek-1017 adds 0/8 terminal sitting success; comparable process counts are not supplied.

Model	Process coverage	Target found	Within 20 cm	Near + sit issued	Success
Astra	8/8	6	5	5	0
Sol	8/8	3	1	1	0
Fable	8/8	0	0	0	0
Qwen-Max	0/8	–	–	–	0
Gemini	6/8	3†	3†	3†	0
Q-397B	0/8	–	–	–	0
Q-27B	0/8	–	–	–	0
Opus 5.5	8/8	4	4	–	0
DeepSeek-1017	–	–	–	–	0

Consecutive goals and task-specific evidence. Table A29 reports auxiliary metrics under both evaluation conditions. CALVIN reports average consecutive goals completed. The other rows average task-normalized native criteria or control evidence, with definitions below.

Table A28. Complete Manipulation success by protocol (%). Baseline totals use 182 assignments; – marks missing records. The last three sources use the same records in both protocols. The common nine-source subset isolates sources with records under both protocols.

Benchmark	*N*	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	41.8	19.8	24.7	9.3	18.7	9.9	11.0
Common nine sources	153	41.2	19.0	22.2	8.5	15.7	10.5	11.8
RGB-expanded								
CALVIN	10	0.0	10.0	40.0	10.0	0.0	0.0	0.0
CLIPort	18	55.6	33.3	50.0	27.8	38.9	22.2	16.7
RLBench	10	40.0	10.0	30.0	10.0	10.0	0.0	0.0
RoboCasa	10	10.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	20.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	42.0	12.0	34.0	14.0	6.0	4.0	6.0
RoboWits	10	10.0	10.0	20.0	20.0	10.0	0.0	10.0
VIMA-Bench	10	100.0	90.0	100.0	70.0	60.0	20.0	20.0
VLABench	25	64.0	48.0	68.0	32.0	52.0	24.0	28.0
ManiSkill	9	11.1	0.0	55.6	0.0	0.0	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	39.6	23.6	44.5	19.2	22.0	8.8	9.9

Continued from previous page.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Common nine sources	153	41.2	23.5	42.5	20.3	20.3	9.2	10.5

Table A29. Manipulation auxiliary metrics under both protocols. CALVIN reports average consecutive goals completed (0–5); other rows average task-normalized criteria or control evidence (%). Definitions and time windows appear in Table A30. – denotes unavailable records; the last three rows are shared.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN goals (0–5)	10	3.10	1.40	0.90	0.80	1.80	0.00	0.10
RoboCasa	4	–	–	–	–	–	–	–
RoboCasa365	7	–	–	–	–	–	–	–
ManiSkill conditions	5	60.00	0.00	–	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67
RGB-expanded								
CALVIN goals (0–5)	10	1.10	0.50	2.90	1.30	0.70	0.00	0.00
RoboCasa	4	25.00	33.33	83.33	50.00	50.00	16.67	25.00
RoboCasa365	7	33.33	23.81	57.14	28.57	47.62	28.57	23.81
ManiSkill conditions	5	0.00	0.00	60.00	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67

Comparisons with matching records. The common nine-source subset excludes ManiSkill and the three shared-current sources, leaving 153 assignments per agent and protocol. Astra has 63 successes in both configurations, but task-specific gains and losses offset; Fable increases from 34 to 65. Appendix E.1 examines the corresponding execution evidence.

Metric definitions and coverage. The 183-case Manipulation inventory contains 182 binary tasks and one continuous-return AnymalC-Spin task. AnymalC-Spin is evaluated separately through continuous return. For each model and configuration, task success is $\mathrm{SR} = 100N^{-1}\sum_i s_i$, where $s_i = 1$ requires native acceptance. Runs without native acceptance receive zero success credit, including timeouts, execution or infrastructure failures, and missing records. Each task contributes one run to the assigned denominator.

CALVIN scores a five-goal sequence as successful only when all five goals are completed in order. Its average number of consecutive goals completed is $\bar{L} = 10^{-1}\sum_{i=1}^{10} L_i$, where $L_i \in \{0, \ldots, 5\}$ is the verified consecutive prefix. Unknown prefixes contribute zero: baseline Qwen-Max has one such record, while expanded Gemini's five available records contribute seven goals over the fixed ten-case denominator, giving 0.70.

For the other auxiliary metrics, case i has K_i scoring items and verified indicators z_{ij}. Its score is $u_i = K_i^{-1}\sum_{j=1}^{K_i} z_{ij}$; the reported mean is $100N^{-1}\sum_i u_i$. Failed or unknown items contribute zero. This weights tasks equally even when K_i differs. The diagnostics summarize recorded control events on fixed task subsets using the scoring items and evidence windows in Table A30.

Table A30. Six auxiliary metrics, averaging task-normalized evidence within each subset.

Source	N	Scoring items	Evidence window
RoboCasa	4	Approach, grasp and lift, stable release	Any verified event during execution
RoboCasa365	7	Same three transport milestones	Any verified event during execution
ManiSkill	5	Two or three native conditions per task	Final verification state
LIBERO-PRO	10	Two task-specific conditions	Final state and limited process records
robosuite	5	Two lifting, three stacking, or four wiping items	Native success and terminal reward-derived evidence
BEHAVIOR-1K	5	Six control-evidence items	Before the first episode-end signal

Transport control versus destination success. RoboCasa uses ArrangeBreadBasket, LoadDishwasher, PickPlaceToasterToCounter, and RecycleBottlesByType. RoboCasa365 uses BreadAndCheese, MakeBananaMilkshake, MicrowaveThawingFridge, MixCakeFrosting, PlaceMeatInMarinade, PlaceVegetablesEvenly, and RemoveCuttingBoardItems. Door, appliance, and navigation-only tasks are outside these subsets. Approach requires executed motion and confirmed arrival; grasp-and-lift requires both grasp and lift confirmation. Stable release requires prior grasp-and-lift of the same canonically named object, followed by confirmed release and stability. Each item scores at most once per case. These events need not cover every task object, and release does not establish the requested destination relation.

Table A31. ManiSkill's five-task diagnostic. Items are evaluated at final verification; grasp history is separate. The other four binary tasks remain in complete-task success.

Task	K_i	Scored conditions
PickCube	2	Object placed at its goal; placed and robot static. The suspended goal requires no release.
PlaceSphere	3	Sphere on its support; on support and released; both with the sphere static.
StackCube	3	Cube A on B; on B and released; both with cube A static.
PlugCharger	2	Goal distance ≤ 0.005 m; distance satisfied and angular error ≤ 0.2 rad.
PegInsertionSide	2	Peg-head $x \geq -0.015$ m in the hole frame; official insertion success.

LiftPegUpright, PullCube, PushCube, and RollBall contribute to the nine-task success rate but not this diagnostic. Dashes mark unavailable Fable baseline records in both sets.

Task-specific conditions in LIBERO-PRO and robosuite. LIBERO-PRO uses fixed init-0 instances: spatial 1/7, object 2/5, goal 1/7, libero_10 1/9, spatial_swap 0, and object_swap 4. Ordinary placement scores recorded grasp and the official target relation; stove opening scores operation of the correct control and an open stove. The two longer tasks score a cup inside a microwave followed by the door being closed, or one food item followed by both items in a basket. Missing grasp/control records score zero; for the longer tasks, necessary preceding relations are confirmed only on official success.

robosuite uses cube lifting with seeds 99/50, cube stacking and restacking with seed 14, and spill wiping with seed 48. Lifting scores recorded grasp and native success; grasp history is missing. Stacking uses the fixed local reward q: $q > 0.125$ supports a grasp/lift/stack condition, $q \geq 0.5$ supports the top cube being lifted, and native success supplies the third item. A restacking zero can mask intermediate events under the two-cube airborne rule; these items remain unknown and receive zero credit. Wiping checks 25%, 50%, 75%, and complete removal. Current records verify the first three only on native complete success, rather than interpreting total reward as a wiped fraction. Consequently, Fable's 100% complete-task success and 80% auxiliary score are compatible: missing intermediate evidence lowers the latter.

Table A32. BEHAVIOR-1K control evidence. Each item records one type of control event and scores once per case; terminal BDDL goal success is evaluated separately.

Item	Positive evidence
Localization	Finite coordinates or pose from public RGB-D localization or grasp sampling.
Candidate	A nonempty pregrasp or grasp candidate set.
Arrival	Measured arrival from approach or recovery feedback for the same arm.
Holding	An explicit in-hand check or holding confirmation inside checked placement.
Held motion	Associated measured motion of at least 5 cm while holding, or movement with a passed holding check inside checked placement.
Release	Holding, active gripper opening, then no longer holding; or confirmed release inside checked placement.

The five tasks are outfit_a_basic_toolbox, can_meat, tidying_bedroom, putting_shoes_on_rack, and chop_an_onion. Records are linked by arm, name, and temporal order; physical-instance IDs are not consistently recorded. Object identity and inside/ontop relations are checked separately by the terminal goal evaluator. Held motion uses a 5 cm threshold. Control-event records cover four tasks for each of four models and the toolbox task for each of the other three. All retain the five-task denominator. Scoring uses the evaluated episodes up to their first end signal and requires direct evidence for each item. Fable's nine verified items yield $100 \times 9/(5 \times 6) = 30\%$, versus Astra's four items and 13.33%. RoboWits reports task success.

E Execution and Multi-Round Review Comparisons

E.1 Manipulation under the RGB-Expanded Protocol

Protocol and denominators. Baseline uses Table 2's source-specific observations and native tools. RGB-expanded adds current RGB at each decision. Both conditions use matched inference budgets, interaction allowances, and native execution-step limits. The comparison covers ten suites and 162 tasks: CALVIN, CLIPort, RLBench, RoboCasa, RoboCasa365, RoboTwin 2.0, RoboWits, VIMA, VLABench, and ManiSkill. Table 3b displays three suites; Table A28 gives all results. Common excludes ManiSkill because Fable's baseline records are missing, retaining nine suites and 153 tasks. LIBERO-PRO, robosuite, and BEHAVIOR-1K reuse 20 current records in both CSVs, bringing the domain total to 182. Missing baseline Fable ManiSkill and RoboCasa/365 stage records appear as dashes; the domain success aggregate credits the missing ManiSkill records with zero successes.

Observation and interaction settings. Baseline and RGB-expanded share the per-source inference and interaction budgets in Table A33, together with the same native execution-step limits. Baseline uses each source's native observations and tools; RGB-expanded provides the current scene's RGB views at every model decision. VIMA also supplies task reference images. Text interaction history and public state/perception tools remain available. Both conditions share an agent loop in which one code response can execute several actions.

Request and token accounting. The 7,378 RGB-expanded Astra, Sol, and Fable requests across ten ordinary groups use temperature zero, with neither explicit reasoning effort nor a separate thinking budget. Of these, 7,372 request 3,600 response tokens; six use smaller limits as the case budget runs low. The configurations use a 3,600-token response ceiling and a 12-minute request timeout. Reported reasoning usage is recorded separately and counted once in the case total, avoiding duplication when included in completion usage; absent usage details remain unknown. Verifier allowance is one more than the model-round limit.

Table A33. Shared execution ceilings for Baseline and RGB-expanded. Times are minutes; token budgets are thousands. A round is a model decision, Tools counts interface calls, and Checks counts verifier calls. Baseline limits apply to all eight agents; the paired RGB-expanded comparison covers the original seven.

Source	Rounds	Tools	Checks	Tokens	Trial	Code	Native
CALVIN	40	7,000	41	1,280	120	10	10
CLIPort	24	256	25	960	60	10	10
RLBench	16	256	17	640	60	10	10
RoboCasa	24	256	25	1,120	60	6	10
RoboCasa365	24	256	25	1,120	60	8	10
RoboTwin 2.0	16	256	17	640	60	10	20
RoboWits	24	256	25	640	60	30	20
VIMA-Bench	16	256	17	640	60	10	10
VLABench	16	256	17	640	60	10	10
ManiSkill	32	1,024	33	640	60	10	10
LIBERO-PRO	24	256	25	640	180	60	60
robosuite	24	256	25	640	180	60	60
BEHAVIOR-1K	24	256	25	640	180	60	60

Tool calls, action calls, and physical steps are distinct counters. Trial time includes model waiting, code execution, and feedback within the loop; Code and Native limit one code execution and one native call, respectively. Request timeouts do not add extra time beyond Trial. Initialization, outer watchdogs, and retry backoff have separate clocks, so Trial is not identical to total wall-clock duration.

VIMA's reference-image count varies by task. The remaining seven ManiSkill inventory cases, including AnymalC-Spin, use one base view. RoboCasa/365 views are rendered scene observations. Request images and process videos are separate records: continuous video is not enabled uniformly for VLABench, LIBERO-PRO, or robosuite, while the later BEHAVIOR revision queue records at 15 fps. Video frame rate does not define model observation frequency.

Table A34. Expanded camera presets, width × height in pixels. The preset inventory covers all 183 assigned cases, including AnymalC-Spin. Multiple views enter as separate images.

Source	Views	Resolution
CALVIN	Static; gripper	200×200; 84×84
CLIPort	Front, left, right; no Oracle camera	640×480
RLBench	Front, wrist, overhead	128×128
RoboCasa/365	Left/right agent view, eye-in-hand	512×512
RoboTwin 2.0	Head, left, right	320×240
RoboWits	Ego, left/right wrist	848×480
VIMA-Bench	Front/top scene views and task reference images	256×128
VLABench	Right, left, forward, Franka wrist	480×480
ManiSkill	Base; additional hand view for PegInsertionSide, PlugCharger, StackCube	128×128
LIBERO-PRO	Agent view, wrist	800×512
robosuite	Agent view, wrist	512×512
BEHAVIOR-1K	Ego, left/right wrist	512×512

CALVIN: interaction versus feedback control. The execution analysis reports Astra's average consecutive goals decreasing from 3.10 to 1.10, Sol's from 1.40 to 0.50, and Fable's increasing from 0.90 to 2.90. Astra's median action-interface calls fall from 1,795 to 480.5, and Sol's from 1,247.5 to 734. Their median cumulative prompt tokens rise from approximately 321k and 363k to 1,206k and 1,217k. These totals accumulate prompt tokens over successive requests. Under RGB-expanded, five Astra and eight Sol episodes exhaust the shared per-case token

Table A35. Native execution-step limits shared by both conditions. Values show environment defaults → effective evaluation limits; control frequency is unchanged.

Source	Base → effective	Counter and termination semantics
CLIPort	5 → 10	Pick-and-place actions
RoboCasa/365	1000 → 2000	Control steps; field does not truncate episodes in the current wrapper
RoboTwin 2.0	400 → 800	Native action calls, not all simulator steps
RoboWits	800 → 1600	Control steps for the checked case
ManiSkill	50 → 100	Control steps; PickCube termination/boundary checks verify the effective limit
LIBERO-PRO	4000 → 8000	Wrapper/control counter; field does not truncate episodes
robosuite	1500 → 3000	Wrapper/control counter; field does not truncate episodes
BEHAVIOR-1K	500 → 1000	Environment control steps; later collision revision retains 1000
CALVIN, RLBench, VIMA	No finite limit	Checked instances remain bounded by rounds, calls, tokens, and time
VLABench	Unlimited	Simulation-time limit remains unlimited

allowance. Sol nevertheless increases full-sequence success from zero to one, illustrating why mean progress and completion need separate reporting.

Figure A2a connects these aggregate changes to controller construction and final goal completion in sequence 0557.

Budget utilization and interface effects. Three of Fable's four RGB-expanded CALVIN successes finish at rounds 22, 24, and 25, within the shared execution allowance. These episodes use later rounds to continue control after establishing feedback. Figure A2b examines controller construction in PickCube. Tool semantics also matter: the RGB tool compares colors on a 0–255 scale, whereas some Astra/Sol calls use 0–1 values and tight tolerances, producing empty detections. Fable obtains the red region with [255,0,0] and tolerance 80. Correct units determine whether the controller receives a usable target.

Delivered observations and execution failures. All 7,378 Astra, Sol, and Fable requests across the ten benchmark groups contain RGB. No archived Astra or Sol response ends at the response-length limit, whereas 396 Fable responses do. RLBench still contains four Astra, two Sol, and four Fable code-execution timeouts. Astra/Sol failures therefore occur even with delivered images and complete responses; code timeouts and interface semantics provide additional execution-level explanations. VIMA improves for all three agents, and Astra improves on RoboTwin while remaining unchanged over the common 153 tasks. The gains vary with the task and how each model uses the additional visual observations.

E.2 Fixed-Input Multi-Round Review

Multi-round review lets agents revisit their initial answers using the same task observations and scoring criteria. Self-review adds reconsideration, Crop review adds image crops, and Tool-assisted review adds specialist visual and geometric measurements. The three conditions permit up to two, two, and six additional responses, respectively, across Geometry, Spatial Reasoning, and Affordance. Table 3a reports results; sampling and tools are specified below.

Shared cases. Astra and Qwen-27B share 46 cases: 19 Geometry, 18 Spatial Reasoning, and nine Affordance. A 72-case pilot covers 24 benchmark/task strata selected by task category and SHA-256 ordering with seed 20260923, independently of scores and answers. It excludes the 60 exact cases from an earlier performance-selected study. One case per stratum forms a 24-case development set; withholding two more cases that share development scenes leaves the 46 evaluation cases. Some scenes recur, and some cases were inspected during development.

Evidence construction. Tool-assisted review adapts GCA's measurement and reference-frame formalization (Chen et al., 2025b) and SpatialClaw's inspectable execution (Cho et al., 2026). It

Table A36. Review conditions for the same cases and native scoring interfaces. Budgets are additional to native execution; response limits count output tokens.

Condition	Evidence available during review	Additional response budget
Initial	Original task observations	None
Self-review	Original observations and initial answer	Two responses, up to 1,800 tokens each
Crop review	Self-review evidence plus image crops	Two responses, up to 1,800 tokens each
Tool-assisted	Original RGB, crops, and specialist measurements	Six responses, up to 4,096 tokens each; up to 16 specialist requests

formulates the measurement before consulting the candidate answer, then uses a persistent Python session. SAM 3 (Carion et al., 2025) supplies regions; SuperPoint (DeTone et al., 2018) and LightGlue (Lindenberger et al., 2023) supply correspondences; GeoCalib (Veicht et al., 2024) estimates calibration. Numerical routines compute relations and estimates. Finite-value and coordinate checks validate the computed outputs. Remaining semantic errors include selecting a whole-object mask for a functional part and ambiguous translation scale.

Observation and execution controls. Specialist tools receive only previously exposed RGB, without predicted depth, learned 3D reconstruction, private calibration, scoring annotations, or extra video frames. Public depth or masks originally supplied by a task remain available to the base model. Review reuses native candidates where available; one Astra Spatial case starts from its public observation because the candidate is missing. Temperature is zero, and total-token and wall-time accounting includes native execution. Incomplete runs may be retried; the first completed result is retained regardless of correctness, yielding one outcome per condition and case.

Outcome interpretation. Geometry counts improved, tied, and worsened results across 24 metric entries from 19 cases; one Qwen-27B crop entry is unavailable. Spatial accuracy retains 18 assigned cases, with missing answers counted as incorrect. Affordance reports box–box IoU and success at IoU≥ 0.5 over nine cases.

E.3 Native Capability and Complete Task Execution

Native capability concerns the decisions a model makes through an execution interface. Comparing robot-agent studies therefore requires identifying the task requirements, the competence supplied by the interface, and the resources available for completion. These distinctions contextualize the arena's findings without changing its central question: can the model complete the task through its available operations?

Context and supplied control competence. Robotic in-context learning uses demonstrations and interaction through context-conditioned policies, geometric transfer, world models, or skill- and agent-based control (Huang et al., 2026a). Fixed model weights can therefore coexist with substantial execution support. Learned skill selection (Ahn et al., 2022), programmatic composition of perception and control APIs (Liang et al., 2022), and spatial value maps (Huang et al., 2023a) assign different responsibilities to the model and executor. GPT-Policy (Cheng et al., 2026) conditions a fixed VLM on demonstrations, action references, and feedback, while a constrained controller checks and executes its requests. Our baseline leaves task interpretation and action selection to the model through source-native helpers; the supplementary review study separately introduces specialist measurements.

Task composition and execution support change the comparison. Our RoboCasa sample contains seven atomic and three composite tasks; RoboCasa365 contains three atomic and seven composite tasks (Appendix A). PyRUA-Lean (Si et al., 2026) evaluates 18 atomic and 32 composite RoboCasa365 task types at five seeds each, with 86.7% and 39.4% success. Its

Target50 selection differs from our pretrain-designated instances. Removing its frozen RLDX-1 policy lowers atomic and composite success to 65.6% and 20.0%, while retaining programmatic composition of other primitives. Execution allowances also differ: our RoboCasa protocols use 24 model rounds and 60 minutes of trial time; PyRUA-Lean uses 40 calls for atomic tasks, 100 for composite tasks, and a two-hour watchdog. Both allow multiple actions per Python response. These settings evaluate different combinations of model reasoning, supplied control, and task demands.

More interaction helps when it supports effective control. GPT-Policy (Cheng et al., 2026) reports 0/3 towel-pickup successes without demonstrations, averaging 96.3 decisions and 24.6 minutes. Human video raises success to 2/3 and lowers those averages to 76.7 decisions and 18.9 minutes, including failed trials. Our CALVIN traces likewise distinguish extended activity from productive control: five Astra and eight Sol RGB-expanded episodes exhaust their token allowance, while three of Fable's four successes finish at rounds 22, 24, and 25 (Appendix E.1). In PickCube, Astra and Sol make 25 and 32 responses without success; Fable succeeds through calibrated feedback control (Figure A2). Round counts describe interaction use, while recorded duration measures elapsed time.

Precision must survive through termination. GPT-Policy's plug task requires the plug to remain fully seated after release. Robot video yields 0/3 successes; adding aligned action references yields 2/3 (Cheng et al., 2026). Our within-task diagnostics expose related final-state requirements: initiation or arrival fails in 36 of Astra's 37 unsuccessful traces, and five HumanCLAW episodes reach the seat and issue a sitting action without satisfying the sitting condition. The common issue is converting locally plausible decisions into the exact state the task requires. This is the ability the arena measures across native VLM agents, together with the interaction and time consumed before completion or termination.

F Case Studies

From spatial judgments to usable contacts. Figure A1 illustrates Section 4.3 with six cases from the first evaluation run.

① *Spatial references.* In these runs, Astra correctly judges placement compatibility but misidentifies object-facing direction and camera translation. ② *Contact versus coverage.* Astra identifies usable tool and furniture regions. For pulling, three models place valid contacts on the handle, but their boxes cover only a fraction of the reference door region.

Controller construction across observation settings. Figure A2 links controller construction to terminal outcomes in CALVIN sequence 0557 and ManiSkill PickCube; execution settings are in Appendix E.1.

CALVIN: Astra's feedback controller emerges later. Baseline Astra builds control from tool-center-point error at round 5; RGB-expanded Astra builds it at round 12. Fable constructs a state-reading movement loop at round 2. Earlier controller construction coincides with longer completed goal sequences in these cases.

PickCube: fewer steps, better organized control. Fable calibrates joint response before constructing reusable numerical feedback control, then succeeds in 36 native steps. Astra and Sol use 100 and 56 steps across 25 and 32 responses without success; their trajectories emphasize direct joint increments and cross-round queries.

Motion and task completion. Figure A3 contrasts visible motion with verified task completion.

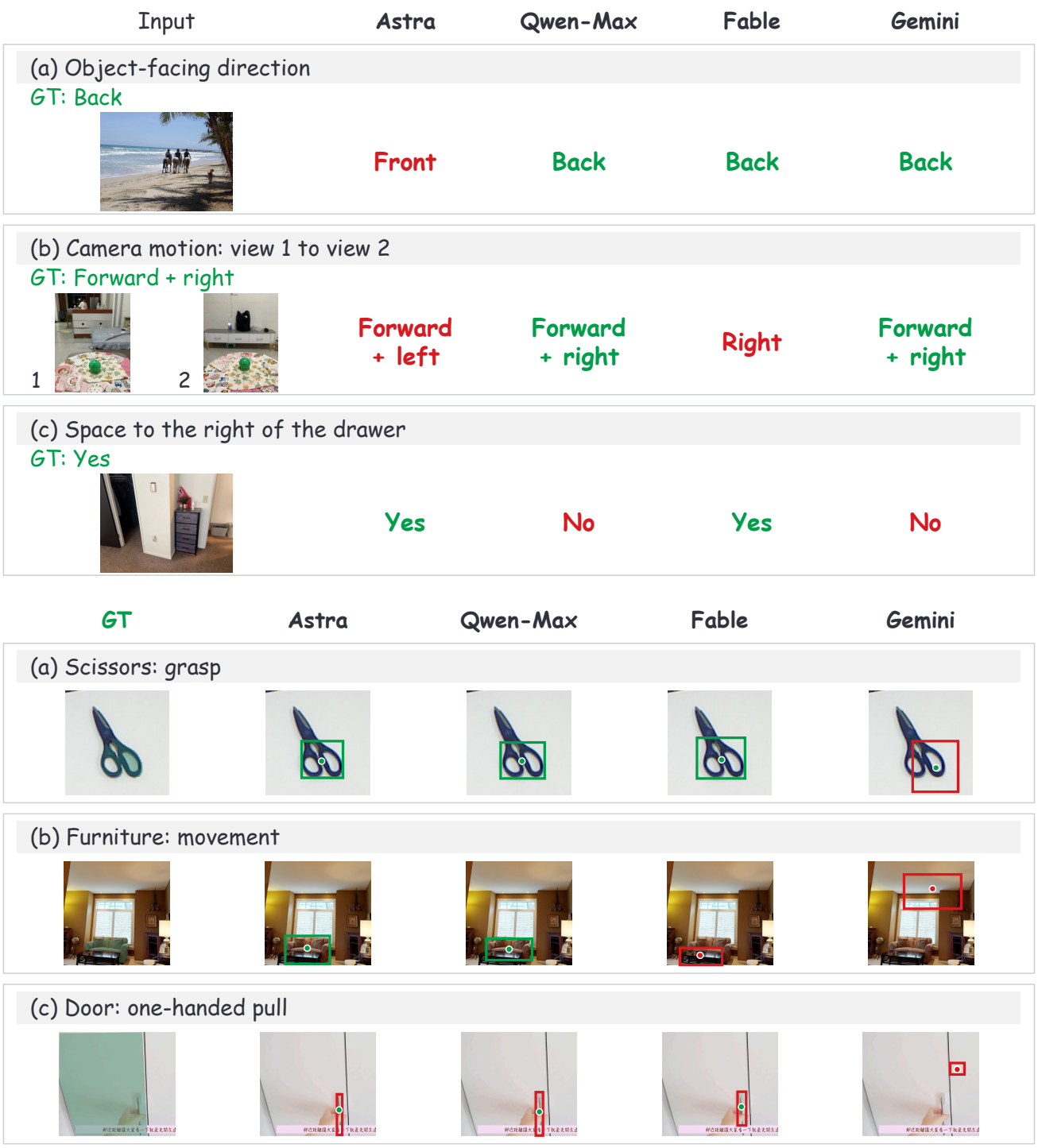


Figure A1. Spatial judgments from the first evaluation run (top: 3DSRBench, MindCube, RoboSpatial) and affordance localization (bottom: UMD, ReasonAff, RAGNet). Top questions concern the dog's facing direction, camera motion, and whether the plant fits beside the drawer; green answers match GT. Below, green points lie inside the GT mask and green boxes have IoU≥ 0.5; red denotes failure.

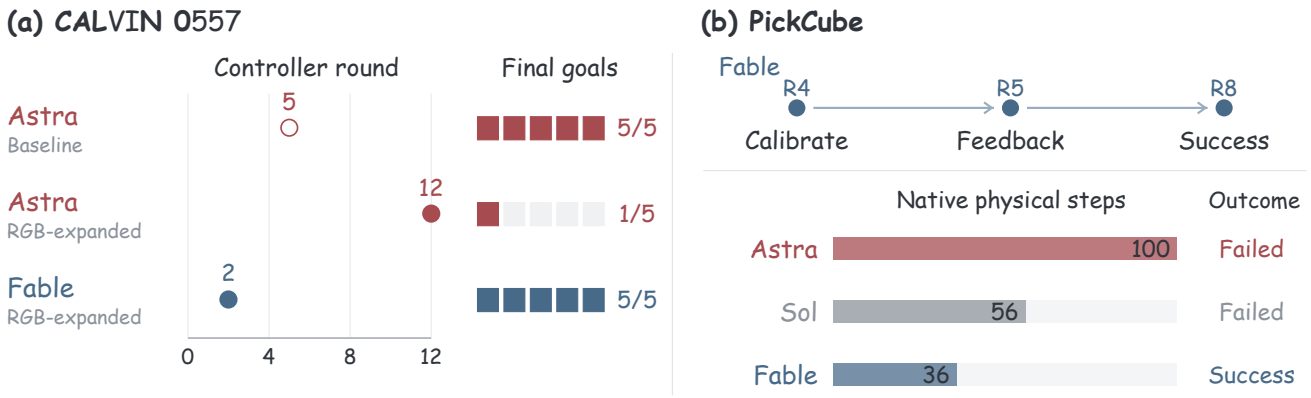


Figure A2. Controller construction and task outcomes. (a) CALVIN dots mark controller construction rounds; filled tiles count final consecutive goals. (b) PickCube shows Fable's ordered calibration, feedback-control, and success events, plus all three agents' physical-step totals and outcomes under RGB-expanded execution. R denotes model rounds; physical steps count environment updates.

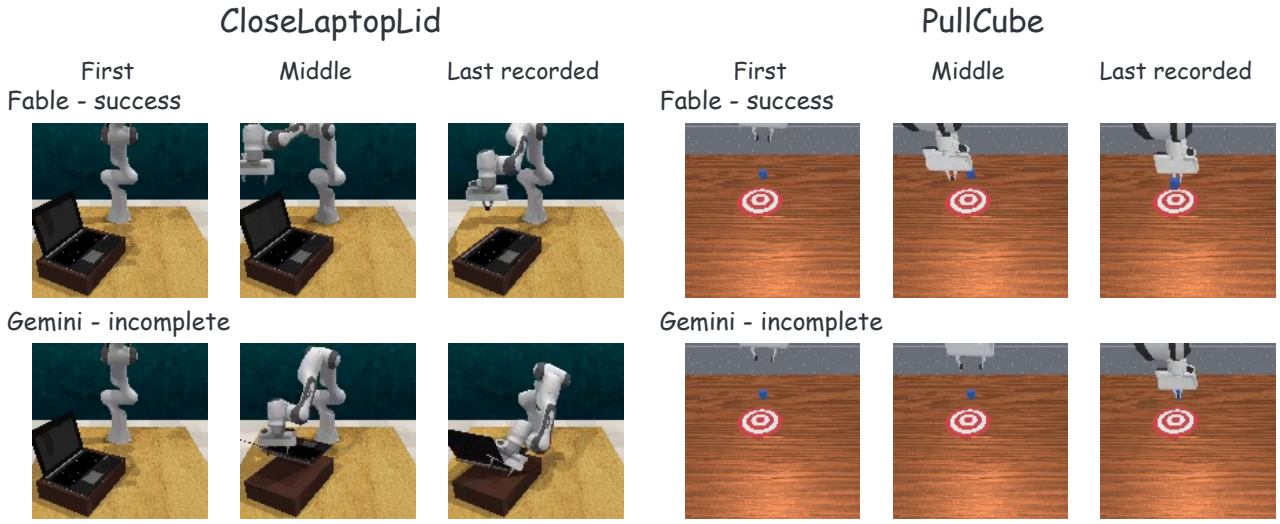


Figure A3. Archived execution contrasts under matched task settings and budgets. CloseLaptopLid uses RLBench's front camera; PullCube uses ManiSkill's base camera. Native records determine success. These earlier episodes are separate from the protocol comparison.

G Evaluation Scales and Public Resources

Table A37 expands Table 1 with instance counts and released artifacts. Task categories, training demonstrations, and source datasets are excluded from the instance count. Our release provides a fixed 1,000-case index across five capability domains, together with observations, episode initializations, and evaluation code.

OpenEQA adds navigation to episodic QA in its active track; Memory denotes cross-session persistence. VLABench's non-interactive track scores action sequences from fixed multi-view images against reference graphs. Hard Vision, Easy Vision varies reasoning effort and tool use; its 55 source benchmarks do not specify an instance count.

Table A37. Reported evaluation populations and released resources. Counts follow the cited papers; artifact availability is checked on 3–4 October 2026. Task-type and source counts describe coverage rather than tested instances. Links lead to author-maintained resources.

Study	Reported evaluation population	Released evaluation resources
OpenEQA (2024)	1,636 QA entries; the 557 HM3D questions also support active navigation. GPT-4V's active evaluation uses a 184-question subset.	Baseline and scoring code; QA manifest and episode-history download instructions.
BEAR (2025)	4,469 multimodal QA entries; robot online validation is additional.	Agent code; benchmark data and scoring scripts.
EmbodiedBench (2025a)	1,128 testing instances in four environments.	Code and configurations; benchmark datasets.
EmbodiedEval (2025)	328 tasks in 125 scenes.	Code; 328-entry test set and scene assets.
Embodied Arena (2025)	22 source benchmarks; a combined evaluated-instance count is not reported.	Platform and configurations; source datasets are downloaded separately.
CaP-X (2026)	Core: seven task types, each with 100 trials per tier. Extensions include 30 LIBERO-PRO task types and two BEHAVIOR tasks with 25 trials each.	Code, environments, and evaluation scripts. The repository lists 39 task types; the paper describes 187 platform tasks.
VLABench (2024)	100 task categories: 60 primitive and 40 composite; episode counts depend on the experiment.	Code and evaluation tracks; fixed-image VLM evaluation data.
WorldLines (2026b)	310 memory QA entries and 21 planning cases; the 62-entry ablation uses a QA subset.	The paper supplies prompts and examples; an author code/data release was not located.
Hard/Easy Vision (2026)	55 benchmarks, 34 capabilities, nine areas; total and per-benchmark evaluated-instance counts are not reported.	Results and qualitative examples; the repository contains a README and overview image, without evaluation scripts or a test manifest.
Embodied Agent Arena	**1,000 cases** from 32 established sources and GeoProbe, including 168 GeoProbe cases.	Harness and evaluation code; case index, observations, and episode initializations.

H Additional Cases Across the Five Capabilities

Figures A4–A8 connect instructions to predictions and execution. The public case atlas covers 181 tasks across 43 benchmark entries and subsets, with ten-model results on each task. Gemini 4 Argon adds 79 matched recordings to the execution gallery. The detailed execution-log analysis retains 143 episodes in 46 same-task groups from the earlier nine-model cohort (Appendix I). The selection spans task types, model disagreements, and successful and unsuccessful outcomes.

(a) GeoProbe | Real Object Motion

Track the green-handled glass jug relative to the table.

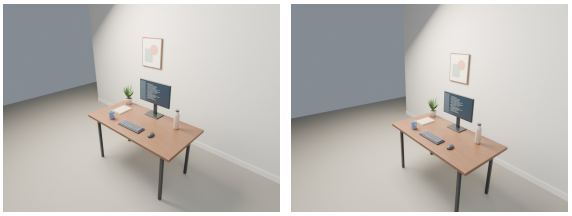
View 1 View 2

Model	Direction / cm
GPT-6 Astra	right / 2
GPT-6 Sol	right / 2
Fable 5.1	right / 4
Gemini 3.8 Flash	none / 0
Qwen 3.8 Max	right / 3
Qwen 3.5 397B	none / 0
Qwen 3.5 27B	left / 7
Opus 5.5	right / 2.5
DeepSeek-1017	up-right / 4

Reference: right, 2 cm

(b) GeoProbe | Camera Translation

Estimate camera B's position in the coordinate frame of camera A.



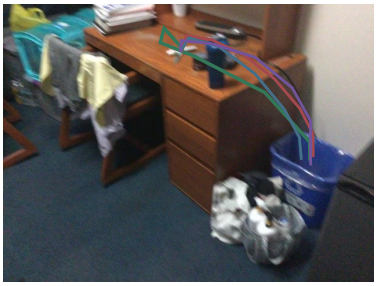
View 1 View 2

Model	Translation (cm)
GPT-6 Astra	[8, 0, -10]
GPT-6 Sol	[10, -10, 0]
Fable 5.1	[6, -3, 0]
Gemini 3.8 Flash	[25, 12, 0]
Qwen 3.8 Max	[10, 4, 0]
Qwen 3.5 397B	[15, 8, 5]
Qwen 3.5 27B	[8, 3, -12]
Opus 5.5	[15, -5, 0]
DeepSeek-1017	[6, -5, 12]

Reference translation: [12, 9, 0] cm

(c) TraceSpatial | 3D Trajectory

Predict a trajectory from the toothpaste to the blue waste bin.



GT Astra Fable Opus 5.5

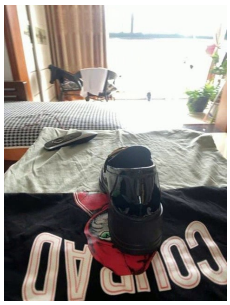
Image projection of 3D paths

Model	3D Fréchet (m)
GPT-6 Astra	0.525
GPT-6 Sol	0.634
Fable 5.1	0.826
Gemini 3.8 Flash	1.148
Qwen 3.8 Max	0.699
Qwen 3.5 397B	0.715
Qwen 3.5 27B	0.497
Opus 5.5	0.549
DeepSeek-1017	0.556

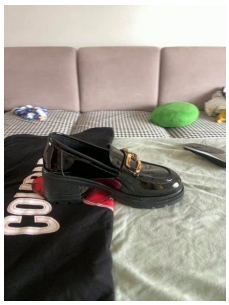
Figure A4. Geometry: displacement, reference frames, and trajectories. Selected first-run predictions. (a) The jug moves right by 2 cm relative to the table. (b) Camera translation uses camera A's frame, with reference $[12, 9, 0]$ cm; orientation is omitted. (c) TraceSpatial paths are projected into the image; the errors are 3D Fréchet distances in meters, distinct from aggregate trajectory RMSE. Reference, Astra, Fable, and Opus 5.5 paths are drawn, with all nine errors listed.

(a) MindCube | Viewpoint Change

In which direction did the camera move between these views?



View 1



View 2

Model	First-run answer
GPT-6 Astra	D
GPT-6 Sol	B
Fable 5.1	A
Gemini 3.8 Flash	A
Qwen 3.8 Max	A
Qwen 3.5 397B	C
Qwen 3.5 27B	A
Opus 5.5	A
DeepSeek-1017	A

Reference A: diagonally forward and right

(b) MMSI-Bench | Relative Direction

If the open cupboard faces north, where is the cooking area?

View 1

View 2

Model	First-run answer
GPT-6 Astra	A
GPT-6 Sol	A
Fable 5.1	A
Gemini 3.8 Flash	D
Qwen 3.8 Max	C
Qwen 3.5 397B	B
Qwen 3.5 27B	B
Opus 5.5	D
DeepSeek-1017	D

Reference A: northwest

(c) 3DSRBench | Object Orientation

Which side of the person is facing the camera?

Model	First-run answer
GPT-6 Astra	B
GPT-6 Sol	B
Fable 5.1	C
Gemini 3.8 Flash	B
Qwen 3.8 Max	B
Qwen 3.5 397B	C
Qwen 3.5 27B	B
Opus 5.5	C
DeepSeek-1017	C

Reference B: left

Figure A5. Spatial Reasoning: errors depend on the reference frame. Selected first-run answers without YOLOE. (a) MindCube asks about camera movement between two views: A is diagonally forward-right, B right, C left, and D diagonally forward-left. (b) MMSI-Bench displays two selected views from its multi-view input and asks for a direction anchored to the open cupboard: A is northwest, B southwest, C north, and D northeast. (c) 3DSRBench asks which side of the person faces the camera: A is front, B left, C back, and D right. Green letters match the source reference; red letters do not. Different agents succeed on different frame transformations.

Figure A6. Affordance: region shape and target identity are separate challenges. Direct polygon outputs on the same UMD, ReasonAff, and RAGNet tasks. Green fill is the reference mask; colored native contours are model predictions. Each row uses one shared crop for the reference, Astra, Fable 5.1, Opus 5.5, and DeepSeek-1017. IoU is computed over the full original image, not the crop. (a) Scissor outlines differ in extent and alignment. (b) Handle contours cover only part of the annotated pulling region. (c) Agents select different objects under the two-hand instruction. These are single-run examples; the project-page gallery provides all ten models, full images, and recorded polygon coordinates.

Boundary precision and task interpretation fail differently. On the scissors, Opus 5.5 reaches 0.627 IoU while DeepSeek-1017 reaches 0.128: both identify the tool, but their outlines recover different parts of its cutting region. On the RAGNet image, Opus 5.5 reaches 0.942 while the other three displayed models score zero, reflecting different target selections rather than a small boundary error.

A plausible contact is not a complete region. All four pulling predictions concentrate on the handle, while the annotation covers a larger region; IoU remains 0.096–0.140. This example explains why contact validity and region overlap measure different aspects of affordance, and why neither alone establishes successful physical execution.

(a) ALFWorld | Object and Light

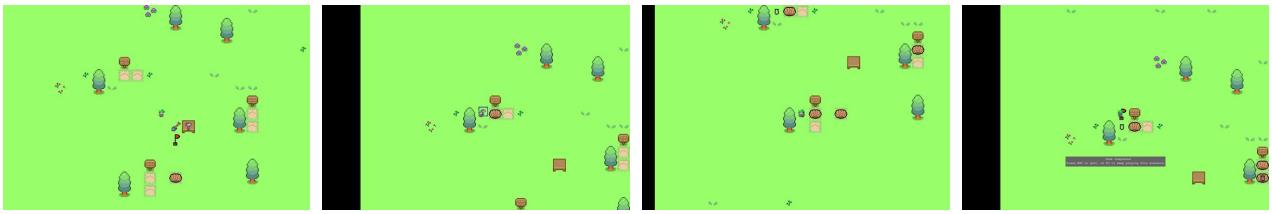
Turn on a lamp while holding the requested object.

obs. 1 obs. 3 obs. 6 obs. 8

Claude Opus 5.5 Reported success | 8 code rounds | 80.9 s

(b) DiscoveryWorld | Archaeology

Date the artifacts and flag the site containing the oldest one.

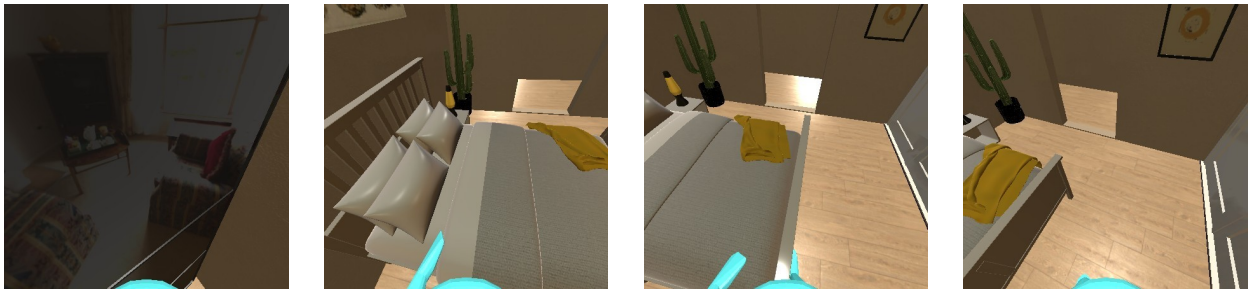


obs. 1 obs. 13 obs. 31 obs. 44

Claude Opus 5.5 Reported success | 7 code rounds | 167.9 s

(c) HumanCLAW | Sitting

Find the couch, approach until touching it, and sit.



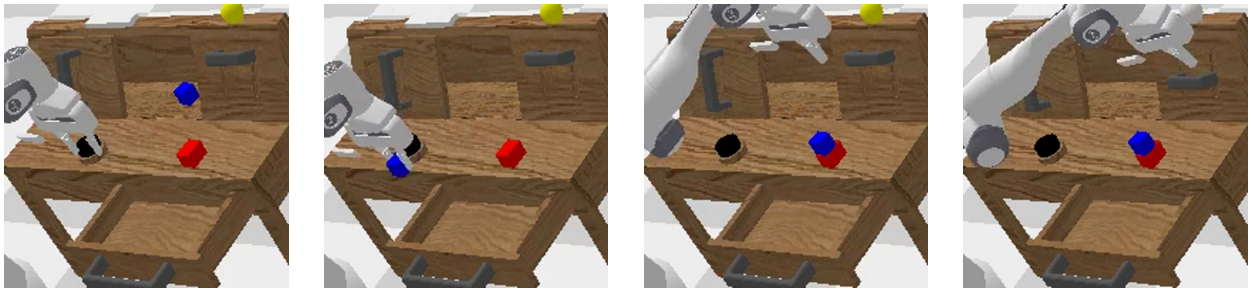
obs. 1 obs. 20 obs. 49 obs. 69

Claude Opus 5.5 Task failure | 70 code rounds | 933.6 s

Figure A7. Task Planning: a terminal episode need not be a completed task. Additional Opus 5.5 runs. (a) ALFWorld reports success after 8 code rounds under ceilings of 20 minutes, 8 rounds, 120 environment steps, 540 primitive calls, and 320k tokens. (b) DiscoveryWorld reports success after 7 rounds under 20 minutes, 8 rounds, 100 steps, 400 calls, and 320k tokens; its text/JSON interaction is illustrated with environment snapshots. (c) HumanCLAW stops after 70 rounds (933.6 s) without completing the couch-sitting task, under ceilings of 180 minutes, 101 rounds, 100 steps, 400 calls, and 2M tokens. The last case is a failed terminal state, not a wall-clock timeout. Frames are sampled saved observations, not continuous video.

(a) CALVIN | Sequence 0526

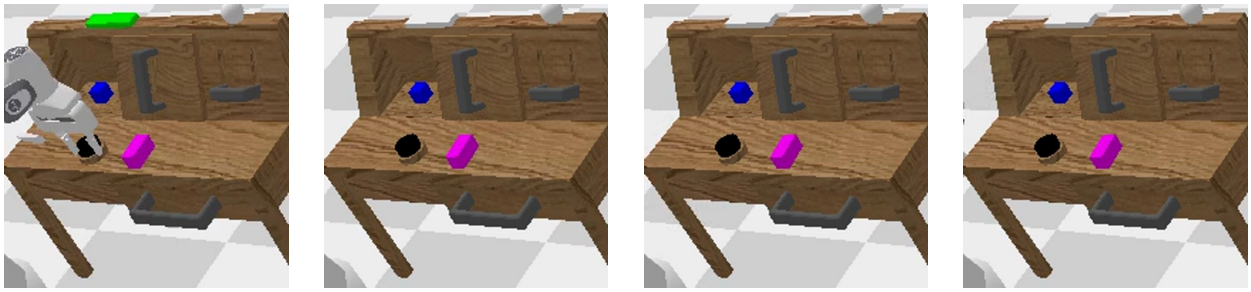
Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Reported success | 40-round ceiling | 229.4 s

(b) CALVIN | Sequence 0557

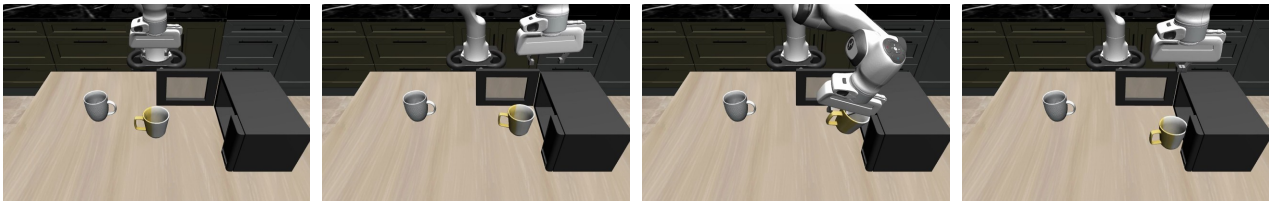
Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Round limit reached | 40 rounds | 978.6 s

(c) LIBERO-PRO | Mug and Microwave

Put the yellow-and-white mug in the microwave and close it.



Claude Opus 5.5 Round limit reached | 24 rounds | 1308.6 s

Figure A8. Manipulation: motion can continue without completing all goals. Additional Opus 5.5 runs through recorded source-native interfaces. (a) CALVIN sequence 0526 reports success; (b) sequence 0557 exhausts 40 model rounds. Both allow 7,000 primitive calls and 1.28M tokens. (c) LIBERO-PRO requires placing a mug inside the microwave and closing it; the run exhausts 24 model rounds, with ceilings of 256 primitive calls and 640k tokens. Outcomes come from the terminal reports. Frame labels index the encoded recording, whereas the seconds beside each outcome are wall-clock runtime. Different task instances and source interfaces are illustrated here, not a matched comparison of models.

H.1 Execution Across Models

Figures A9 and A10 pair task instructions and chronological frames with outcomes, model rounds, and elapsed runtime for the earlier nine-model cohort. ALFRED covers household goals; CALVIN requires five ordered manipulation subgoals. Tasks are selected separately for each model. DeepSeek-1017 shares the spray-bottle task with Astra and the CALVIN 0526 sequence with five other agents.

Completion, interaction, and time. Table A38 groups identical task identifiers. Fable completes the RoboCasa drawer task in 20 rounds; four other agents exhaust their budgets at 24, spanning 215.5–3,223.2 s. On LIBERO-PRO, Qwen-397B exhausts its budget while three agents complete the placement. On ALFWorld, Sol succeeds in fewer rounds than Fable (6 versus 7), but takes more than twice as long (433.8 versus 197.5 s). Thus, neither interaction count nor elapsed time alone establishes complete, timely task execution.

Execution settings. ALFRED and ALFWorld use recorded Baseline interfaces. DeepSeek-1017 CALVIN uses RGB-expanded observations with the source-specific ceilings in Table A15. CALVIN's original seven agents use RGB-expanded; the Opus 5.5, LIBERO-PRO, and RoboCasa records identify native primitives without a verified per-round image schedule. These episodes remain separate from Table 2's Baseline aggregate and the paired observation experiment. Runtime includes model and tool execution, not hardware-normalized inference alone. Budget exhaustion need not coincide with the model-round ceiling; the recorded termination is retained without attributing it to an unverified limit.

Table A38. Shared-task execution records. Rows within each block use the same task identifier and source interface. Rounds count recorded model responses; time is elapsed wall-clock runtime. ALFWorld allows 8 rounds, 120 environment steps, and 20 minutes. LIBERO-PRO and RoboCasa allow 24 rounds and 256 primitive calls, with token ceilings of 640k and 1.12M, respectively. These are selected episodes, not repeated-run averages.

Model	Rounds	Time (s)	Recorded outcome
ALFWorld: put a cleaned bar of soap on a counter			
GPT-6 Sol	6	433.8	Success
Claude Fable 5.1	7	197.5	Success
LIBERO-PRO: put the tomato sauce in the basket			
Claude Fable 5.1	16	1,553.5	Success
Gemini 3.8 Flash	15	429.0	Success
Qwen 3.5 397B-A17B	24	934.6	Budget exhausted
Claude Opus 5.5	20	2,376.8	Success
RoboCasa: open the left drawer			
GPT-6 Sol	24	735.5	Budget exhausted
Claude Fable 5.1	20	618.1	Success
Gemini 3.8 Flash	24	246.3	Budget exhausted
Qwen 3.8 Max	24	3,223.2	Budget exhausted
Qwen 3.5 27B	24	215.5	Budget exhausted

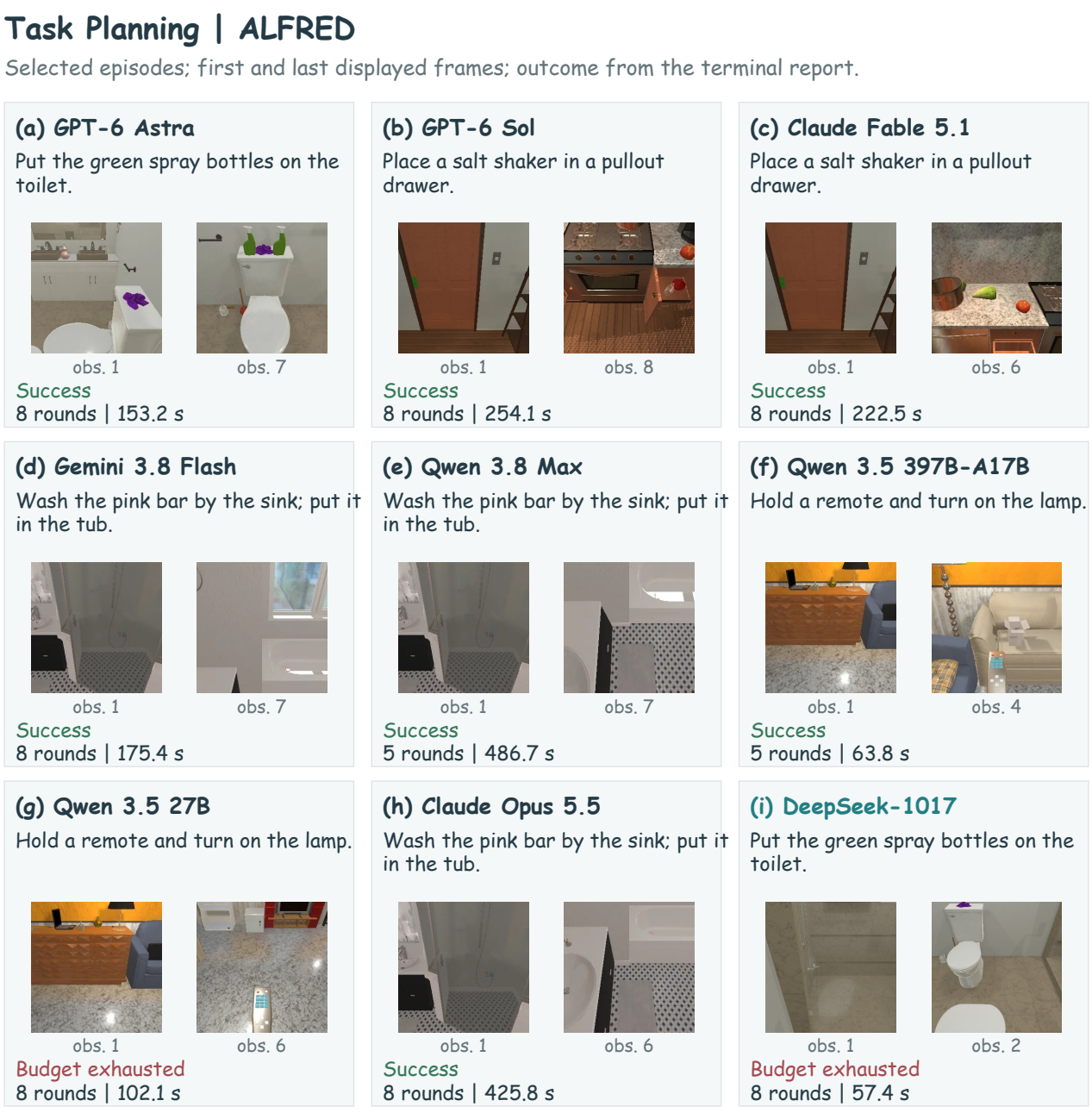


Figure A9. Nine-model task execution in ALFRED. Each panel shows the first and last displayed observations from an individual Baseline episode, with a shortened instruction, terminal outcome, rounds, and wall-clock runtime. All runs allow 8 model rounds, 120 environment steps, and 20 minutes. DeepSeek-1017 exhausts 8 rounds on the same spray-bottle task as Astra. The two Qwen models in panels (f,g) share the remote-and-lamp task: Qwen-397B reports success after 5 rounds, whereas Qwen-27B exhausts its budget at 8. Other panels cover separately selected tasks. Saved observations preserve temporal order with unequal intervals.

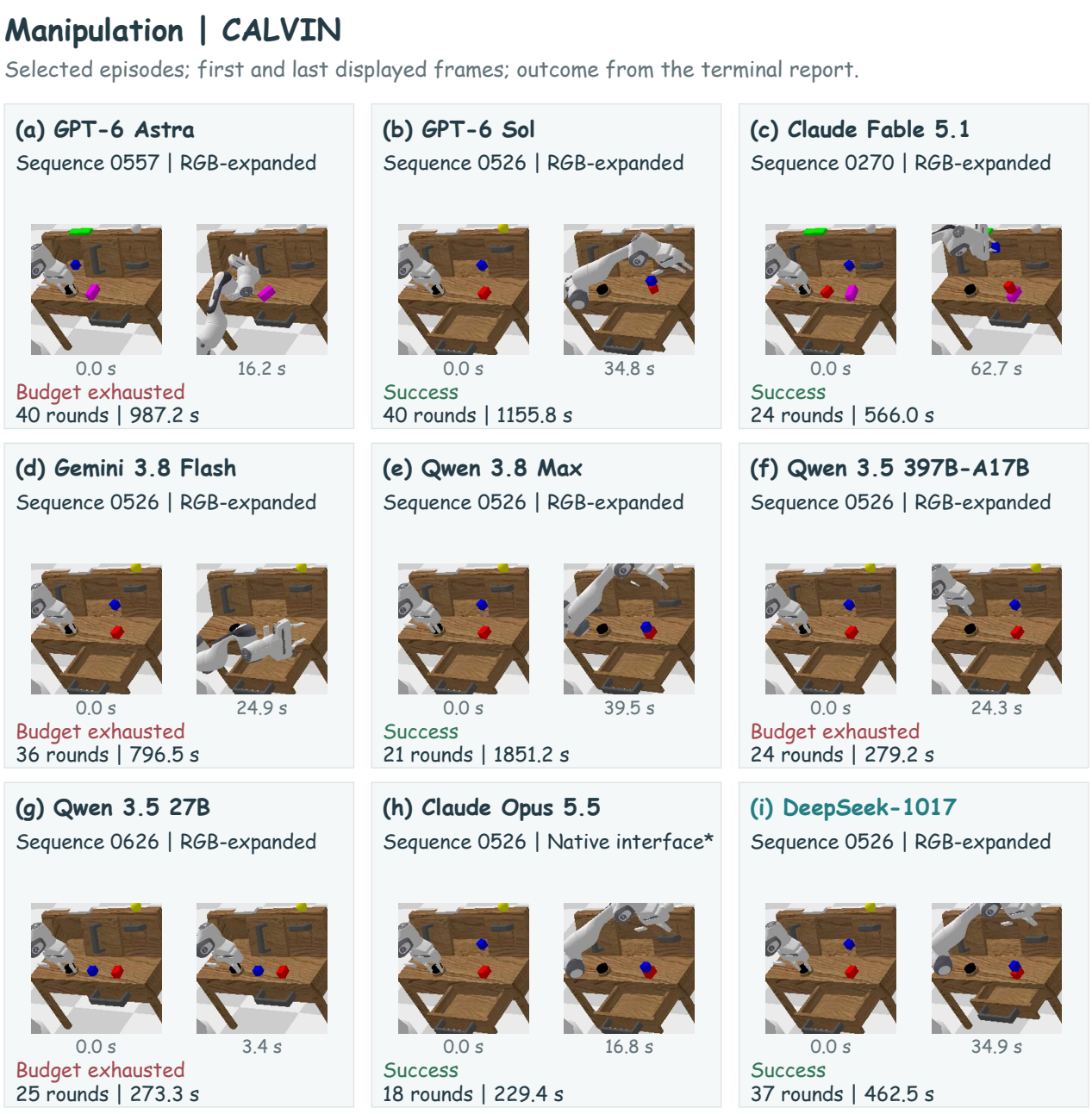


Figure A10. Nine-model execution of CALVIN sequences. Each sequence requires five ordered subgoals. Sequence 0526 requires retrieving the blue block from the sliding cabinet, storing it in the drawer, retrieving and stacking it, and switching off the bulb. Sol, Qwen Max, Opus 5.5, and DeepSeek-1017 report success on this sequence; Gemini and Qwen-397B report budget exhaustion. Each run allows 40 model rounds, 7,000 primitive calls, and 1.28M tokens. Panels (a–g,i) use RGB-expanded; *Opus 5.5 records native primitives without a verified per-round image schedule. DeepSeek-1017 completes the sequence in 37 recorded responses and 462.5 s, within a 120-minute episode ceiling. The other eight selected episodes retain their recorded limits. Each panel shows two selected frames; their labels give encoded-video time, while the outcome line gives wall-clock runtime. Tasks and observation settings vary across panels.

I From Recorded Actions to Complete Tasks

This analysis connects the aggregate scores to task overlap, resource allocation, and the state changes inside an episode. We use the earlier nine-model cohort with detailed logs: 1,413 Planning records, one episode per model on the same 157 assignments. The online analysis provides source filters, model-pair comparisons, and downloadable episode metrics. Its execution explorer links 143 recorded episodes in 46 same-task groups to model decisions, primitive actions, returned feedback, and terminal outcomes.

I.1 Complementary Strengths and Shared Failures

Astra completes 126 tasks. Taking the union of all nine models' successes covers 138 tasks, adding 12 beyond Astra; 19 tasks remain unsuccessful for every model. Only eight tasks are completed by all nine (Figure A11a). The common failure set contains four ALFRED, four ALFWorld, nine HumanCLAW, and two ScienceWorld tasks. The HumanCLAW subset includes all eight sitting tasks and one television task. This overlap distinguishes tasks on which models complement one another from those on which model selection alone does not resolve the observed failure. This union describes complementarity after execution, without testing a model-selection policy.

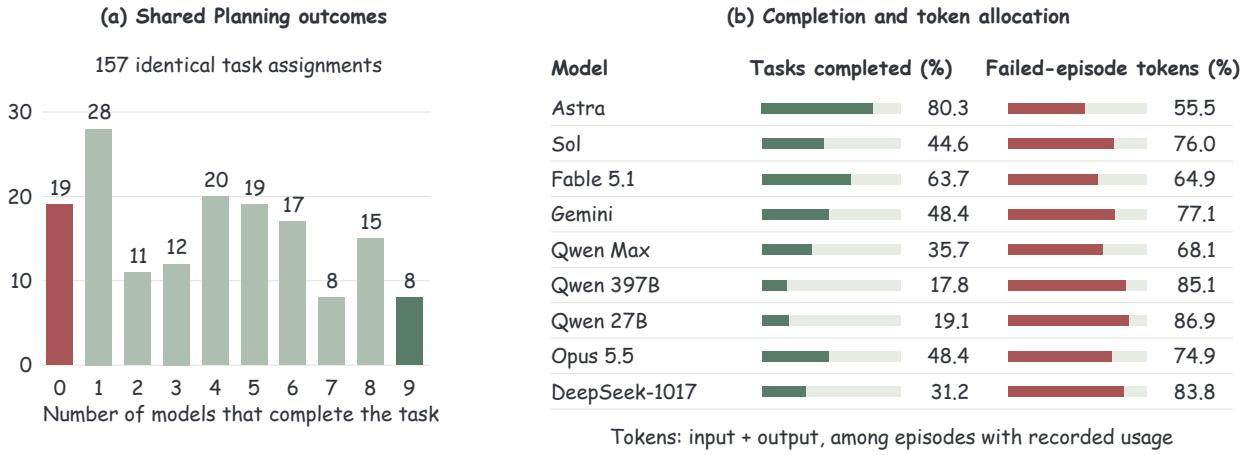


Figure A11. Shared Planning failures and the resources spent on them. (a) Number of tasks completed by exactly zero through nine models, over the same 157 assignments. (b) Completed-task percentage and the share of recorded tokens spent in unsuccessful episodes. Tokens sum recorded input and output usage; Table A39 gives usage coverage. All model names use the identifiers in Appendix B.2.

I.2 Completion, Latency, and Token Allocation

Unsuccessful episodes consume 55.5–86.9% of each model's recorded Planning tokens (Figure A11b). Astra completes 80.3% of its assignments, yet its 31 unsuccessful episodes account for 55.5% of recorded tokens. These attempts dominate resource use despite being a minority of its tasks. The median episode duration ranges from 57.1 s for DeepSeek-1017 to 280.7 s for Sol (Table A39). Short duration has to be read with completion: it can describe an efficient solution or an early unsuccessful stop.

The same HumanCLAW chair-navigation task illustrates why model rounds and elapsed time are separate measures. Astra completes in 18 rounds and 470.7 s, whereas DeepSeek-1017 takes 65 rounds but 405.7 s. In a selected BEHAVIOR-1K toolbox task, all eight recorded models fail after approximately 12–133 minutes. These episodes stop at token or round limits despite

Table A39. Planning resources and recovery over 157 tasks per model. Time is episode wall time in seconds; IQR spans the 25th–75th percentiles. Failed tokens is the percentage of recorded tokens spent in unsuccessful episodes; usage gives the number of episodes with token records, out of 157. Recovery counts completed episodes after an execution error, divided by episodes containing an error.

Model	Completed	Median (s)	IQR (s)	Failed tokens (%)	Usage	Recovery
Astra	126/157	231.7	153.2–361.8	55.5	157	1/1
Sol	70/157	280.7	199.9–427.1	76.0	157	1/6
Fable	100/157	109.5	62.0–201.4	64.9	157	6/25
Gemini	76/157	61.8	28.6–175.9	77.1	157	4/24
Qwen-Max	56/157	273.6	192.7–407.3	68.1	133	10/50
Qwen-397B	28/157	98.7	58.2–136.4	85.1	144	2/55
Qwen-27B	30/157	77.9	55.5–127.8	86.9	145	3/43
Opus 5.5	76/157	135.6	105.3–211.0	74.9	157	48/105
DeepSeek-1017	49/157	57.1	36.1–78.5	83.8	157	9/50

a 180-minute time ceiling. Increasing wall time alone would leave the observed stopping limits unchanged. The useful diagnostic is therefore the resource that actually ends an episode, together with the unsatisfied goal conditions.

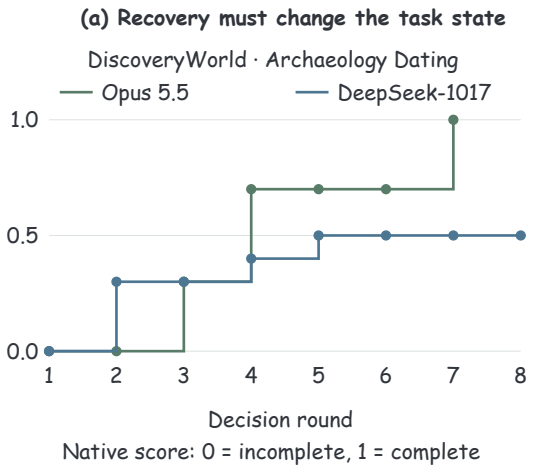
I.3 Recovery Matters When It Changes the Task State

Opus 5.5 completes 48 of the 105 Planning episodes containing a recorded execution error; these recoveries account for 48 of its 76 successes. Error records include malformed code and rejected API use in the evaluated interface. The Archaeology Dating task shows what successful recovery entails (Figure A12a). Opus 5.5's first two rounds fail before any environment action. It subsequently collects the tools, excavates the sites, reads the artifact ages, and places the flag at the oldest site. The native score reaches 1.0 in round seven. DeepSeek-1017 reaches 0.5 by round five, but its last three rounds end in code errors without further environment steps. A corrected response advances the episode only when the environment state moves toward the goal.

I.4 Physical Activity and Verified Completion

On CALVIN sequence 0526, Opus 5.5 completes five subgoals with 338 action submissions, 18 model calls, and 229.4 s. Gemini submits 501 actions and queries the state 464 times, yet ends at zero of five subgoals after 36 model calls and 796.5 s, exhausting its token budget. The retained subgoal queries show Opus 5.5 advancing through retrieval, storage, stacking, and the light switch; Gemini remains on the initial lifting goal. The common ceilings are 40 rounds, 7,000 tool calls, 1.28M tokens, and 120 minutes. Recorded API waiting accounts for 88.6% and 92.5% of their respective episode times. Action count, observation count, and model latency explain different parts of the execution.

ALFRED separates these quantities from goal satisfaction in another way. For the same instruction to clean a soap bar and place it in the bathtub, Gemini satisfies all three conditions, while DeepSeek-1017 ends with only one satisfied. Both make eight model calls. On RoboCasa's left-drawer task, Fable completes in 20 rounds, whereas Sol exhausts 24 rounds without the required final state. The manipulation recordings use their retained observation schedules and task settings; they explain individual executions without replacing the Baseline aggregates in Table 2. Together, the cases show why a generalist needs to preserve achieved conditions, verify the remaining ones, and direct further interaction toward them.



(b) More activity need not finish a sequence

CALVIN · Sequence 0526

Recorded quantity	Opus 5.5	Gemini
Action submissions	338	501
State observations	2	464
Model calls	18	36
Wall time (s)	229.4	796.5
Completed subgoals	5/5	0/5

40 rounds · 1.28M tokens · 7,000 tool calls

Figure A12. Verified progress separates recovery from repeated activity. (a) Native score after each recorded Archaeology Dating round; Opus 5.5 recovers from two initial code errors, while DeepSeek-1017 stalls at 0.5. (b) CALVIN sequence 0526: Opus 5.5 completes all five subgoals with fewer action submissions, state queries, model calls, and less elapsed time than Gemini. Action submissions count `submit_calvin_action`; observations count `observe_calvin_state`. These are recorded episode comparisons through source-native interfaces. Full settings, actions, and videos are linked in the online explorer.

I.5 Task Alignment and Measurement

Planning assignments are aligned by the shared task ID. HumanCLAW wrapper-specific IDs are checked against native task identity in all 117 records. Episode time uses the reported total duration, or the last recorded loop time when no total is retained. Token fractions exclude missing usage rather than substituting zero. Recovery requires native final success after an execution error; budget guards and timed-out rounds are excluded from the error count.

In the video explorer, code cells are matched to recorded rounds using timestamps and execution status. Of 1,506 retained cells, 1,468 match a round boundary; the other 38 remain separately inspectable. The 143 episodes contain 2,264 recorded rounds and 13,167 retained primitive calls. Per-round curves show native scores or satisfied goal conditions where those quantities are recorded. Encoded-video time and episode wall time remain distinct: recordings illustrate motion, while trace timestamps locate decisions and measured progress. Each displayed comparison links to its original reports, task IDs, and execution settings.

Reading the broader case collection. Static panels show first-run predictions; spatial cases exclude YOLOE. In point-and-box panels, contact labels test reference-mask membership and box IoU compares rectangles. Direct-polygon panels compare rasterized predictions with reference masks (Appendix C.5). Execution panels retain source-native interfaces and observation schedules; the selected Opus 5.5 reports do not establish the image schedule needed for the paired RGB intervention. Its main-table Baseline setting was confirmed separately. Four earlier HumanCLAW recordings with conflicting labels and feedback, and the Qwen-397B proximity demonstration, are excluded from evaluation. Opus 5.5 HumanCLAW statistics use the final 13 records in Appendix C.3. Planning panels show ordered saved observations; DiscoveryWorld snapshots illustrate text/JSON interaction. Manipulation frame labels use encoded-video time.

J Gemini 4 Argon: Coverage and Execution Settings

Gemini 4 Argon uses `gemini-4-argon`, recorded on 20–21 October 2026. The release contains 3,192 adopted trial records covering all 1,000 arena cases, plus 70 polygon submissions. Non-tracing Geometry, Spatial Reasoning, and point/box Affordance use five repetitions; tracing, polygons, Planning, and Manipulation use one. Continuous metrics average valid outputs within each run and then weight the five runs equally. Spatial inputs use corrected 3DSRBench image flips and native labels, without YOLOE. Tables 2 and A8 include these results alongside the unchanged earlier models.

Argon scores 71.6% ± 0.7 Spatial Pass@1, 0.356 ± 0.009 box–mask IoU, and 0.441 polygon IoU. Its 3D path RMSE is 0.310 m, with 24/56 joint successes (42.9%). Stronger geometric and regional predictions therefore still leave most 3D trace constraints unsatisfied. In Planning it completes 110/157 tasks (70.1%): 14/20 ALFRED, 16/20 ALFWorld, 6/10 DiscoveryWorld, 2/13 HumanCLAW, 52/73 ScienceWorld, and 20/21 VirtualHome. Both HumanCLAW successes are navigation; all eight sitting tasks remain incomplete.

Manipulation uses current RGB at every decision and the source-specific budgets below. It completes 80/182 binary tasks (44.0%); Anymal's continuous-reward case remains separate. CALVIN averages 3.30 consecutive goals with 4/10 complete sequences. These results extend the main evaluation, while the paired observation experiments in Table 3 retain their original model cohort. The execution-log analysis in Appendix I covers the earlier nine models.

Table A40. Gemini 4 Argon Manipulation: task success and per-episode ceilings. Tokens include reasoning and context; rounds are decision interactions. One episode per binary assignment.

Source	Success	SR (%)	Rounds	Tokens (k)	Time (s)
CALVIN	4/10	40.0	40	1,280	7,200
BEHAVIOR-1K	0/5	0.0	24	640	10,800
LIBERO-PRO	8/10	80.0	24	640	10,800
robosuite	5/5	100.0	24	640	10,800
CLIPort	7/18	38.9	24	960	3,600
ManiSkill	5/9	55.6	32	640	3,600
RLBench	8/10	80.0	16	640	3,600
RoboCasa	2/10	20.0	24	1,120	3,600
RoboCasa365	1/10	10.0	24	1,120	3,600
RoboTwin 2.0	14/50	28.0	16	640	3,600
RoboWits	1/10	10.0	24	640	3,600
VIMA	9/10	90.0	16	640	3,600
VLABench	16/25	64.0	16	640	3,600